

# JESS Thermodynamic Database v8.9

## Thorium

24-Jan-23

| Reaction No. 2751               |    |       |         |                 |   |       |        |
|---------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Th+4 + H+1_F-1 = Th+4_F-1 + H+1 |    |       |         |                 |   |       |        |
| No                              | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 25 | 0.5   | NaClO4  | lgK:4.70(3SF)   | 5 | MRX   | 366    |
| 2                               | 25 | 0.5   | NaClO4  | lgK:4.7(0.05SD) | 5 | CMN   | 437    |
| 3                               | 25 | 0.5   | NaClO4  | lgK:4.65(3SF)   | 5 | MRX   | 463    |
| 4                               | 25 | 0.5   | Unknown | lgK:4.63(3SF)   | 3 | MDS   | 437    |
| 5                               | 25 | 1     | NaClO4  | lgK:4.58(3SF)   | 0 | MIS   | 437    |
| 6                               | 25 | 3     | NaClO4  | lgK:4.52(3SF)   | 5 | MIS   | 5398   |
| 7                               | 20 | 4     | HClO4   | lgR:4.62(3SF)   | 5 | MDS   | 5398   |
| 8                               | 20 | 4     | HClO4   | lgR:4.68(3SF)   | 5 | MIS   | 5398   |
| 9                               | 25 | 1     | HClO4   | lgR:4.49(3SF)   | 3 | MDS   | 437    |
| 10                              | 25 | 1     | HClO4   | lgR:4.51(3SF)   | 3 | MIS   | 437    |
| 11                              | 25 | 2     | HClO4   | lgR:4.43(3SF)   | 4 | MDS   | 437    |
| 12                              | 25 | 4     | NaClO4  | dH:3.4(0.1SD)   | 5 | MCL   | 2261   |

| Reaction No. 2752                      |    |       |         |                 |   |       |        |
|----------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Th+4_F-1 + H+1_F-1 = Th+4_F-1(2) + H+1 |    |       |         |                 |   |       |        |
| No                                     | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                      | 25 | 0.5   | NaClO4  | lgK:2.76(3SF)   | 5 | MRX   | 366    |
| 2                                      | 25 | 0.5   | NaClO4  | lgK:2.8(0.05SD) | 5 | CMN   | 437    |
| 3                                      | 25 | 0.5   | NaClO4  | lgK:2.81(3SF)   | 5 | MRX   | 463    |
| 4                                      | 25 | 0.5   | Unknown | lgK:2.86(3SF)   | 3 | MDS   | 437    |
| 5                                      | 25 | 1     | NaClO4  | lgK:2.76(3SF)   | 0 | MIS   | 437    |
| 6                                      | 25 | 3     | NaClO4  | lgK:2.66(3SF)   | 4 | MIS   | 437    |
| 7                                      | 20 | 4     | HClO4   | lgR:2.97(3SF)   | 5 | MIS   | 5398   |
| 8                                      | 25 | 4     | NaClO4  | dH:3.1(0.1SD)   | 5 | MCL   | 2261   |

| Reaction No. 2753                         |    |       |        |                 |   |       |        |
|-------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Th+4_F-1(2) + H+1_F-1 = Th+4_F-1(3) + H+1 |    |       |        |                 |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                         | 25 | 0.5   | NaClO4 | lgK:1.5(0.07SD) | 5 | CMN   | 437    |

|   |    |     |                    |               |   |     |     |
|---|----|-----|--------------------|---------------|---|-----|-----|
| 2 | 25 | 0.5 | NaClO <sub>4</sub> | lgK:1.51(3SF) | 3 | MRX | 463 |
| 3 | 25 | 1   | NaClO <sub>4</sub> | lgK:1.41(3SF) | 3 | MIS | 437 |
| 4 | 25 | 3   | NaClO <sub>4</sub> | lgK:1.6(2SF)  | 3 | MIS | 437 |
| 5 | 25 | 1   | HClO <sub>4</sub>  | lgR:1.41(3SF) | 3 | MIS | 437 |
| 6 | 25 | 3   | HClO <sub>4</sub>  | lgR:1.6(2SF)  | 3 | MIS | 437 |

| Reaction No. 2845                     |       |       |                    |                  |   |       |        |
|---------------------------------------|-------|-------|--------------------|------------------|---|-------|--------|
| Th+4 + H+1_CrO4-2 = Th+4_CrO4-2 + H+1 |       |       |                    |                  |   |       |        |
| No                                    | t     | I-Str | Medium             | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 11    | 0.2   | LiClO <sub>4</sub> | lgK:0.53(2SF)    | 5 | MGL   | 590    |
| 2                                     | 17.3  | 0.2   | LiClO <sub>4</sub> | lgK:0.59(2SF)    | 5 | MGL   | 590    |
| 3                                     | 25    | 0     | Inf. Dilution      | lgK:0.7(1SF)     | 1 | ESC   |        |
| 4                                     | 25.7  | 0.2   | LiClO <sub>4</sub> | lgK:0.67(0.14SD) | 5 | MSP   | 590    |
| 5                                     | 11:26 | 0.2   | LiClO <sub>4</sub> | dH:3.67(0.04SD)  | 5 | MSP   | 590    |
| 6                                     | 11:26 | 1     | LiClO <sub>4</sub> | dH:3.91(0.36SD)  | 5 | MCL   | 590    |

| Reaction No. 3082                         |    |       |         |               |   |       |        |
|-------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Th+4_NTA-3 + 2<OH-1> = Th+4_OH-1(2)_NTA-3 |    |       |         |               |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | Unknown | lgK:19.0(3SF) | 4 | ENS   | 773    |

| Reaction No. 3158               |    |       |                  |                   |   |       |        |
|---------------------------------|----|-------|------------------|-------------------|---|-------|--------|
| HOEDTA-3 + Th+4 = Th+4_HOEDTA-3 |    |       |                  |                   |   |       |        |
| No                              | t  | I-Str | Medium           | Value (Dev)       | W | Tech. | Ref. # |
| 1                               | 20 | 0.1   | Unknown          | lgK:19.24(0.01SD) | 6 | MIX   | 999    |
| 2                               | 25 | 0.1   | KNO <sub>3</sub> | lgK:18.5(3SF)     | 4 | MSP   | 999    |
| 3                               | 25 | 0.1   | KNO <sub>3</sub> | dH:-1.05(3SF)     | 5 | MTD   | 3800   |

| Reaction No. 3159                         |    |       |         |              |   |       |        |
|-------------------------------------------|----|-------|---------|--------------|---|-------|--------|
| Th+4_HOEDTA-3 + OH-1 = Th+4_OH-1_HOEDTA-3 |    |       |         |              |   |       |        |
| No                                        | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | Unknown | lgK:8.4(2SF) | 4 | ENS   | 773    |

| Reaction No. 3433             |    |       |                    |                |   |       |        |
|-------------------------------|----|-------|--------------------|----------------|---|-------|--------|
| Oxine-1 + Th+4 = Th+4_Oxine-1 |    |       |                    |                |   |       |        |
| No                            | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 20 | 0.1   | NaClO <sub>4</sub> | lgK:11.70(4SF) | 0 | MGL   | 6788   |

| Note (1): Low wgt for internal consistency |    |     |             |                  |   |     |      |
|--------------------------------------------|----|-----|-------------|------------------|---|-----|------|
| 2                                          | 25 | 0.1 | ClO4-1 salt | lgK:10.45(4SF)   | 5 | MDS | 999  |
| 3                                          | 25 | 0.1 | Unknown     | lgK:10.5(3SF)    | 4 | ENS | 773  |
| 4                                          | 25 | 5   | NaCl        | lgK:10.5(0.09SD) | 5 | MPS | 7144 |

| Reaction No. 3434                   |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 2<Oxine-1> + Th+4 = Th+4_Oxine-1(2) |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | NaClO4  | lgK:21.3(3SF) | 4 | MSL   | 999    |
| 2                                   | 25 | 0.1   | Unknown | lgK:20.4(3SF) | 4 | ENS   | 773    |

| Reaction No. 3435                   |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 3<Oxine-1> + Th+4 = Th+4_Oxine-1(3) |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | Unknown | lgK:29.9(3SF) | 4 | ENS   | 773    |

| Reaction No. 3436                   |    |       |         |               |   |       |        |
|-------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 4<Oxine-1> + Th+4 = Th+4_Oxine-1(4) |    |       |         |               |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | Unknown | lgK:38.8(3SF) | 4 | ENS   | 773    |

| Reaction No. 3466               |    |       |        |               |   |       |        |
|---------------------------------|----|-------|--------|---------------|---|-------|--------|
| Sulfox-2 + Th+4 = Th+4_Sulfox-2 |    |       |        |               |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | KNO3   | lgK:9.56(3SF) | 6 | MGL   | 999    |

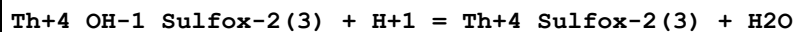
| Reaction No. 3467                     |    |       |         |                |   |       |        |
|---------------------------------------|----|-------|---------|----------------|---|-------|--------|
| 2<Sulfox-2> + Th+4 = Th+4_Sulfox-2(2) |    |       |         |                |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | Unknown | lgK:18.29(4SF) | 6 | ENS   | 773    |

| Reaction No. 3468                     |    |       |         |                |   |       |        |
|---------------------------------------|----|-------|---------|----------------|---|-------|--------|
| 3<Sulfox-2> + Th+4 = Th+4_Sulfox-2(3) |    |       |         |                |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | Unknown | lgK:25.91(4SF) | 6 | ENS   | 773    |

| Reaction No. 3469                     |  |  |  |  |  |  |  |
|---------------------------------------|--|--|--|--|--|--|--|
| 4<Sulfox-2> + Th+4 = Th+4_Sulfox-2(4) |  |  |  |  |  |  |  |

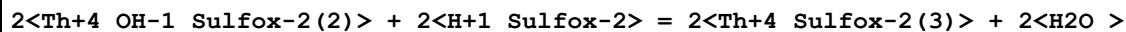
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:32.02 (4SF) | 6 | ENS   | 773    |

## Reaction No. 3470



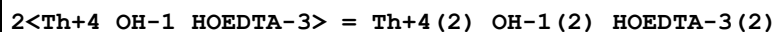
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:5.0 (2SF) | 1 | ELC   |        |
| 2  | 25 | 0.1   | KNO3          | lgK:6.2 (2SF) | 0 | MGL   | 999    |

## Reaction No. 3471



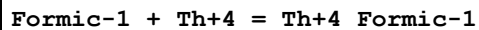
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:8.9 (2SF) | 4 | ENS   | 773    |

## Reaction No. 3519



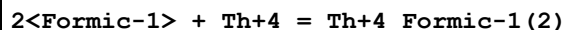
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:5.2 (2SF) | 4 | ENS   | 773    |

## Reaction No. 3673



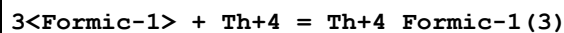
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:3.09 (0.03SD) | 4 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.1 (2SF)     | 1 | ESC   |        |

## Reaction No. 3674



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:5.1 (0.05SD) | 4 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:5.1 (2SF)    | 1 | ESC   |        |

## Reaction No. 3675



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:6.7 (0.08SD) | 4 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:6.7 (2SF)    | 1 | ESC   |        |

## Reaction No. 3720

| Oxalic-2 + Th+4 = Th+4_Oxalic-2            |    |       |               |                  |   |       |        |
|--------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| No                                         | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                          | 23 | 0.5   | NaNO3         | lgK:8.45 (3SF)   | 5 | MSP   | 906    |
| 2                                          | 25 | 0     | Inf. Dilution | lgK:11.05 (4SF)  | 1 | EOR   |        |
| 3                                          | 25 | 0     | Inf. Dilution | lgK:7.16 (3SF)   | 0 | MKN   | 931    |
| 4                                          | 25 | 0     | Inf. Dilution | lgK:10.6 (3SF)   | 3 | MSL   | 931    |
| 5                                          | 25 | 0     | Inf. Dilution | lgK:10.60 (4SF)  | 4 | CRV   | 31301  |
| 6                                          | 25 | 0.05  | Unknown       | lgK:8.80 (0.1SD) | 0 | MGL   | 906    |
| Note (6): Low wgt for internal consistency |    |       |               |                  |   |       |        |
| 7                                          | 25 | 0.05  | Unknown       | lgK:8.81 (0.1SD) | 0 | MHG   | 906    |
| Note (7): Low wgt for internal consistency |    |       |               |                  |   |       |        |
| 8                                          | 25 | 0.1   | Unknown       | lgK:9.22 (3SF)   | 5 | MSL   | 906    |
| 9                                          | 25 | 1     | HClO4/m       | lgK:8.00 (3SF)   | 3 | EOD   | 6366   |
| 10                                         | 25 | 1     | NH4ClO4       | lgK:8.23 (3SF)   | 4 | MSL   | 931    |
| 11                                         | 25 | 1     | NaClO4        | lgK:7.857 (4SF)  | 3 | MDS   | 4706   |
| 12                                         | 25 | 1     | HClO4/m       | dH:-3.38 (3SF)   | 3 | EOD   | 6366   |

| Reaction No. 3721                     |    |       |               |                  |   |       |        |
|---------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<Oxalic-2> + Th+4 = Th+4_Oxalic-2(2) |    |       |               |                  |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 23 | 0.5   | NaNO3         | lgK:15.43 (4SF)  | 0 | MSP   | 906    |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:18.17 (4SF)  | 1 | EOR   |        |
| 3                                     | 25 | 0     | Inf. Dilution | lgK:20.2 (3SF)   | 3 | MSL   | 931    |
| 4                                     | 25 | 0     | Inf. Dilution | lgK:20.20 (4SF)  | 4 | CRV   | 31301  |
| 5                                     | 25 | 1     | (H,Na)ClO4    | lgK:14.117 (5SF) | 0 | MDS   | 4706   |
| 6                                     | 25 | 1     | NH4ClO4       | lgK:16.8 (3SF)   | 4 | MSL   | 931    |

| Reaction No. 3722                     |    |       |               |                  |   |       |        |
|---------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 3<Oxalic-2> + Th+4 = Th+4_Oxalic-2(3) |    |       |               |                  |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:27.69 (4SF)  | 1 | EOR   |        |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:26.4 (3SF)   | 3 | MSL   | 931    |
| 3                                     | 25 | 0     | Inf. Dilution | lgK:26.40 (4SF)  | 4 | CRV   | 31301  |
| 4                                     | 25 | 1     | NH4ClO4       | lgK:22.8 (3SF)   | 4 | MSL   | 931    |
| 5                                     | 25 | 1     | NaClO4        | lgK:19.939 (5SF) | 0 | MDS   | 4706   |
| 6                                     | 30 | 0.4   | Unknown       | lgK:24.48 (4SF)  | 2 | MGL   | 140    |

| Reaction No. 3723                       |    |       |         |                 |   |       |        |
|-----------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Th+4 + H+1_Oxalic-2 = Th+4_H+1_Oxalic-2 |    |       |         |                 |   |       |        |
| No                                      | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                       | 25 | 0.05  | Unknown | lgK:7.36(0.1SD) | 4 | MGL   | 906    |

| Reaction No. 3752               |    |       |               |                 |   |       |        |
|---------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| ClAcet-1 + Th+4 = Th+4_ClAcet-1 |    |       |               |                 |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 20 | 1     | NaClO4        | lgK:2.8(0.03SD) | 4 | MQH   | 2109   |
| 2                               | 25 | 0     | Inf. Dilution | lgK:3.0(2SF)    | 1 | ESC   |        |
| 3                               | 25 | 0.5   | HClO4         | lgK:2.98(3SF)   | 6 | ENS   | 999    |

| Reaction No. 3753                     |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 2<ClAcet-1> + Th+4 = Th+4_ClAcet-1(2) |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 20 | 1     | NaClO4        | lgK:4.6(0.1SD) | 4 | MQH   | 2109   |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:4.6(2SF)   | 1 | ESC   |        |

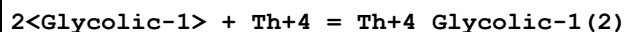
| Reaction No. 3754                     |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 3<ClAcet-1> + Th+4 = Th+4_ClAcet-1(3) |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 20 | 1     | NaClO4        | lgK:5.7(0.2SD) | 4 | MQH   | 2109   |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:5.8(2SF)   | 1 | ESC   |        |

| Reaction No. 3755                     |    |       |               |               |   |       |        |
|---------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 4<ClAcet-1> + Th+4 = Th+4_ClAcet-1(4) |    |       |               |               |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 20 | 1     | NaClO4        | lgK:6.79(3SF) | 6 | MQH   | 2109   |
| 2                                     | 20 | 1     | Unknown       | lgK:6.8(2SF)  | 4 | ENS   | 773    |
| 3                                     | 25 | 0     | Inf. Dilution | lgK:7.0(2SF)  | 1 | ESC   |        |

| Reaction No. 3780                   |    |       |               |                 |   |       |        |
|-------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Glycolic-1 + Th+4 = Th+4_Glycolic-1 |    |       |               |                 |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 20 | 1     | NaClO4        | lgK:4.0(0.03SD) | 4 | MQH   | 3132   |
| 2                                   | 25 | 0     | Inf. Dilution | lgK:4.97(3SF)   | 4 | EBU   | 6372   |
| 3                                   | 25 | 1     | NaClO4        | lgK:4.11(3SF)   | 5 | MGL   | 7703   |

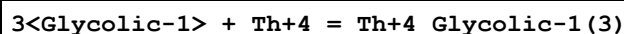
|   |    |   |                    |                  |   |     |      |
|---|----|---|--------------------|------------------|---|-----|------|
| 4 | 25 | 1 | NaClO <sub>4</sub> | dH:2.09 (3SF) kJ | 5 | MCL | 7703 |
|---|----|---|--------------------|------------------|---|-----|------|

## Reaction No. 3781



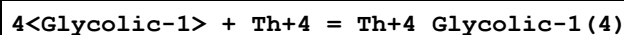
| No | t  | I-Str | Medium             | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO <sub>4</sub> | lgK:7.4 (0.04SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution      | lgK:8.86 (3SF)   | 4 | EBU   | 6372   |
| 3  | 25 | 1     | NaClO <sub>4</sub> | lgK:7.45 (3SF)   | 5 | MGL   | 7703   |
| 4  | 25 | 1     | NaClO <sub>4</sub> | dH:1.26 (3SF) kJ | 5 | MCL   | 7703   |

## Reaction No. 3782



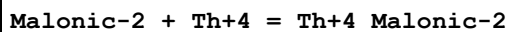
| No | t  | I-Str | Medium             | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------------------|-------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO <sub>4</sub> | lgK:9.9 (0.04SD)  | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution      | lgK:11.91 (4SF)   | 4 | EBU   | 6372   |
| 3  | 25 | 1     | NaClO <sub>4</sub> | lgK:10.18 (4SF)   | 5 | MGL   | 7703   |
| 4  | 25 | 1     | NaClO <sub>4</sub> | dH:-1.71 (3SF) kJ | 5 | MCL   | 7703   |

## Reaction No. 3783



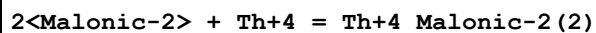
| No | t  | I-Str | Medium             | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------------------|-------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO <sub>4</sub> | lgK:12.0 (0.1SD)  | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution      | lgK:14.23 (4SF)   | 3 | EBU   | 6372   |
| 3  | 25 | 1     | NaClO <sub>4</sub> | lgK:11.97 (4SF)   | 5 | MGL   | 7703   |
| 4  | 25 | 1     | NaClO <sub>4</sub> | dH:-5.61 (3SF) kJ | 5 | MCL   | 7703   |

## Reaction No. 3828



| No | t  | I-Str | Medium             | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------------------|-------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO <sub>4</sub> | lgK:7.4 (0.04SD)  | 5 | MGL   | 3675   |
| 2  | 25 | 0     | Inf. Dilution      | lgK:7.25 (0.01SD) | 6 | MKN   | 999    |
| 3  | 25 | 1     | NaClO <sub>4</sub> | lgK:7.474 (4SF)   | 6 | MGL   | 4670   |
| 4  | 25 | 1     | NaClO <sub>4</sub> | dH:11.88 (4SF) kJ | 5 | MCL   | 1684   |
| 5  | 25 | 1     | NaClO <sub>4</sub> | dH:2.8 (0.03SD)   | 4 | MCL   | 4670   |

## Reaction No. 3829



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:12.7(0.08SD) | 5 | MGL   | 3675   |
| 2  | 25 | 0     | Inf. Dilution | lgK:13.8(3SF)    | 1 | ELC   |        |
| 3  | 25 | 1     | KCl           | lgK:11.82(4SF)   | 0 | MGL   | 1021   |
| 4  | 25 | 1     | NaClO4        | lgK:12.792(5SF)  | 5 | MGL   | 4670   |
| 5  | 25 | 1     | NaClO4        | dH:2.0(0.1SD)    | 4 | MCL   | 4670   |

## Reaction No. 3857

Lactic-1 + Th+4 = Th+4\_Lactic-1

| No | t  | I-Str | Medium                                         | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|------------------------------------------------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO4                                         | lgK:4.16(3SF)    | 5 | MGL   | 6788   |
| 2  | 20 | 1     | NaClO4                                         | lgK:4.21(0.02SD) | 4 | MQH   | 3132   |
| 3  | 25 | 0     | Inf. Dilution                                  | lgK:5.49(3SF)    | 5 | MPT   | 906    |
| 4  | 25 | 0.05  | NaClO4                                         | lgK:4.8(0.2SD)   | 4 | MPT   | 906    |
| 5  | 25 | 0     | Ethanol:Water<br>19.8:80.2Wgt<br>Inf. Dilution | lgK:6.27(3SF)    | 5 | MPT   | 906    |
| 6  | 25 | 0.05  | Ethanol:Water<br>19.8:80.2Wgt<br>NaClO4        | lgK:5.(0.2SD)    | 5 | MPT   | 906    |
| 7  | 25 | 0     | Ethanol:Water<br>40:60Wgt<br>Inf. Dilution     | lgK:6.91(3SF)    | 5 | MPT   | 906    |
| 8  | 25 | 0.05  | Ethanol:Water<br>40:60Wgt<br>NaClO4            | lgK:5.8(0.1SD)   | 5 | MPT   | 906    |

## Reaction No. 3869

Maleic-2 + Th+4 = Th+4\_Maleic-2

| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:6.3(0.03SD) | 6 | MGL   | 3675   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.0(2SF)    | 1 | ELC   |        |

## Reaction No. 3870

2&lt;Maleic-2&gt; + Th+4 = Th+4\_Maleic-2(2)

| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:10.6(0.06SD) | 6 | MGL   | 3675   |
| 2  | 25 | 0     | Inf. Dilution | lgK:13.5(3SF)    | 1 | ELC   |        |

## Reaction No. 3896



| Succinic-2 + Th+4 = Th+4_Succinic-2 |    |       |               |                  |   |       |        |
|-------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| No                                  | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 20 | 1     | NaClO4        | lgK:6.23(0.01SD) | 6 | MGL   | 3675   |
| 2                                   | 25 | 0     | Inf. Dilution | lgK:8.8(2SF)     | 1 | EDH   |        |
| 3                                   | 25 | 0.5   | Unknown       | lgK:8.38(3SF)    | 0 | MSL   | 906    |
| 4                                   | 25 | 1     | NaClO4        | lgK:6.44(3SF)    | 6 | MGL   | 1078   |
| 5                                   | 25 | 1     | NaClO4        | dH:18.6(3SF) kJ  | 5 | MCL   | 1078   |

| Reaction No. 3993                             |    |       |         |               |   |       |        |
|-----------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Th+4 + H+1_Salicylic-2 = Th+4_H+1_Salicylic-2 |    |       |         |               |   |       |        |
| No                                            | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | Unknown | lgK:4.25(3SF) | 6 | ENS   | 773    |

| Reaction No. 4006                 |    |       |               |                   |   |       |        |
|-----------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| SulfSal-3 + Th+4 = Th+4_SulfSal-3 |    |       |               |                   |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 20 | 0.1   | NaClO4        | lgK:11.97(4SF)    | 6 | MGL   | 6788   |
| 2                                 | 20 | 0.1   | Unknown       | lgK:14.30(4SF)    | 0 | ENS   | 1539   |
| 3                                 | 20 | 1     | NaClO4        | lgK:12.30(0.02SD) | 5 | MGL   | 906    |
| 4                                 | 25 | 0     | Inf. Dilution | lgK:12.5(3SF)     | 1 | ESC   |        |

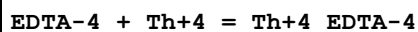
| Reaction No. 4039                   |    |       |               |                 |   |       |        |
|-------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Phthalic-2 + Th+4 = Th+4_Phthalic-2 |    |       |               |                 |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 20 | 1     | NaClO4        | lgK:5.9(0.03SD) | 5 | MGL   | 3675   |
| 2                                   | 25 | 0     | Inf. Dilution | lgK:5.9(2SF)    | 1 | ESC   |        |

| Reaction No. 4040                         |    |       |               |                  |   |       |        |
|-------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<Phthalic-2> + Th+4 = Th+4_Phthalic-2(2) |    |       |               |                  |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 20 | 1     | NaClO4        | lgK:10.1(0.07SD) | 5 | MGL   | 3675   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:10.1(3SF)    | 1 | ESC   |        |

| Reaction No. 4061                   |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Mandelic-1 + Th+4 = Th+4_Mandelic-1 |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 20 | 1     | NaClO4 | lgK:3.88(0.02SD) | 4 | MQH   | 3132   |

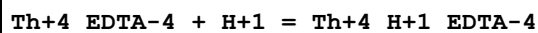
|   |    |   |               |              |   |     |  |
|---|----|---|---------------|--------------|---|-----|--|
| 2 | 25 | 0 | Inf. Dilution | lgK:3.9(2SF) | 1 | ESC |  |
|---|----|---|---------------|--------------|---|-----|--|

## Reaction No. 4196



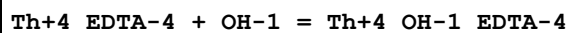
| No | t  | I-Str | Medium             | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------------------|------------------|---|-------|--------|
| 1  | 20 | 0.1   | KClO <sub>4</sub>  | lgK:23.3(0.04SD) | 5 | MDS   | 903    |
| 2  | 20 | 0.1   | KNO <sub>3</sub>   | lgK:23.2(3SF)    | 3 | MPL   | 136    |
| 3  | 20 | 0.1   | KNO <sub>3</sub>   | lgK:23.25(4SF)   | 3 | ENS   | 2373   |
| 4  | 20 | 0.1   | Many Media         | lgK:23.3(0.1SD)  | 7 | EIU   | 903    |
| 5  | 20 | 0.1   | NaClO <sub>4</sub> | lgK:24.45(4SF)   | 0 | MRX   | 903    |
| 6  | 20 | 0.1   | NaClO <sub>4</sub> | lgK:25.3(3SF)    | 0 | MRX   | 931    |
| 7  | 20 | 0.1   | NaClO <sub>4</sub> | lgK:21.17(4SF)   | 0 | MGL   | 6788   |
| 8  | 20 | 0.5   | NaCl               | lgK:21.8(0.04SD) | 0 | MGL   | 903    |
| 9  | 21 | 0.2   | NaClO <sub>4</sub> | lgK:25.1(3SF)    | 0 | MSP   | 2884   |
| 10 | 22 | 0.5   | HCl                | lgK:21.3(3SF)    | 0 | MDS   | 903    |
| 11 | 25 | 0     | Inf. Dilution      | lgK:26.63(4SF)   | 1 | EOR   |        |
| 12 | 25 | 0     | Inf. Dilution      | lgK:26.63(4SF)   | 4 | CRV   | 31301  |
| 13 | 25 | 0.5   | Na+1 salt          | lgK:22.3(3SF)    | 2 | CRV   | 552    |
| 14 | 25 | 0.1   | KNO <sub>3</sub>   | dH:-12.9(3SF) kJ | 5 | MCL   | 1888   |
| 15 | 25 | 0.1   | KNO <sub>3</sub>   | dH:-2.75(3SF)    | 5 | MTD   | 3800   |
| 16 | 25 | 0.1   | KNO <sub>3</sub>   | dH:-3.1(0.2SD)   | 5 | MTD   | 3800   |
| 17 | 25 | 0.1   | Unknown            | dH:-2.9(2SF)     | 6 | CRV   | 552    |

## Reaction No. 4197



| No | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO <sub>4</sub> | lgK:1.98(3SF)  | 5 | MRX   | 142    |
| 2  | 20 | 0.1   | NaClO <sub>4</sub> | lgK:2.0(0.1SD) | 7 | EIU   | 903    |
| 3  | 25 | 0     | Inf. Dilution      | lgK:2.0(2SF)   | 1 | ESC   |        |
| 4  | 25 | 0.1   | KNO <sub>3</sub>   | dH:0.3(1SF) kJ | 5 | MCL   | 1888   |

## Reaction No. 4198



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:6.74(3SF) | 3 | ENS   | 773    |

## Reaction No. 4256

| 2<Th+4_EDTA-4> = Th+4(2)_EDTA-4(2) |    |       |         |              |   |       |        |
|------------------------------------|----|-------|---------|--------------|---|-------|--------|
| No                                 | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                  | 25 | 0.1   | Unknown | lgK:4.3(2SF) | 4 | ENS   | 773    |

| Reaction No. 4272                   |    |       |         |                  |   |       |        |
|-------------------------------------|----|-------|---------|------------------|---|-------|--------|
| Th+4_DTPA-5 + H+1 = Th+4_H+1_DTPA-5 |    |       |         |                  |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | NaClO4  | lgK:2.16(3SF)    | 6 | MRX   | 999    |
| 2                                   | 20 | 0.1   | NaClO4  | lgK:2.16(0.05SD) | 5 | CRV   | 13242  |
| 3                                   | 25 | 0.1   | Unknown | dH:0.0(1SF)      | 4 | CRV   | 2556   |

| Reaction No. 4273                     |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_DTPA-5 + OH-1 = Th+4_OH-1_DTPA-5 |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 20 | 0.1   | NaClO4        | lgK:4.9(2SF)   | 4 | MRX   | 999    |
| 2                                     | 20 | 0.1   | NaClO4        | lgK:4.9(0.2SD) | 5 | CRV   | 13242  |
| 3                                     | 25 | 0     | Inf. Dilution | lgK:4.9(2SF)   | 1 | ESC   |        |

| Reaction No. 4320                     |    |       |         |              |   |       |        |
|---------------------------------------|----|-------|---------|--------------|---|-------|--------|
| Th+4_EGTA-4 + OH-1 = Th+4_OH-1_EGTA-4 |    |       |         |              |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)  | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | Unknown | lgK:6.5(2SF) | 4 | ENS   | 773    |

| Reaction No. 4371                   |    |       |               |               |   |       |        |
|-------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| TTHA-6 + Th+4 = Th+4_TTHA-6         |    |       |               |               |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | Inf. Dilution | lgK:33.(2SF)  | 1 | EES   |        |
| Note (1): Based also on La+3 & Ga+3 |    |       |               |               |   |       |        |
| 2                                   | 25 | 0.1   | KNO3          | lgK:31.9(3SF) | 4 | MRX   | 999    |

| Reaction No. 4372                   |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4_TTHA-6 + H+1 = Th+4_H+1_TTHA-6 |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KNO3   | lgK:3.05(3SF)  | 6 | MGL   | 999    |
| 2                                   | 25 | 0.1   | KNO3   | lgK:3.1(0.2SD) | 5 | CRV   | 13242  |

| Reaction No. 4540 |  |  |  |  |  |  |  |
|-------------------|--|--|--|--|--|--|--|
|-------------------|--|--|--|--|--|--|--|

| Acac-1 + Th+4 = Th+4_Acac-1 |    |       |                    |               |   |       |        |
|-----------------------------|----|-------|--------------------|---------------|---|-------|--------|
| No                          | t  | I-Str | Medium             | Value (Dev)   | W | Tech. | Ref. # |
| 1                           | 25 | 0.01  | Unknown            | lgK:7.85(3SF) | 6 | MDS   | 999    |
| 2                           | 25 | 0.1   | NaClO <sub>4</sub> | lgK:7.7(2SF)  | 4 | MDS   | 999    |
| 3                           | 30 | 0     | Inf. Dilution      | lgK:8.8(2SF)  | 4 | MGL   | 999    |

| Reaction No. 4541                 |    |       |                    |                |   |       |        |
|-----------------------------------|----|-------|--------------------|----------------|---|-------|--------|
| 2<Acac-1> + Th+4 = Th+4_Acac-1(2) |    |       |                    |                |   |       |        |
| No                                | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 25 | 0.01  | NaClO <sub>4</sub> | lgK:15.57(4SF) | 6 | MSL   | 999    |
| 2                                 | 25 | 0.1   | Unknown            | lgK:14.9(3SF)  | 4 | ENS   | 773    |
| 3                                 | 30 | 0     | Unknown            | lgK:16.2(3SF)  | 4 | ENS   | 773    |

| Reaction No. 4542                 |    |       |         |               |   |       |        |
|-----------------------------------|----|-------|---------|---------------|---|-------|--------|
| 3<Acac-1> + Th+4 = Th+4_Acac-1(3) |    |       |         |               |   |       |        |
| No                                | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 25 | 0.1   | Unknown | lgK:20.8(3SF) | 4 | ENS   | 773    |
| 2                                 | 30 | 0     | Unknown | lgK:22.5(3SF) | 4 | ENS   | 773    |

| Reaction No. 4543                 |    |       |               |               |   |       |        |
|-----------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 4<Acac-1> + Th+4 = Th+4_Acac-1(4) |    |       |               |               |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:26.7(3SF) | 5 | MGL   | 2360   |
| 2                                 | 25 | 0.1   | Unknown       | lgK:25.8(3SF) | 4 | ENS   | 773    |

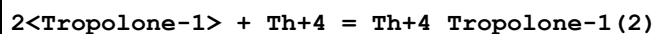
| Reaction No. 4640                                                |    |       |         |               |   |       |        |
|------------------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| 2<Th+4_Tiron-4> + 2<OH-1> + Tiron-4 = Th+4(2)_OH-1(2)_Tiron-4(3) |    |       |         |               |   |       |        |
| No                                                               | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                | 25 | 0.1   | Unknown | lgK:35.8(3SF) | 4 | ENS   | 773    |

| Reaction No. 4641                                                 |    |       |                  |               |   |       |        |
|-------------------------------------------------------------------|----|-------|------------------|---------------|---|-------|--------|
| Th+4(2)_OH-1(2)_Tiron-4(3) + 2<H+1> = Th+4(2)_Tiron-4(3) + 2<H2O> |    |       |                  |               |   |       |        |
| No                                                                | t  | I-Str | Medium           | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0.1   | KNO <sub>3</sub> | lgK:12.8(3SF) | 4 | MGL   | 999    |

| Reaction No. 4685                     |  |  |  |  |  |  |  |
|---------------------------------------|--|--|--|--|--|--|--|
| Tropolone-1 + Th+4 = Th+4_Tropolone-1 |  |  |  |  |  |  |  |

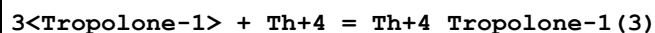
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:9.61 (3SF) | 6 | MDS   | 999    |

## Reaction No. 4686



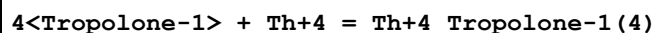
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:18.24 (4SF) | 6 | ENS   | 773    |

## Reaction No. 4687



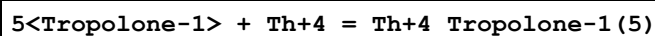
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:25.89 (4SF) | 6 | ENS   | 773    |

## Reaction No. 4688



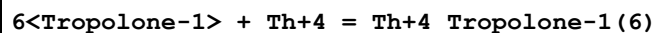
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:32.56 (4SF) | 6 | MDS   | 999    |

## Reaction No. 4689



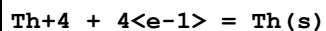
| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:34.85 (4SF) | 6 | ENS   | 773    |

## Reaction No. 4690



| No | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:36.72 (4SF) | 6 | ENS   | 773    |

## Reaction No. 4910



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-128.4 (4SF) | 0 | EOD   | 22560  |

Note (1): From Latimer WM, The Oxidation States of the Elements, 1952

|   |    |   |               |                   |   |     |       |
|---|----|---|---------------|-------------------|---|-----|-------|
| 2 | 25 | 0 | Inf. Dilution | lgK:-123.46 (5SF) | 4 | CRV | 32224 |
| 3 | 25 | 1 | HCl           | lgK:-123.5 (4SF)  | 2 | EES |       |

Note (3): Based on the rel. constancy of dH (which otherwise should be 180 kcal)

|   |    |   |               |                  |   |     |     |
|---|----|---|---------------|------------------|---|-----|-----|
| 4 | 25 | 0 | Inf. Dilution | dG:168.5 (0.1SD) | 0 | ENS | 802 |
|---|----|---|---------------|------------------|---|-----|-----|

|                                                                        |    |   |               |                 |   |     |       |
|------------------------------------------------------------------------|----|---|---------------|-----------------|---|-----|-------|
| 5                                                                      | 25 | 0 | Inf. Dilution | dG:705.1(4SF)kJ | 4 | CRV | 7421  |
| 6                                                                      | 25 | 0 | Inf. Dilution | dG:168.500(6SF) | 3 | CRV | 9029  |
| 7                                                                      | 25 | 0 | Inf. Dilution | E:-1.90(0.01SD) | 0 | MDG | 775   |
| Note (7): Low wgt for internal consistency                             |    |   |               |                 |   |     |       |
| 8                                                                      | 25 | 0 | Inf. Dilution | E:-1.899(4SF)   | 0 | CRV | 6330  |
| 9                                                                      | 25 | 0 | Inf. Dilution | E:-1.826(4SF)   | 4 | CRV | 7761  |
| 10                                                                     | 25 | 0 | Inf. Dilution | E:-1.899(4SF)   | 0 | CRV | 22519 |
| 11                                                                     | 25 | 0 | Inf. Dilution | E:-2.56(3SF)    | 0 | CRV | 22551 |
| 12                                                                     | 25 | 0 | Inf. Dilution | E:-1.900(4SF)   | 0 | EOD | 22560 |
| Note (12): From Latimer WM, The Oxidation States of the Elements, 1952 |    |   |               |                 |   |     |       |
| 13                                                                     | 25 | 0 | Inf. Dilution | dH:183.8(0.1SD) | 3 | ENS | 802   |
| 14                                                                     | 25 | 0 | Inf. Dilution | dH:768.8(4SF)kJ | 3 | CRV | 7761  |
| 15                                                                     | 25 | 0 | Inf. Dilution | dH:183.800(6SF) | 3 | CRV | 9029  |
| 16                                                                     | 25 | 1 | HCl           | dH:184.(0.4SD)  | 0 | ENS | 3006  |
| 17                                                                     | 25 | 6 | HCl           | dH:182.(0.5SD)  | 2 | ENS | 3006  |

| Reaction No. 5218                                      |    |       |         |               |   |       |        |
|--------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Th+4 + 2<H+1_Salicylic-2> = Th+4_H+1(2)_Salicylic-2(2) |    |       |         |               |   |       |        |
| No                                                     | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.1   | Unknown | lgK:7.60(3SF) | 5 | ENS   | 773    |

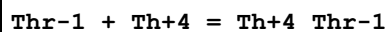
| Reaction No. 5219                                      |    |       |         |                |   |       |        |
|--------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Th+4 + 3<H+1_Salicylic-2> = Th+4_H+1(3)_Salicylic-2(3) |    |       |         |                |   |       |        |
| No                                                     | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.1   | Unknown | lgK:10.05(4SF) | 5 | ENS   | 773    |

| Reaction No. 5220                                      |    |       |         |                |   |       |        |
|--------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Th+4 + 4<H+1_Salicylic-2> = Th+4_H+1(4)_Salicylic-2(4) |    |       |         |                |   |       |        |
| No                                                     | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                      | 25 | 0.1   | Unknown | lgK:11.60(4SF) | 5 | ENS   | 773    |

| Reaction No. 5340                                 |    |       |               |                   |   |       |        |
|---------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 2<ISaccharinic-1> + Th+4 = Th+4_ISaccharinic-1(2) |    |       |               |                   |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                 | 15 | 1     | NaClO4        | lgK:18.26(0.17SD) | 5 | MSE   | 22899  |
| 2                                                 | 25 | 0     | Inf. Dilution | lgK:19.4(3SF)     | 1 | ESC   |        |
| 3                                                 | 25 | 1     | NaClO4        | lgK:19.38(0.35SD) | 5 | MSE   | 22899  |

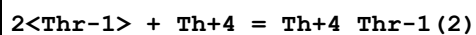
|   |    |   |                    |                    |   |     |       |
|---|----|---|--------------------|--------------------|---|-----|-------|
| 4 | 35 | 1 | NaClO <sub>4</sub> | lgK:19.47 (0.13SD) | 5 | MSE | 22899 |
|---|----|---|--------------------|--------------------|---|-----|-------|

## Reaction No. 5576



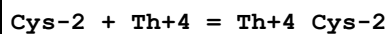
| No | t  | I-Str | Medium           | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|------------------|-------------------|---|-------|--------|
| 1  | 20 | 0.5   | Unknown          | lgK:8.00 (3SF)    | 0 | MIX   | 905    |
| 2  | 25 | 0.1   | KNO <sub>3</sub> | lgK:7.21 (3SF)    | 2 | MGL   | 1115   |
| 3  | 25 | 0.5   | Unknown          | lgK:7.97 (0.02SD) | 2 | MPT   | 999    |

## Reaction No. 5577



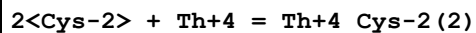
| No | t  | I-Str | Medium           | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|------------------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO <sub>3</sub> | lgK:14.01 (4SF) | 5 | MGL   | 1115   |

## Reaction No. 5634



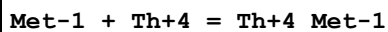
| No | t  | I-Str | Medium           | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|------------------|-----------------|---|-------|--------|
| 1  | 15 | 0.1   | KNO <sub>3</sub> | lgK:14.30 (4SF) | 0 | MGL   | 905    |
| 2  | 25 | 0.1   | KNO <sub>3</sub> | lgK:7.51 (3SF)  | 4 | MGL   | 1115   |
| 3  | 30 | 0.1   | KNO <sub>3</sub> | lgK:14.05 (4SF) | 0 | MGL   | 905    |
| 4  | 35 | 0.1   | Unknown          | lgK:13.61 (4SF) | 0 | MEP   | 6223   |

## Reaction No. 5635



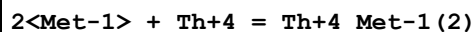
| No | t  | I-Str | Medium           | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|------------------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO <sub>3</sub> | lgK:14.80 (4SF) | 4 | MGL   | 1115   |

## Reaction No. 5706



| No | t  | I-Str | Medium           | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|------------------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO <sub>3</sub> | lgK:6.8 (0.04SD) | 5 | MGL   | 1115   |

## Reaction No. 5707



| No | t  | I-Str | Medium           | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|------------------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO <sub>3</sub> | lgK:13.5 (0.06SD) | 5 | MGL   | 1115   |

## Reaction No. 5745

| Ser-1 + Th+4 = Th+4_Ser-1 |    |       |         |                 |   |       |        |
|---------------------------|----|-------|---------|-----------------|---|-------|--------|
| No                        | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                         | 20 | 0.5   | Unknown | lgK:8.10(3SF)   | 4 | MIX   | 905    |
| 2                         | 25 | 0.1   | KNO3    | lgK:8.3(0.03SD) | 4 | MGL   | 1115   |
| 3                         | 25 | 0.5   | KNO3    | lgK:8.07(3SF)   | 5 | MGL   | 2261   |

| Reaction No. 6114                               |    |       |        |                  |   |       |        |
|-------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Th+4 + H+1(2)_Maleic-2 - 2<H+1> = Th+4_Maleic-2 |    |       |        |                  |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 1     | NaClO4 | lgK:-0.8(0.03SD) | 4 | MGL   | 915    |

| Reaction No. 6115                                     |    |       |        |              |   |       |        |
|-------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Th+4 + 2<H+1(2)_Maleic-2> - 4<H+1> = Th+4_Maleic-2(2) |    |       |        |              |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                     | 25 | 1     | NaClO4 | lgK:-5.(1SF) | 6 | MGL   | 915    |

| Reaction No. 6116                                     |    |       |        |                  |   |       |        |
|-------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Th+4 + 3<H+1(2)_Maleic-2> - 6<H+1> = Th+4_Maleic-2(3) |    |       |        |                  |   |       |        |
| No                                                    | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                     | 25 | 1     | NaClO4 | lgK:-7.6(0.03SD) | 4 | MGL   | 915    |

| Reaction No. 6117                                             |    |       |        |                    |   |       |        |
|---------------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Th+4 + 2<H+1(2)_Maleic-2> - 6<H+1> = Th+4_H+1(-2)_Maleic-2(2) |    |       |        |                    |   |       |        |
| No                                                            | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                             | 25 | 1     | NaClO4 | lgK:-13.65(0.02SD) | 6 | MGL   | 915    |

| Reaction No. 6118                                       |    |       |        |                 |   |       |        |
|---------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Th+4 + H+1(2)_Maleic-2 - 6<H+1> = Th+4_H+1(-4)_Maleic-2 |    |       |        |                 |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                       | 25 | 1     | NaClO4 | lgK:-23.(0.7SD) | 3 | MGL   | 915    |

| Reaction No. 6610                     |    |       |               |               |   |       |        |
|---------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 4<Oxalic-2> + Th+4 = Th+4_Oxalic-2(4) |    |       |               |               |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:29.6(4SF) | 1 | EOR   |        |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:29.6(3SF) | 4 | MSL   | 931    |
| 3                                     | 25 | 1     | NH4ClO4       | lgK:27.2(3SF) | 4 | MSL   | 931    |



| Reaction No. 6628               |    |       |               |                  |   |       |        |
|---------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Acetic-1 + Th+4 = Th+4_Acetic-1 |    |       |               |                  |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                               | 20 | 0.1   | NaClO4        | lgK:3.88(3SF)    | 3 | MGL   | 6788   |
| 2                               | 20 | 1     | NaClO4        | lgK:3.88(0.02SD) | 2 | MQH   | 2109   |
| 3                               | 25 | 0     | Inf. Dilution | lgK:4.569(4SF)   | 1 | EOR   |        |
| 4                               | 25 | 0     | Inf. Dilution | lgK:5.55(3SF)    | 3 | EBU   | 6372   |
| 5                               | 25 | 0     | Inf. Dilution | lgK:4.73(0.16SD) | 2 | EDT   | 31309  |
| 6                               | 25 | 0.3   | NaClO4        | lgK:3.66(3SF)    | 5 | MEP   | 31309  |
| 7                               | 25 | 0.5   | KNO3          | lgK:3.1(0.2SD)   | 2 | MPT   | 906    |
| 8                               | 25 | 1     | NaClO4        | lgK:3.86(3SF)    | 4 | MGL   | 1078   |
| 9                               | 25 | 1     | Unknown       | lgK:1.02(3SF)    | 0 | MPT   | 377    |
| 10                              | 25 | 1     | Unknown       | lgK:1.15(3SF)    | 0 | MSP   | 906    |
| 11                              | 25 | 1     | NaClO4        | dH:11.3(3SF)kJ   | 4 | MCL   | 1078   |
| 12                              | 25 | 1     | NaClO4        | dH:2.7(0.04SD)   | 5 | MGL   | 3099   |

| Reaction No. 6629                     |    |       |               |                  |   |       |        |
|---------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<Acetic-1> + Th+4 = Th+4_Acetic-1(2) |    |       |               |                  |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 20 | 1     | NaClO4        | lgK:6.9(0.03SD)  | 4 | MQH   | 2109   |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:9.057(4SF)   | 1 | EOR   |        |
| 3                                     | 25 | 0     | Inf. Dilution | lgK:10.09(4SF)   | 0 | EBU   | 6372   |
| 4                                     | 25 | 0     | Inf. Dilution | lgK:8.92(0.09SD) | 2 | EDT   | 31309  |
| 5                                     | 25 | 0.3   | NaClO4        | lgK:7.04(3SF)    | 5 | MEP   | 31309  |
| 6                                     | 25 | 0.5   | KNO3          | lgK:3.2(0.2SD)   | 0 | MPT   | 906    |
| 7                                     | 25 | 1     | NaClO4        | lgK:6.97(3SF)    | 5 | MGL   | 3099   |
| 8                                     | 25 | 1     | NaClO4        | dH:1.1(0.03SD)   | 5 | MCL   | 3099   |

| Reaction No. 6630                     |    |       |               |                   |   |       |        |
|---------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 3<Acetic-1> + Th+4 = Th+4_Acetic-1(3) |    |       |               |                   |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                     | 20 | 1     | NaClO4        | lgK:9.1(0.05SD)   | 3 | MQH   | 2109   |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:11.74(4SF)    | 1 | EOR   |        |
| 3                                     | 25 | 0     | Inf. Dilution | lgK:13.85(4SF)    | 0 | EBU   | 6372   |
| 4                                     | 25 | 0     | Inf. Dilution | lgK:12.16(0.11SD) | 2 | EDT   | 31309  |
| 5                                     | 25 | 0.3   | NaClO4        | lgK:9.75(3SF)     | 5 | MEP   | 31309  |

|   |    |   |        |                 |   |     |      |
|---|----|---|--------|-----------------|---|-----|------|
| 6 | 25 | 1 | NaClO4 | lgK:8.94 (3SF)  | 4 | MGL | 3099 |
| 7 | 25 | 1 | NaClO4 | dH:3.3 (0.09SD) | 5 | MCL | 3099 |

## Reaction No. 7241

Propanoic-1 + Th+4 = Th+4\_Propanoic-1

| No | t  | I-Str | Medium  | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------|-------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4  | lgK:3.94 (0.03SD) | 4 | MQH   | 2109   |
| 2  | 25 | 1     | Unknown | lgK:1.42 (3SF)    | 0 | MPT   | 377    |
| 3  | 25 | 1     | Unknown | lgK:1.31 (3SF)    | 0 | MSP   | 906    |

## Reaction No. 7242

2&lt;Propanoic-1&gt; + Th+4 = Th+4\_Propanoic-1(2)

| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:7.2 (0.07SD) | 4 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.2 (2SF)    | 1 | ESC   |        |

## Reaction No. 7243

3&lt;Propanoic-1&gt; + Th+4 = Th+4\_Propanoic-1(3)

| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:9.44 (3SF) | 6 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.5 (2SF)  | 1 | ESC   |        |

## Reaction No. 7244

4&lt;Propanoic-1&gt; + Th+4 = Th+4\_Propanoic-1(4)

| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:11.2 (0.1SD) | 3 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:11.2 (3SF)   | 1 | ESC   |        |

## Reaction No. 7418

2&lt;Lactic-1&gt; + Th+4 = Th+4\_Lactic-1(2)

| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:7.8 (0.03SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution | lgK:8.7 (2SF)    | 1 | ELC   |        |

## Reaction No. 7419

3&lt;Lactic-1&gt; + Th+4 = Th+4\_Lactic-1(3)

| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|   |    |   |                    |                   |   |     |      |
|---|----|---|--------------------|-------------------|---|-----|------|
| 1 | 20 | 1 | NaClO <sub>4</sub> | lgK:10.5 (0.06SD) | 4 | MQH | 3132 |
| 2 | 25 | 0 | Inf. Dilution      | lgK:10.5 (3SF)    | 1 | ESC |      |

| Reaction No. 7420                     |    |       |                    |                   |   |       |        |
|---------------------------------------|----|-------|--------------------|-------------------|---|-------|--------|
| 4<Lactic-1> + Th+4 = Th+4_Lactic-1(4) |    |       |                    |                   |   |       |        |
| No                                    | t  | I-Str | Medium             | Value (Dev)       | W | Tech. | Ref. # |
| 1                                     | 20 | 1     | NaClO <sub>4</sub> | lgK:12.9 (0.08SD) | 4 | MQH   | 3132   |
| 2                                     | 25 | 0     | Inf. Dilution      | lgK:12.9 (3SF)    | 1 | ESC   |        |

| Reaction No. 9195         |    |       |                    |                  |   |       |        |
|---------------------------|----|-------|--------------------|------------------|---|-------|--------|
| NTA-3 + Th+4 = Th+4_NTA-3 |    |       |                    |                  |   |       |        |
| No                        | t  | I-Str | Medium             | Value (Dev)      | W | Tech. | Ref. # |
| 1                         | 20 | 0.1   | Inert              | lgK:16.9 (3SF)   | 6 | CMN   | 1118   |
| 2                         | 20 | 0.1   | NaClO <sub>4</sub> | lgK:16.9 (3SF)   | 4 | MRX   | 142    |
| 3                         | 25 | 0.1   | KNO <sub>3</sub>   | lgK:12.4 (3SF)   | 0 | MGL   | 6508   |
| 4                         | 25 | 0.1   | NaClO <sub>4</sub> | lgK:13.3 (0.2SD) | 0 | MEF   | 931    |
| 5                         | 35 | 0.1   | Unknown            | lgK:8.17 (3SF)   | 0 | MEP   | 6223   |
| 6                         | 25 | 0.1   | KNO <sub>3</sub>   | dH:1.4 (0.2SD)   | 6 | MTD   | 3800   |

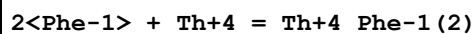
| Reaction No. 9502                          |    |       |                  |                  |   |       |        |
|--------------------------------------------|----|-------|------------------|------------------|---|-------|--------|
| Ala-1 + Th+4 = Th+4_Ala-1                  |    |       |                  |                  |   |       |        |
| No                                         | t  | I-Str | Medium           | Value (Dev)      | W | Tech. | Ref. # |
| 1                                          | 25 | 0     | Inf. Dilution    | lgK:9.983 (4SF)  | 1 | EOR   |        |
| 2                                          | 25 | 0.1   | KNO <sub>3</sub> | lgK:7.2 (0.08SD) | 0 | MGL   | 1115   |
| Note (2): Low wgt for internal consistency |    |       |                  |                  |   |       |        |
| 3                                          | 25 | 0.5   | KNO <sub>3</sub> | lgK:8.8 (0.04SD) | 4 | MGL   | 2392   |
| 4                                          | 25 | 0.5   | KNO <sub>3</sub> | lgK:8.8 (0.06SD) | 4 | MIX   | 2392   |

| Reaction No. 9503               |    |       |                  |                   |   |       |        |
|---------------------------------|----|-------|------------------|-------------------|---|-------|--------|
| 2<Ala-1> + Th+4 = Th+4_Ala-1(2) |    |       |                  |                   |   |       |        |
| No                              | t  | I-Str | Medium           | Value (Dev)       | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution    | lgK:15.96 (4SF)   | 1 | EOR   |        |
| 2                               | 25 | 0.1   | KNO <sub>3</sub> | lgK:14.5 (0.03SD) | 6 | MGL   | 1115   |

| Reaction No. 9504         |   |       |        |             |   |       |        |
|---------------------------|---|-------|--------|-------------|---|-------|--------|
| Phe-1 + Th+4 = Th+4_Phe-1 |   |       |        |             |   |       |        |
| No                        | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

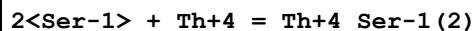
|   |    |     |      |                  |   |     |      |
|---|----|-----|------|------------------|---|-----|------|
| 1 | 25 | 0.1 | KNO3 | lgK:7.84(0.01SD) | 2 | MGL | 1115 |
| 2 | 25 | 0.5 | KNO3 | lgK:8.2(0.03SD)  | 2 | MPT | 999  |

## Reaction No. 9505



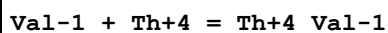
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:14.5(0.03SD) | 6 | MGL   | 1115   |

## Reaction No. 9506



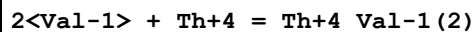
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:16.8(0.06SD) | 4 | MGL   | 1115   |

## Reaction No. 9507



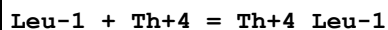
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 20 | 0.5   | KNO3   | lgK:8.6(0.1SD)   | 4 | MIX   | 2392   |
| 2  | 25 | 0.1   | KNO3   | lgK:8.30(0.01SD) | 3 | MGL   | 1115   |
| 3  | 25 | 0.5   | KNO3   | lgK:8.6(0.05SD)  | 4 | MPL   | 2392   |

## Reaction No. 9508



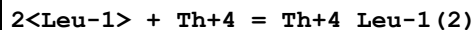
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:14.2(0.04SD) | 6 | MGL   | 1115   |

## Reaction No. 9509



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:8.3(0.05SD) | 0 | MGL   | 1115   |
| 2  | 25 | 0.5   | KNO3   | lgK:8.7(0.05SD) | 5 | MGL   | 2392   |
| 3  | 35 | 0.1   | NaClO4 | lgK:8.80(3SF)   | 4 | MEP   | 2392   |

## Reaction No. 9510



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:14.1(0.09SD) | 6 | MGL   | 1115   |

## Reaction No. 9511

| Ile-1 + Th+4 = Th+4_Ile-1 |    |       |        |                  |   |       |        |
|---------------------------|----|-------|--------|------------------|---|-------|--------|
| No                        | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                         | 25 | 0.1   | KNO3   | lgK:8.26(0.01SD) | 6 | MGL   | 1115   |

| Reaction No. 9512               |    |       |        |                   |   |       |        |
|---------------------------------|----|-------|--------|-------------------|---|-------|--------|
| 2<Ile-1> + Th+4 = Th+4_Ile-1(2) |    |       |        |                   |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | KNO3   | lgK:14.22(0.02SD) | 6 | MGL   | 1115   |

| Reaction No. 9843                 |    |       |        |                   |   |       |        |
|-----------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4 + H+1_Asp-2 = Th+4_H+1_Asp-2 |    |       |        |                   |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 25 | 1     | NaClO4 | lgK:4.2(0.03SD)   | 4 | MGL   | 1078   |
| 2                                 | 25 | 1     | NaClO4 | dH:10.9(0.4SD) kJ | 5 | MCL   | 1078   |

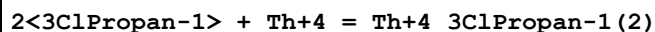
| Reaction No. 9844                           |    |       |        |                |   |       |        |
|---------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4 + H+1_Succinic-2 = Th+4_H+1_Succinic-2 |    |       |        |                |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 1     | NaClO4 | lgK:3.60(3SF)  | 6 | MGL   | 1078   |
| 2                                           | 25 | 1     | NaClO4 | dH:8.5(2SF) kJ | 5 | MCL   | 1078   |

| Reaction No. 9845                                            |    |       |               |                 |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Gly-1 + Th+4 = Th+4_Gly-1                                    |    |       |               |                 |   |       |        |
| No                                                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                            | 20 | 0.1   | NaClO4        | lgK:9.68(3SF)   | 3 | MGL   | 6788   |
| 2                                                            | 20 | 0.5   | KNO3          | lgK:8.9(0.1SD)  | 7 | MIX   | 1555   |
| 3                                                            | 25 | 0     | Inf. Dilution | lgK:10.11(4SF)  | 1 | EOR   |        |
| 4                                                            | 25 | 0     | Inf. Dilution | lgK:10.10(4SF)  | 4 | EBU   | 6372   |
| 5                                                            | 25 | 0.5   | KNO3          | lgK:8.9(0.05SD) | 7 | MGL   | 1555   |
| 6                                                            | 25 | 1     | NaClO4        | lgK:2.55(3SF)   | 0 | MGL   | 3841   |
| 7                                                            | 35 | 0.01  | NaClO4        | lgK:7.82(3SF)   | 0 | MEP   | 11516  |
| Note (7): Yadava H et al., Bull. Soc. Chim. Fr., 1984, , 314 |    |       |               |                 |   |       |        |

| Reaction No. 10255                    |    |       |        |                 |   |       |        |
|---------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 3ClPropan-1 + Th+4 = Th+4_3ClPropan-1 |    |       |        |                 |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 20 | 1     | NaClO4 | lgK:3.5(0.04SD) | 4 | MGL   | 2109   |

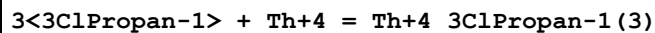
|   |    |   |               |               |   |     |  |
|---|----|---|---------------|---------------|---|-----|--|
| 2 | 25 | 0 | Inf. Dilution | lgK:3.5 (2SF) | 1 | ESC |  |
|---|----|---|---------------|---------------|---|-----|--|

## Reaction No. 10256



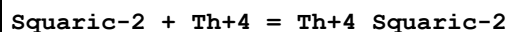
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:5.98 (3SF) | 6 | MGL   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:6.0 (2SF)  | 1 | ESC   |        |

## Reaction No. 10257



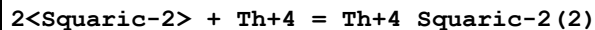
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:8.17 (3SF) | 6 | MGL   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:8.2 (2SF)  | 1 | ESC   |        |

## Reaction No. 10399



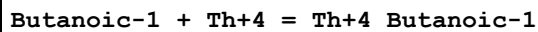
| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 1     | NH4ClO4 | lgK:4.1 (0.07SD) | 4 | MIX   | 999    |

## Reaction No. 10400



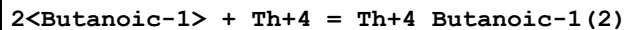
| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 1     | NH4ClO4 | lgK:7.3 (0.04SD) | 4 | MIX   | 999    |

## Reaction No. 10668



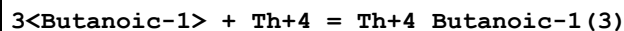
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:3.90 (0.02SD) | 4 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.9 (2SF)     | 1 | ESC   |        |

## Reaction No. 10669



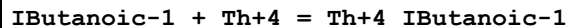
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:7.0 (0.03SD) | 4 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.0 (2SF)    | 1 | ESC   |        |

## Reaction No. 10670



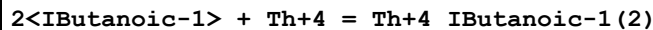
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:9.7(0.08SD) | 3 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.7(2SF)    | 1 | ESC   |        |

## Reaction No. 10690



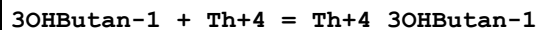
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4 | lgK:3.85(0.04SD) | 4 | MQH   | 2109   |

## Reaction No. 10691



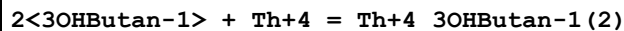
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:7.3(0.07SD) | 4 | MQH   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.3(2SF)    | 1 | ESC   |        |

## Reaction No. 10779



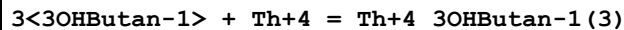
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:3.87(0.02SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.9(2SF)     | 1 | ESC   |        |

## Reaction No. 10780



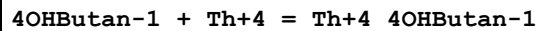
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:6.8(0.04SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution | lgK:6.8(2SF)    | 1 | ESC   |        |

## Reaction No. 10781



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:9.0(0.07SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.0(2SF)    | 1 | ESC   |        |

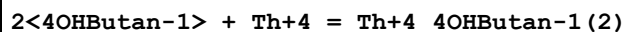
## Reaction No. 10798



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4 | lgK:3.8(0.03SD) | 4 | MQH   | 3132   |

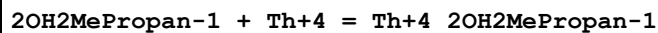
|   |    |   |               |              |   |     |  |
|---|----|---|---------------|--------------|---|-----|--|
| 2 | 25 | 0 | Inf. Dilution | lgK:3.8(2SF) | 1 | ESC |  |
|---|----|---|---------------|--------------|---|-----|--|

## Reaction No. 10799



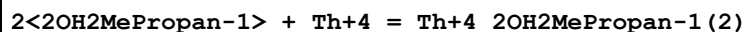
| No | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------------------|-----------------|---|-------|--------|
| 1  | 20 | 1     | NaClO <sub>4</sub> | lgK:6.7(0.07SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution      | lgK:6.7(2SF)    | 1 | ESC   |        |

## Reaction No. 10807



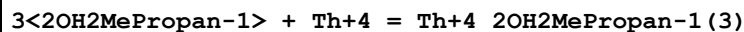
| No | t  | I-Str | Medium             | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO <sub>4</sub> | lgK:4.43(0.03SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution      | lgK:4.5(2SF)     | 1 | ESC   |        |

## Reaction No. 10808



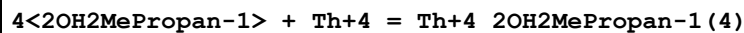
| No | t  | I-Str | Medium             | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------------------|------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO <sub>4</sub> | lgK:8.15(0.04SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution      | lgK:8.2(2SF)     | 1 | ESC   |        |

## Reaction No. 10809



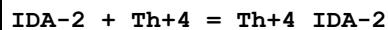
| No | t  | I-Str | Medium             | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------------------|-------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO <sub>4</sub> | lgK:11.06(0.06SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution      | lgK:11.1(3SF)     | 1 | ESC   |        |

## Reaction No. 10810



| No | t  | I-Str | Medium             | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------------------|-------------------|---|-------|--------|
| 1  | 20 | 1     | NaClO <sub>4</sub> | lgK:13.60(0.09SD) | 4 | MQH   | 3132   |
| 2  | 25 | 0     | Inf. Dilution      | lgK:13.6(3SF)     | 1 | ESC   |        |

## Reaction No. 11187



| No | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------------------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO <sub>4</sub> | lgK:11.15(4SF) | 0 | MGL   | 6788   |
| 2  | 25 | 0.1   | KCl                | lgK:10.15(4SF) | 3 | MGL   | 2261   |



|   |    |     |         |                   |   |     |      |
|---|----|-----|---------|-------------------|---|-----|------|
| 3 | 25 | 0.1 | KNO3    | lgK:10.7 (0.06SD) | 3 | MGL | 2287 |
| 4 | 25 | 0.5 | Unknown | lgK:9.3 (0.04SD)  | 0 | MPT | 999  |
| 5 | 25 | 1   | NaClO4  | lgK:9.69 (0.01SD) | 6 | MGL | 2747 |
| 6 | 25 | 1   | NaClO4  | dH:6.5 (1.SD) kJ  | 6 | MCL | 2747 |

| Reaction No. 11441                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_Acac-1 + Acac-1 = Th+4_Acac-1(2) |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.01  | NaClO4        | lgK:7.43 (3SF) | 6 | MSL   | 999    |
| 2                                     | 25 | 0.01  | Unknown       | lgK:7.73 (3SF) | 6 | MDS   | 999    |
| 3                                     | 25 | 0.1   | NaClO4        | lgK:7.2 (2SF)  | 4 | MDS   | 999    |
| 4                                     | 30 | 0     | Inf. Dilution | lgK:7.4 (2SF)  | 4 | MGL   | 999    |

| Reaction No. 11442                       |    |       |               |                |   |       |        |
|------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_Acac-1(2) + Acac-1 = Th+4_Acac-1(3) |    |       |               |                |   |       |        |
| No                                       | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                        | 25 | 0.01  | NaClO4        | lgK:6.15 (3SF) | 6 | MSL   | 999    |
| 2                                        | 25 | 0.01  | Unknown       | lgK:6.28 (3SF) | 6 | MDS   | 999    |
| 3                                        | 25 | 0.1   | NaClO4        | lgK:5.90 (3SF) | 6 | MDS   | 999    |
| 4                                        | 30 | 0     | Inf. Dilution | lgK:6.3 (2SF)  | 4 | MGL   | 999    |

| Reaction No. 11443                       |    |       |               |                |   |       |        |
|------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_Acac-1(3) + Acac-1 = Th+4_Acac-1(4) |    |       |               |                |   |       |        |
| No                                       | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                        | 25 | 0.01  | NaClO4        | lgK:5.14 (3SF) | 6 | MSL   | 999    |
| 2                                        | 25 | 0.01  | Unknown       | lgK:5.00 (3SF) | 6 | MDS   | 999    |
| 3                                        | 25 | 0.1   | NaClO4        | lgK:4.95 (3SF) | 6 | MDS   | 999    |
| 4                                        | 30 | 0     | Inf. Dilution | lgK:4.2 (2SF)  | 4 | MGL   | 999    |

| Reaction No. 11489                  |    |       |                                      |                  |   |       |        |
|-------------------------------------|----|-------|--------------------------------------|------------------|---|-------|--------|
| Glutaric-2 + Th+4 = Th+4_Glutaric-2 |    |       |                                      |                  |   |       |        |
| No                                  | t  | I-Str | Medium                               | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.05  | Unknown                              | lgK:7.44 (3SF)   | 2 | MGL   | 906    |
| 2                                   | 25 | 0.5   | Unknown                              | lgK:8.76 (3SF)   | 2 | MSL   | 999    |
| 3                                   | 25 | 0.05  | Ethanol:Water<br>25:75Vol<br>Unknown | lgK:7.7 (0.04SD) | 4 | MGL   | 906    |

|   |    |      |                                      |                   |   |     |     |
|---|----|------|--------------------------------------|-------------------|---|-----|-----|
| 4 | 25 | 0.05 | Ethanol:Water<br>40:60Vol<br>Unknown | lgK:8.24 (0.02SD) | 4 | MGL | 906 |
| 5 | 25 | 0.05 | Ethanol:Water<br>50:50Vol<br>Unknown | lgK:9.0 (0.07SD)  | 4 | MGL | 906 |

| Reaction No. 11490                          |    |       |               |                  |   |       |        |
|---------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th+4 + H+1_Glutaric-2 = Th+4_H+1_Glutaric-2 |    |       |               |                  |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 20 | 1     | NaClO4        | lgK:3.5 (0.05SD) | 4 | MQH   | 999    |
| 2                                           | 25 | 0     | Inf. Dilution | lgK:3.5 (2SF)    | 1 | ESC   |        |

| Reaction No. 11491                                   |    |       |               |                |   |       |        |
|------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4 + 2<H+1_Glutaric-2> = Th+4_H+1(2)_Glutaric-2(2) |    |       |               |                |   |       |        |
| No                                                   | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                    | 20 | 1     | NaClO4        | lgK:6.14 (3SF) | 6 | MQH   | 999    |
| 2                                                    | 25 | 0     | Inf. Dilution | lgK:6.2 (2SF)  | 1 | ESC   |        |

| Reaction No. 12246              |    |       |               |                |   |       |        |
|---------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Adipic-2 + Th+4 = Th+4_Adipic-2 |    |       |               |                |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:7.0 (2SF)  | 1 | ESC   |        |
| 2                               | 25 | 0.5   | Unknown       | lgK:8.42 (3SF) | 1 | MSL   | 906    |
| 3                               | 40 | 0.1   | NaClO4        | lgK:5.2 (2SF)  | 2 | MEP   | 4506   |

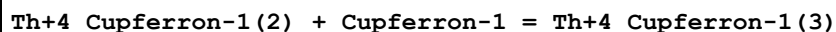
| Reaction No. 12460                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 4<Cupferron-1> + Th+4 = Th+4_Cupferron-1(4) |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | NaClO4 | lgK:27.00 (4SF) | 6 | MDS   | 999    |

| Reaction No. 12461                    |    |       |         |                |   |       |        |
|---------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Cupferron-1 + Th+4 = Th+4_Cupferron-1 |    |       |         |                |   |       |        |
| No                                    | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | Unknown | lgK:7.35 (3SF) | 6 | MDS   | 999    |

| Reaction No. 12462                                   |   |       |        |             |   |       |        |
|------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Th+4_Cupferron-1 + Cupferron-1 = Th+4_Cupferron-1(2) |   |       |        |             |   |       |        |
| No                                                   | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

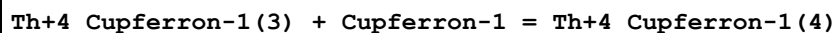
|   |    |     |         |               |   |     |     |
|---|----|-----|---------|---------------|---|-----|-----|
| 1 | 25 | 0.1 | Unknown | lgK:6.95(3SF) | 6 | MDS | 999 |
|---|----|-----|---------|---------------|---|-----|-----|

## Reaction No. 12463



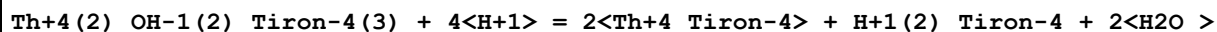
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:6.55(3SF) | 6 | MDS   | 999    |

## Reaction No. 12464



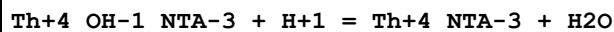
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:6.15(3SF) | 6 | MDS   | 999    |

## Reaction No. 12516



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:11.9(3SF) | 4 | MGL   | 999    |

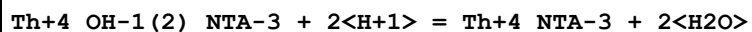
## Reaction No. 12570



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.62(0.01SD) | 6 | ENS   | 1118   |
| 2  | 25 | 0.1   | KNO3          | lgK:8.6(2SF)     | 0 | MGL   | 999    |

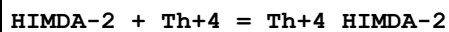
Note (2): Low wgt for internal consistency

## Reaction No. 12586



| No | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|--------|--------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:8.2(2SF) | 0 | MGL   | 6508   |

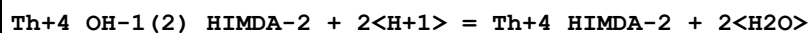
## Reaction No. 12792



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:10.7(3SF) | 0 | MGL   | 140    |
| 2  | 25 | 0.1   | KNO3   | lgK:10.7(3SF) | 3 | EOD   | 22560  |

Note (2): Courtney et al., JACS, 1958, 80, 2121

## Reaction No. 12793



| No                                              | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
|-------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| 1                                               | 25 | 0.1   | KNO3   | lgK:7.8(2SF) | 0 | MGL   | 140    |
| 2                                               | 25 | 0.1   | KNO3   | lgK:7.8(2SF) | 3 | EOD   | 22560  |
| Note (2): Courtney et al., JACS, 1958, 80, 2121 |    |       |        |              |   |       |        |

| Reaction No. 12977                                   |    |       |        |               |   |       |        |
|------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Tropolone-1 + Tropolone-1 = Th+4_Tropolone-1(2) |    |       |        |               |   |       |        |
| No                                                   | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                    | 25 | 0.1   | NaClO4 | lgK:8.63(3SF) | 6 | MDS   | 999    |

| Reaction No. 12978                                      |    |       |        |               |   |       |        |
|---------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Tropolone-1(2) + Tropolone-1 = Th+4_Tropolone-1(3) |    |       |        |               |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | NaClO4 | lgK:7.65(3SF) | 6 | MDS   | 999    |

| Reaction No. 12979                                      |    |       |        |               |   |       |        |
|---------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Tropolone-1(4) + Tropolone-1 = Th+4_Tropolone-1(5) |    |       |        |               |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | NaClO4 | lgK:2.29(3SF) | 6 | MDS   | 999    |

| Reaction No. 12980                                      |    |       |        |               |   |       |        |
|---------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Tropolone-1(5) + Tropolone-1 = Th+4_Tropolone-1(6) |    |       |        |               |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | NaClO4 | lgK:1.87(3SF) | 6 | MDS   | 999    |

| Reaction No. 13511                        |    |       |               |                  |   |       |        |
|-------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<Mandelic-1> + Th+4 = Th+4_Mandelic-1(2) |    |       |               |                  |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 20 | 1     | NaClO4        | lgK:6.89(0.03SD) | 4 | MQH   | 3132   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:6.9(2SF)     | 1 | ESC   |        |

| Reaction No. 13512                        |    |       |               |                  |   |       |        |
|-------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 3<Mandelic-1> + Th+4 = Th+4_Mandelic-1(3) |    |       |               |                  |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                         | 20 | 1     | NaClO4        | lgK:9.69(0.05SD) | 4 | MQH   | 3132   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:9.7(2SF)     | 1 | ESC   |        |

| Reaction No. 13513 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| 4<Mandelic-1> + Th+4 = Th+4_Mandelic-1(4) |    |       |                    |                   |   |       |        |
|-------------------------------------------|----|-------|--------------------|-------------------|---|-------|--------|
| No                                        | t  | I-Str | Medium             | Value (Dev)       | W | Tech. | Ref. # |
| 1                                         | 20 | 1     | NaClO <sub>4</sub> | lgK:11.98(0.08SD) | 4 | MQH   | 3132   |
| 2                                         | 25 | 0     | Inf. Dilution      | lgK:11.8(3SF)     | 1 | ESC   |        |

| Reaction No. 13736                          |    |       |                          |               |   |       |        |
|---------------------------------------------|----|-------|--------------------------|---------------|---|-------|--------|
| Th+4_Oxine-1(2) + Oxine-1 = Th+4_Oxine-1(3) |    |       |                          |               |   |       |        |
| No                                          | t  | I-Str | Medium                   | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | ClO <sub>4</sub> -1 salt | lgK:9.45(3SF) | 6 | MDS   | 999    |
| 2                                           | 25 | 0.1   | NaClO <sub>4</sub>       | lgK:9.42(3SF) | 6 | MSL   | 999    |

| Reaction No. 13737                          |    |       |                          |               |   |       |        |
|---------------------------------------------|----|-------|--------------------------|---------------|---|-------|--------|
| Th+4_Oxine-1(3) + Oxine-1 = Th+4_Oxine-1(4) |    |       |                          |               |   |       |        |
| No                                          | t  | I-Str | Medium                   | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | ClO <sub>4</sub> -1 salt | lgK:8.95(3SF) | 6 | MDS   | 999    |
| 2                                           | 25 | 0.1   | NaClO <sub>4</sub>       | lgK:8.41(3SF) | 6 | MSL   | 999    |

| Reaction No. 13738                          |    |       |                    |               |   |       |        |
|---------------------------------------------|----|-------|--------------------|---------------|---|-------|--------|
| Th+4_Oxine-1(4) + Oxine-1 = Th+4_Oxine-1(5) |    |       |                    |               |   |       |        |
| No                                          | t  | I-Str | Medium             | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | NaClO <sub>4</sub> | lgK:3.18(3SF) | 6 | MSL   | 999    |

| Reaction No. 13739                       |    |       |                          |               |   |       |        |
|------------------------------------------|----|-------|--------------------------|---------------|---|-------|--------|
| Th+4_Oxine-1 + Oxine-1 = Th+4_Oxine-1(2) |    |       |                          |               |   |       |        |
| No                                       | t  | I-Str | Medium                   | Value (Dev)   | W | Tech. | Ref. # |
| 1                                        | 25 | 0.1   | ClO <sub>4</sub> -1 salt | lgK:9.95(3SF) | 6 | MDS   | 999    |

| Reaction No. 13819                          |    |       |                  |               |   |       |        |
|---------------------------------------------|----|-------|------------------|---------------|---|-------|--------|
| Th+4_Sulfox-2 + Sulfox-2 = Th+4_Sulfox-2(2) |    |       |                  |               |   |       |        |
| No                                          | t  | I-Str | Medium           | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KNO <sub>3</sub> | lgK:8.73(3SF) | 6 | MGL   | 999    |

| Reaction No. 13820                             |    |       |                  |               |   |       |        |
|------------------------------------------------|----|-------|------------------|---------------|---|-------|--------|
| Th+4_Sulfox-2(2) + Sulfox-2 = Th+4_Sulfox-2(3) |    |       |                  |               |   |       |        |
| No                                             | t  | I-Str | Medium           | Value (Dev)   | W | Tech. | Ref. # |
| 1                                              | 25 | 0.1   | KNO <sub>3</sub> | lgK:7.62(3SF) | 6 | MGL   | 999    |

| Reaction No. 13821 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Th+4_Sulfox-2(3) + Sulfox-2 = Th+4_Sulfox-2(4) |    |       |        |               |   |       |        |
|------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| No                                             | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                              | 25 | 0.1   | KNO3   | lgK:6.12(3SF) | 6 | MGL   | 999    |

| Reaction No. 13822                                               |    |       |        |              |   |       |        |
|------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| 2<Th+4_OH-1_Sulfox-2(3)> + 2<H+1> = 2<Th+4_Sulfox-2(3)> + 2<H2O> |    |       |        |              |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                | 25 | 0.1   | KNO3   | lgK:8.9(2SF) | 4 | MGL   | 999    |

| Reaction No. 14066                         |    |       |         |               |   |       |        |
|--------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Th+4_OH-1_EDTA-4 + H+1 = Th+4_EDTA-4 + H2O |    |       |         |               |   |       |        |
| No                                         | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                          | 20 | 0.1   | Unknown | lgK:7.00(3SF) | 5 | ENS   | 1539   |
| 2                                          | 25 | 0.1   | KNO3    | lgK:7.04(3SF) | 5 | MGL   | 140    |

| Reaction No. 14067                                           |    |       |         |                |   |       |        |
|--------------------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Th+4(2)_OH-1(2)_EDTA-4(2) + 2<H+1> = 2<Th+4_EDTA-4> + 2<H2O> |    |       |         |                |   |       |        |
| No                                                           | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                            | 20 | 0.1   | Unknown | lgK:10.02(4SF) | 6 | CRV   | 552    |
| 2                                                            | 25 | 0.1   | KNO3    | lgK:9.82(3SF)  | 4 | MGL   | 903    |

| Reaction No. 14133                             |    |       |        |              |   |       |        |
|------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Th+4_OH-1_HOEDTA-3 + H+1 = Th+4_HOEDTA-3 + H2O |    |       |        |              |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                              | 25 | 0.1   | KNO3   | lgK:5.4(2SF) | 4 | MGL   | 999    |

| Reaction No. 14134                                         |    |       |               |               |   |       |        |
|------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 2<Th+4_OH-1_HOEDTA-3> + 2<H+1> = 2<Th+4_HOEDTA-3> + 2<H2O> |    |       |               |               |   |       |        |
| No                                                         | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                          | 25 | 0     | Inf. Dilution | lgK:10.8(3SF) | 1 | ELC   |        |
| 2                                                          | 25 | 0.1   | KNO3          | lgK:5.6(2SF)  | 0 | MGL   | 999    |

| Reaction No. 14664          |    |       |        |                |   |       |        |
|-----------------------------|----|-------|--------|----------------|---|-------|--------|
| CDTA-4 + Th+4 = Th+4_CDTA-4 |    |       |        |                |   |       |        |
| No                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                           | 10 | 0.1   | KNO3   | lgK:23.79(4SF) | 3 | MEF   | 999    |
| 2                           | 20 | 0.1   | KNO3   | lgK:23.78(4SF) | 3 | MEF   | 999    |

|   |    |     |                    |                 |   |     |      |
|---|----|-----|--------------------|-----------------|---|-----|------|
| 3 | 20 | 0.1 | NaClO <sub>4</sub> | lgK:29.95 (4SF) | 0 | MEF | 142  |
| 4 | 20 | 0.1 | NaClO <sub>4</sub> | lgK:29.25 (4SF) | 0 | MRX | 999  |
| 5 | 20 | 0.1 | Unknown            | lgK:25.60 (4SF) | 0 | ENS | 1539 |
| 6 | 25 | 0.5 | Unknown            | lgK:24.5 (3SF)  | 2 | CRV | 552  |
| 7 | 30 | 0.1 | KNO <sub>3</sub>   | lgK:23.77 (4SF) | 3 | MEF | 999  |

| Reaction No. 14665                  |    |       |                    |                |   |       |        |
|-------------------------------------|----|-------|--------------------|----------------|---|-------|--------|
| Th+4_CDTA-4 + H+1 = Th+4_H+1_CDTA-4 |    |       |                    |                |   |       |        |
| No                                  | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | NaClO <sub>4</sub> | lgK:2.50 (3SF) | 6 | MRX   | 999    |
| 2                                   | 20 | 0.1   | Unknown            | lgK:2.50 (3SF) | 1 | CRV   | 7271   |
| 3                                   | 25 | 0     | Inf. Dilution      | lgK:2.4 (2SF)  | 1 | ESC   |        |

| Reaction No. 14666                                      |    |       |                    |                |   |       |        |
|---------------------------------------------------------|----|-------|--------------------|----------------|---|-------|--------|
| Th+4(2)_CDTA-4(2) + 2<OH-1> = Th+4(2)_OH-1(2)_CDTA-4(2) |    |       |                    |                |   |       |        |
| No                                                      | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 20 | 0.1   | NaClO <sub>4</sub> | lgK:5.70 (3SF) | 6 | MRX   | 999    |
| 2                                                       | 25 | 0     | Inf. Dilution      | lgK:5.6 (2SF)  | 1 | ESC   |        |

| Reaction No. 14699                                      |    |       |                  |                |   |       |        |
|---------------------------------------------------------|----|-------|------------------|----------------|---|-------|--------|
| Th+4_OH-1_CDTA-4 + H+1 = Th+4_CDTA-4 + H <sub>2</sub> O |    |       |                  |                |   |       |        |
| No                                                      | t  | I-Str | Medium           | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 | 0     | Inf. Dilution    | lgK:7.5 (2SF)  | 1 | EOR   |        |
| 2                                                       | 25 | 0.1   | KNO <sub>3</sub> | lgK:7.85 (3SF) | 3 | MGL   | 999    |
| 3                                                       | 25 | 0.1   | Unknown          | lgK:7.85 (3SF) | 0 | CRV   | 7271   |

| Reaction No. 14701                                |    |       |                  |               |   |       |        |
|---------------------------------------------------|----|-------|------------------|---------------|---|-------|--------|
| 2<Th+4_OH-1_CDTA-4> = Th+4(2)_OH-1(2)_CDTA-4(2)   |    |       |                  |               |   |       |        |
| No                                                | t  | I-Str | Medium           | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | KNO <sub>3</sub> | lgK:4.3 (2SF) | 4 | MGL   | 999    |
| 2                                                 | 25 | 0.1   | KNO <sub>3</sub> | lgK:4.3 (2SF) | 1 | EOD   | 22560  |
| Note (2): Bogucki & Martell, JACS, 1958, 80, 4170 |    |       |                  |               |   |       |        |

| Reaction No. 14731              |    |       |                  |                |   |       |        |
|---------------------------------|----|-------|------------------|----------------|---|-------|--------|
| DGEDTA-4 + Th+4 = Th+4_DGEDTA-4 |    |       |                  |                |   |       |        |
| No                              | t  | I-Str | Medium           | Value (Dev)    | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | KNO <sub>3</sub> | lgK:14.0 (3SF) | 4 | MGL   | 1977   |

| Reaction No. 14749          |    |       |                    |                    |   |       |        |
|-----------------------------|----|-------|--------------------|--------------------|---|-------|--------|
| DTPA-5 + Th+4 = Th+4_DTPA-5 |    |       |                    |                    |   |       |        |
| No                          | t  | I-Str | Medium             | Value (Dev)        | W | Tech. | Ref. # |
| 1                           | 20 | 0.1   | NaClO <sub>4</sub> | lgK:28.78 (4SF)    | 0 | MRX   | 999    |
| 2                           | 20 | 0.1   | NaClO <sub>4</sub> | lgK:28.8 (0.1SD)   | 5 | CRV   | 13242  |
| 3                           | 20 | 0.1   | Unknown            | lgK:30.3 (0.03SD)  | 0 | MIX   | 999    |
| 4                           | 20 | 0.5   | NaCl               | lgK:26.6 (0.03SD)w | 0 | MPH   | 999    |
| 5                           | 20 | 0.5   | Unknown            | lgK:26.6 (0.03SD)w | 3 | MPH   | 999    |
| 6                           | 25 | 0.1   | KNO <sub>3</sub>   | lgK:27. (2SF)      | 0 | MGL   | 999    |
| 7                           | 25 | 0.1   | KNO <sub>3</sub>   | dH:-2.94 (3SF)     | 3 | MTD   | 3800   |

| Reaction No. 14904                                      |    |       |                  |                |   |       |        |
|---------------------------------------------------------|----|-------|------------------|----------------|---|-------|--------|
| Th+4_OH-1_EGTA-4 + H+1 = Th+4_EGTA-4 + H <sub>2</sub> O |    |       |                  |                |   |       |        |
| No                                                      | t  | I-Str | Medium           | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | KNO <sub>3</sub> | lgK:7.30 (3SF) | 6 | MGL   | 999    |

| Reaction No. 15290                                                                                      |    |       |                                 |                   |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|---------------------------------|-------------------|---|-------|--------|
| [18]N <sub>2</sub> O <sub>4</sub> :MeCOO*2-2 + Th+4 = Th+4_[18]N <sub>2</sub> O <sub>4</sub> :MeCOO*2-2 |    |       |                                 |                   |   |       |        |
| No                                                                                                      | t  | I-Str | Medium                          | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0.1   | Me <sub>4</sub> NC <sub>1</sub> | lgK:14.0 (0.07SD) | 5 | MPT   | 1473   |

| Reaction No. 18046                  |    |       |               |                 |   |       |        |
|-------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Th+4 + H+1_CDTA-4 = Th+4_H+1_CDTA-4 |    |       |               |                 |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | Unknown       | lgK:16.30 (4SF) | 0 | ENS   | 1539   |
| 2                                   | 25 | 0     | Inf. Dilution | lgK:17.1 (3SF)  | 1 | ELC   |        |
| 3                                   | 25 | 0     | Inf. Dilution | lgK:16.6 (3SF)  | 0 | ESC   |        |

| Reaction No. 18080                  |    |       |               |                 |   |       |        |
|-------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Th+4 + H+1_EDTA-4 = Th+4_H+1_EDTA-4 |    |       |               |                 |   |       |        |
| No                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | Unknown       | lgK:17.00 (4SF) | 0 | ENS   | 1539   |
| 2                                   | 25 | 0     | Inf. Dilution | lgK:17.5 (3SF)  | 1 | ELC   |        |
| 3                                   | 25 | 0     | Inf. Dilution | lgK:17.37 (4SF) | 4 | CRV   | 31301  |

| Reaction No. 18112 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|



| Th+4 + H+1_DTPA-5 = Th+4_H+1_DTPA-5 |    |       |               |                |   |       |        |
|-------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| No                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 20 | 0.1   | Unknown       | lgK:20.30(4SF) | 6 | ENS   | 1539   |
| 2                                   | 25 | 0     | Inf. Dilution | lgK:20.7(3SF)  | 1 | ESC   |        |

| Reaction No. 18113                         |    |       |         |                |   |       |        |
|--------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Th+4_DTPA-5 + H2O = Th+4_OH-1_DTPA-5 + H+1 |    |       |         |                |   |       |        |
| No                                         | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 20 | 0.1   | Unknown | lgK:-9.10(3SF) | 6 | ENS   | 1539   |
| 2                                          | 25 | 0.1   | KNO3    | lgK:-8.9(2SF)  | 6 | MGL   | 999    |

| Reaction No. 18176              |    |       |               |                  |   |       |        |
|---------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Citric-3 + Th+4 = Th+4_Citric-3 |    |       |               |                  |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                               | 20 | 0.1   | Unknown       | lgK:13.00(4SF)   | 0 | ENS   | 1539   |
| 2                               | 25 | 0     | Inf. Dilution | lgK:14.32(4SF)   | 1 | EOR   |        |
| 3                               | 25 | 0     | Inf. Dilution | lgK:14.13(4SF)   | 5 | MGL   | 3900   |
| 4                               | 25 | 0     | Inf. Dilution | lgK:11.29(4SF)   | 0 | CRV   | 31301  |
| 5                               | 25 | 0.1   | NaCl          | lgK:11.6(0.04SD) | 5 | MGL   | 3900   |
| 6                               | 25 | 0.5   | Unknown       | lgK:13.0(3SF)    | 0 | MGL   | 999    |

| Reaction No. 18177                    |    |       |               |                  |   |       |        |
|---------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<Citric-3> + Th+4 = Th+4_Citric-3(2) |    |       |               |                  |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 20 | 0.1   | Unknown       | lgK:21.00(4SF)   | 6 | ENS   | 1539   |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:24.34(4SF)   | 1 | EOR   |        |
| 3                                     | 25 | 0     | Inf. Dilution | lgK:24.29(4SF)   | 5 | MGL   | 3900   |
| 4                                     | 25 | 0.1   | NaCl          | lgK:21.1(0.06SD) | 5 | MGL   | 3900   |
| 5                                     | 25 | 0.5   | Unknown       | lgK:20.97(4SF)   | 6 | MGL   | 999    |

| Reaction No. 18215                             |    |       |               |                |   |       |        |
|------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_Sulfox-2 + H2O = Th+4_OH-1_Sulfox-2 + H+1 |    |       |               |                |   |       |        |
| No                                             | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                              | 20 | 0.1   | Unknown       | lgK:-6.20(3SF) | 6 | ENS   | 1539   |
| 2                                              | 25 | 0     | Inf. Dilution | lgK:-6.0(2SF)  | 1 | ESC   |        |

| Reaction No. 18551 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Th+4 + H+1_Gly-1 = Th+4_H+1_Gly-1 |    |       |        |                   |   |       |        |
|-----------------------------------|----|-------|--------|-------------------|---|-------|--------|
| No                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 25 | 1     | NaClO4 | lgK:2.55 (0.02SD) | 6 | MGL   | 3841   |
| 2                                 | 25 | 1     | NaClO4 | dH:4.2 (2SF) kJ   | 4 | MCL   | 1078   |
| 3                                 | 25 | 1     | NaClO4 | dH:4.2 (0.1SD) kJ | 5 | MCL   | 3841   |

| Reaction No. 18766                        |    |       |        |                    |   |       |        |
|-------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Th+4 + THEDACH + OH-1 = Th+4_THEDACH_OH-1 |    |       |        |                    |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | NaNO3  | lgK:19.36 (0.02SD) | 6 | MGL   | 1711   |

| Reaction No. 18767                               |    |       |        |                   |   |       |        |
|--------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4_THEDACH_OH-1 + OH-1 = Th+4_THEDACH_OH-1 (2) |    |       |        |                   |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                | 25 | 0.1   | NaNO3  | lgK:9.46 (0.01SD) | 6 | MGL   | 1711   |

| Reaction No. 18882                  |    |       |        |                     |   |       |        |
|-------------------------------------|----|-------|--------|---------------------|---|-------|--------|
| Diglycol-2 + Th+4 = Th+4_Diglycol-2 |    |       |        |                     |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)         | W | Tech. | Ref. # |
| 1                                   | 25 | 1     | NaClO4 | lgK:8.2 (0.08SD)    | 3 | MGL   | 1684   |
| 2                                   | 25 | 1     | NaClO4 | lgK:7. (1SF)        | 0 | MPT   | 2747   |
| 3                                   | 25 | 1     | NaClO4 | dH:8.37 (0.50SD) kJ | 3 | MCL   | 1684   |

| Reaction No. 18883                                 |    |       |        |                       |   |       |        |
|----------------------------------------------------|----|-------|--------|-----------------------|---|-------|--------|
| Th+4_Diglycol-2 + Diglycol-2 = Th+4_Diglycol-2 (2) |    |       |        |                       |   |       |        |
| No                                                 | t  | I-Str | Medium | Value (Dev)           | W | Tech. | Ref. # |
| 1                                                  | 25 | 1     | NaClO4 | lgK:6.7 (0.1SD)       | 4 | MGL   | 1684   |
| 2                                                  | 25 | 1     | NaClO4 | lgK:6.4 (0.09SD)      | 6 | MPT   | 2747   |
| 3                                                  | 25 | 1     | NaClO4 | dH:-11.46 (0.63SD) kJ | 5 | MCL   | 1684   |

| Reaction No. 18884                                     |    |       |        |                      |   |       |        |
|--------------------------------------------------------|----|-------|--------|----------------------|---|-------|--------|
| Th+4_Diglycol-2 (2) + Diglycol-2 = Th+4_Diglycol-2 (3) |    |       |        |                      |   |       |        |
| No                                                     | t  | I-Str | Medium | Value (Dev)          | W | Tech. | Ref. # |
| 1                                                      | 25 | 1     | NaClO4 | lgK:3.34 (3SF)       | 6 | MGL   | 1684   |
| 2                                                      | 25 | 1     | NaClO4 | lgK:3.2 (0.04SD)     | 6 | MPT   | 2747   |
| 3                                                      | 25 | 1     | NaClO4 | dH:35.90 (0.63SD) kJ | 5 | MCL   | 1684   |

| Reaction No. 18885                |    |       |        |                    |   |       |        |
|-----------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Th+4 + H+1_IDA-2 = Th+4_H+1_IDA-2 |    |       |        |                    |   |       |        |
| No                                | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                 | 25 | 1     | NaClO4 | lgK:2.91(0.01SD)   | 6 | MGL   | 2747   |
| 2                                 | 25 | 1     | NaClO4 | dH:7.41(0.88SD) kJ | 5 | MCL   | 2747   |

| Reaction No. 18886                      |    |       |        |                   |   |       |        |
|-----------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| ThioDiAcet-2 + Th+4 = Th+4_ThioDiAcet-2 |    |       |        |                   |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                       | 25 | 1     | NaClO4 | lgK:5.60(0.01SD)  | 6 | MGL   | 2747   |
| 2                                       | 25 | 1     | NaClO4 | dH:20.5(1.1SD) kJ | 5 | MCL   | 2747   |

| Reaction No. 18887                              |    |       |        |                   |   |       |        |
|-------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4 + H+1_ThioDiAcet-2 = Th+4_H+1_ThioDiAcet-2 |    |       |        |                   |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                               | 25 | 1     | NaClO4 | lgK:3.8(0.05SD)   | 6 | MGL   | 2747   |
| 2                                               | 25 | 1     | NaClO4 | dH:12.4(1.5SD) kJ | 5 | MCL   | 2747   |

| Reaction No. 18888                                      |    |       |        |                   |   |       |        |
|---------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4_ThioDiAcet-2 + ThioDiAcet-2 = Th+4_ThioDiAcet-2(2) |    |       |        |                   |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                       | 25 | 1     | NaClO4 | lgK:4.3(0.06SD)   | 6 | MGL   | 2747   |
| 2                                                       | 25 | 1     | NaClO4 | dH:14.9(1.3SD) kJ | 5 | MCL   | 2747   |

| Reaction No. 18889                                          |    |       |        |                   |   |       |        |
|-------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4 + H+1_Succinic-2 + Succinic-2 = Th+4_H+1_Succinic-2(2) |    |       |        |                   |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                           | 25 | 1     | NaClO4 | lgK:8.9(0.09SD)   | 3 | MGL   | 1684   |
| 2                                                           | 25 | 1     | NaClO4 | dH:24.7(2.5SD) kJ | 4 | MCL   | 1684   |

| Reaction No. 18890                             |    |       |        |                 |   |       |        |
|------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Th+4_Malonic-2 + Malonic-2 = Th+4_Malonic-2(2) |    |       |        |                 |   |       |        |
| No                                             | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                              | 20 | 1     | NaClO4 | lgK:5.26(3SF)   | 5 | MEF   | 11036  |
| 2                                              | 25 | 1     | NaClO4 | lgK:5.32(3SF)   | 6 | MGL   | 1684   |
| 3                                              | 25 | 1     | NaClO4 | dH:8.49(3SF) kJ | 5 | MCL   | 1684   |

| Reaction No. 18891                                |    |       |        |                 |   |       |        |
|---------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Th+4_Malonic-2(2) + Malonic-2 = Th+4_Malonic-2(3) |    |       |        |                 |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                 | 25 | 1     | NaClO4 | lgK:3.49(3SF)   | 6 | MGL   | 1684   |
| 2                                                 | 25 | 1     | NaClO4 | dH:4.60(3SF) kJ | 5 | MCL   | 1684   |

| Reaction No. 19617        |    |       |         |                  |   |       |        |
|---------------------------|----|-------|---------|------------------|---|-------|--------|
| Asp-2 + Th+4 = Th+4_Asp-2 |    |       |         |                  |   |       |        |
| No                        | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
| 1                         | 25 | 0.1   | NaClO4  | lgK:9.23(3SF)    | 2 | MGL   | 906    |
| 2                         | 25 | 0.5   | Unknown | lgK:10.5(0.03SD) | 1 | MPT   | 999    |
| 3                         | 25 | 1     | NaClO4  | lgK:4.21(3SF)    | 0 | MGL   | 1078   |

| Reaction No. 19669        |    |       |         |                 |   |       |        |
|---------------------------|----|-------|---------|-----------------|---|-------|--------|
| Hyp-1 + Th+4 = Th+4_Hyp-1 |    |       |         |                 |   |       |        |
| No                        | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                         | 25 | 0.5   | Unknown | lgK:8.2(0.03SD) | 4 | MPT   | 999    |

| Reaction No. 19684        |    |       |        |               |   |       |        |
|---------------------------|----|-------|--------|---------------|---|-------|--------|
| Gln-1 + Th+4 = Th+4_Gln-1 |    |       |        |               |   |       |        |
| No                        | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                         | 25 | 0.1   | NaClO4 | lgK:8.30(3SF) | 6 | MGL   | 999    |

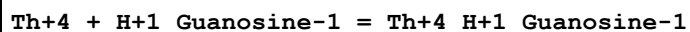
| Reaction No. 19685                 |    |       |        |               |   |       |        |
|------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Gln-1 + Gln-1 = Th+4_Gln-1(2) |    |       |        |               |   |       |        |
| No                                 | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                  | 25 | 0.1   | NaClO4 | lgK:7.61(3SF) | 6 | MGL   | 999    |

| Reaction No. 19686                    |    |       |        |               |   |       |        |
|---------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Gln-1(2) + Gln-1 = Th+4_Gln-1(3) |    |       |        |               |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | NaClO4 | lgK:7.55(3SF) | 6 | MGL   | 999    |

| Reaction No. 20731                    |    |       |               |              |   |       |        |
|---------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Th+4_CDTA-4 + OH-1 = Th+4_OH-1_CDTA-4 |    |       |               |              |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:6.5(2SF) | 1 | ELC   |        |

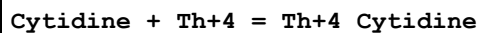
|   |    |     |         |               |   |     |     |
|---|----|-----|---------|---------------|---|-----|-----|
| 2 | 25 | 0.1 | Unknown | lgK:5.93(3SF) | 0 | ENS | 773 |
|---|----|-----|---------|---------------|---|-----|-----|

## Reaction No. 20962



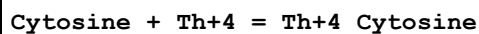
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 1     | NaNO3         | lgK:0.9(0.1SD) | 4 | MPT   | 1760   |
| 2  | 25 | 0     | Inf. Dilution | lgK:0.9(1SF)   | 1 | ESC   |        |

## Reaction No. 20991



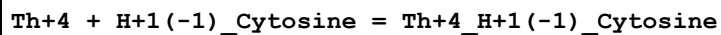
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:4.8(2SF)     | 1 | ESC   |        |
| 2  | 35 | 0.1   | KNO3          | lgK:4.62(0.02SD) | 4 | MPT   | 1760   |

## Reaction No. 21000



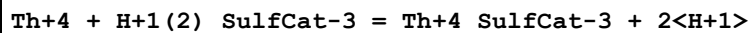
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:5.7(2SF)   | 1 | ESC   |        |
| 2  | 35 | 0.1   | KNO3          | lgK:5.5(0.2SD) | 5 | MGL   | 3855   |

## Reaction No. 21005



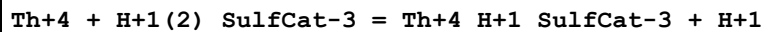
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:12.6(3SF)   | 1 | ESC   |        |
| 2  | 35 | 0.1   | KNO3          | lgK:12.4(0.1SD) | 4 | MPT   | 1760   |

## Reaction No. 21488



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 |       |               | lgK:-3.50(3SF) | 0 | MGL   | 999    |
| 2  | 25 | 0     | Inf. Dilution | lgK:-3.5(2SF)  | 1 | ESC   |        |

## Reaction No. 21489



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 |       |               | lgK:-1.37(3SF) | 0 | MGL   | 999    |
| 2  | 25 | 0     | Inf. Dilution | lgK:-1.4(2SF)  | 1 | ESC   |        |

| Reaction No. 23740                          |    |       |        |                  |   |       |        |
|---------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Th+4_EDTA-4 + Tiron-4 = Th+4_EDTA-4_Tiron-4 |    |       |        |                  |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KNO3   | lgK:13.4 (0.1SD) | 4 | MGL   | 931    |

| Reaction No. 23741                                        |    |       |        |                    |   |       |        |
|-----------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Th+4_EDTA-4 + Chromotropic-4 = Th+4_Chromotropic-4_EDTA-4 |    |       |        |                    |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.1   | KNO3   | lgK:13.66 (0.02SD) | 5 | MGL   | 931    |

| Reaction No. 23742                              |    |       |        |                   |   |       |        |
|-------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4_EDTA-4 + SulfSal-3 = Th+4_SulfSal-3_EDTA-4 |    |       |        |                   |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | KNO3   | lgK:9.29 (0.02SD) | 4 | MGL   | 931    |

| Reaction No. 23743                      |    |       |        |                   |   |       |        |
|-----------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4_EDTA-4 + Cat-2 = Th+4_Cat-2_EDTA-4 |    |       |        |                   |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | KNO3   | lgK:12.9 (0.05SD) | 4 | MGL   | 931    |

| Reaction No. 23744                            |    |       |        |                   |   |       |        |
|-----------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4_EDTA-4 + Sulfox-2 = Th+4_Sulfox-2_EDTA-4 |    |       |        |                   |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KNO3   | lgK:6.98 (0.02SD) | 4 | MGL   | 931    |

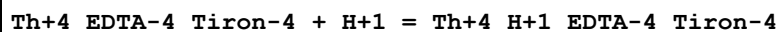
| Reaction No. 23745                      |    |       |        |                   |   |       |        |
|-----------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4_EDTA-4 + IDA-2 = Th+4_IDA-2_EDTA-4 |    |       |        |                   |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | KNO3   | lgK:6.70 (0.02SD) | 5 | MGL   | 931    |

| Reaction No. 23746                                |    |       |        |                   |   |       |        |
|---------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4_EDTA-4 + Phthalic-2 = Th+4_Phthalic-2_EDTA-4 |    |       |        |                   |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | KNO3   | lgK:3.09 (0.02SD) | 4 | MGL   | 931    |

| Reaction No. 23747                                  |   |       |        |             |   |       |        |
|-----------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Th+4_EDTA-4 + H+1_Tiron-4 = Th+4_H+1_EDTA-4_Tiron-4 |   |       |        |             |   |       |        |
| No                                                  | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

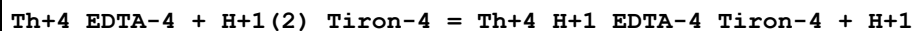
|   |    |     |      |                 |   |     |     |
|---|----|-----|------|-----------------|---|-----|-----|
| 1 | 25 | 0.1 | KNO3 | lgK:5.3(0.05SD) | 4 | MGL | 931 |
|---|----|-----|------|-----------------|---|-----|-----|

## Reaction No. 23748



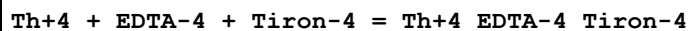
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:4.46(3SF)   | 4 | MGL   | 140    |
| 2  | 25 | 0.1   | KNO3   | lgK:4.5(0.05SD) | 4 | MGL   | 931    |

## Reaction No. 23758



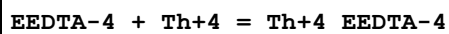
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:-2.26(3SF) | 4 | MGL   | 140    |

## Reaction No. 23759



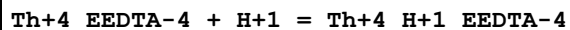
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:36.7(3SF) | 4 | MGL   | 140    |

## Reaction No. 24029



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO4        | lgK:24.9(3SF) | 4 | MRX   | 999    |
| 2  | 25 | 0     | Inf. Dilution | lgK:24.9(3SF) | 1 | ESC   |        |

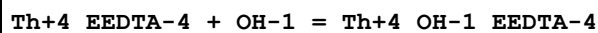
## Reaction No. 24030



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO4        | lgK:2.09(3SF) | 6 | MRX   | 999    |
| 2  | 25 | 0     | Inf. Dilution | lgK:2.5(2SF)  | 2 | EES   |        |

Note (2): Entered to have an ambient reference point

## Reaction No. 24031



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO4        | lgK:7.44(3SF) | 6 | MRX   | 999    |
| 2  | 25 | 0     | Inf. Dilution | lgK:7.5(2SF)  | 2 | EES   |        |

Note (2): Entered to have an ambient reference point

| Reaction No. 24045                           |    |       |        |                |   |       |        |
|----------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4_OH-1_EEDTA-4 + H+1 = Th+4_EEDTA-4 + H2O |    |       |        |                |   |       |        |
| No                                           | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                            | 25 | 0.1   | KNO3   | lgK:6.35 (3SF) | 6 | MGL   | 999    |

| Reaction No. 24087            |    |       |               |                |   |       |        |
|-------------------------------|----|-------|---------------|----------------|---|-------|--------|
| TEDTA-4 + Th+4 = Th+4_TEDTA-4 |    |       |               |                |   |       |        |
| No                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 20 | 0.1   | NaClO4        | lgK:19.8 (3SF) | 4 | MRX   | 999    |
| 2                             | 25 | 0     | Inf. Dilution | lgK:19.8 (3SF) | 1 | ESC   |        |

| Reaction No. 24088                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_TEDTA-4 + H+1 = Th+4_H+1_TEDTA-4 |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 20 | 0.1   | NaClO4        | lgK:2.43 (3SF) | 6 | MRX   | 999    |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:2.4 (2SF)  | 1 | ESC   |        |

| Reaction No. 24089                      |    |       |               |                |   |       |        |
|-----------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_TEDTA-4 + OH-1 = Th+4_OH-1_TEDTA-4 |    |       |               |                |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                       | 20 | 0.1   | NaClO4        | lgK:7.24 (3SF) | 6 | MRX   | 999    |
| 2                                       | 25 | 0     | Inf. Dilution | lgK:7.1 (2SF)  | 1 | ESC   |        |

| Reaction No. 24166                          |    |       |        |                 |   |       |        |
|---------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Th+4_CDTA-4 + Tiron-4 = Th+4_CDTA-4_Tiron-4 |    |       |        |                 |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KNO3   | lgK:13. (0.7SD) | 3 | MGL   | 999    |

| Reaction No. 24167                                        |    |       |        |                    |   |       |        |
|-----------------------------------------------------------|----|-------|--------|--------------------|---|-------|--------|
| Th+4_CDTA-4 + Chromotropic-4 = Th+4_CDTA-4_Chromotropic-4 |    |       |        |                    |   |       |        |
| No                                                        | t  | I-Str | Medium | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                         | 25 | 0.1   | KNO3   | lgK:13.13 (0.02SD) | 4 | MGL   | 999    |

| Reaction No. 24168                              |    |       |        |                  |   |       |        |
|-------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Th+4_CDTA-4 + SulfSal-3 = Th+4_SulfSal-3_CDTA-4 |    |       |        |                  |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | KNO3   | lgK:8.9 (0.08SD) | 3 | MGL   | 999    |



| Reaction No. 24169                      |    |       |        |                   |   |       |        |
|-----------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4_CDTA-4 + Cat-2 = Th+4_Cat-2_CDTA-4 |    |       |        |                   |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | KNO3   | lgK:12.3 (0.05SD) | 4 | MGL   | 999    |

| Reaction No. 24170                            |    |       |        |                   |   |       |        |
|-----------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| Th+4_CDTA-4 + Sulfox-2 = Th+4_Sulfox-2_CDTA-4 |    |       |        |                   |   |       |        |
| No                                            | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                             | 25 | 0.1   | KNO3   | lgK:6.69 (0.02SD) | 6 | MGL   | 999    |

| Reaction No. 24171                      |    |       |        |                  |   |       |        |
|-----------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Th+4_CDTA-4 + IDA-2 = Th+4_IDA-2_CDTA-4 |    |       |        |                  |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | KNO3   | lgK:6.1 (0.08SD) | 4 | MGL   | 999    |

| Reaction No. 24172                                |    |       |        |                  |   |       |        |
|---------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Th+4_CDTA-4 + Phthalic-2 = Th+4_Phthalic-2_CDTA-4 |    |       |        |                  |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | KNO3   | lgK:2.6 (0.03SD) | 4 | MGL   | 999    |

| Reaction No. 27494                  |    |       |        |                  |   |       |        |
|-------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Th+4_IDA-2 + IDA-2 = Th+4_IDA-2 (2) |    |       |        |                  |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KNO3   | lgK:9.1 (0.07SD) | 5 | MGL   | 2287   |

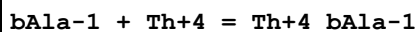
| Reaction No. 27497                          |    |       |        |                   |   |       |        |
|---------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| N2COOPhenIDA-3 + Th+4 = Th+4_N2COOPhenIDA-3 |    |       |        |                   |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                           | 25 | 0.1   | KNO3   | lgK:12.9 (0.06SD) | 3 | MGL   | 2287   |

| Reaction No. 27498                                             |    |       |        |                  |   |       |        |
|----------------------------------------------------------------|----|-------|--------|------------------|---|-------|--------|
| Th+4_N2COOPhenIDA-3 + N2COOPhenIDA-3 = Th+4_N2COOPhenIDA-3 (2) |    |       |        |                  |   |       |        |
| No                                                             | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                              | 25 | 0.1   | KNO3   | lgK:8.4 (0.04SD) | 3 | MGL   | 2287   |

| Reaction No. 27500          |   |       |        |             |   |       |        |
|-----------------------------|---|-------|--------|-------------|---|-------|--------|
| EGTA-4 + Th+4 = Th+4_EGTA-4 |   |       |        |             |   |       |        |
| No                          | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

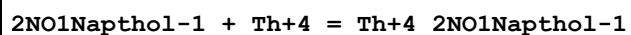
|   |    |     |      |                 |   |     |      |
|---|----|-----|------|-----------------|---|-----|------|
| 1 | 25 | 0.1 | KNO3 | lgK:9.9(0.03SD) | 3 | MGL | 2287 |
|---|----|-----|------|-----------------|---|-----|------|

## Reaction No. 29308



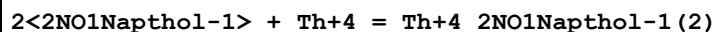
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 20 | 0.5   | KNO3          | lgK:9.8(0.05SD)  | 6 | MIX   | 2392   |
| 2  | 25 | 0     | Inf. Dilution | lgK:10.3(3SF)    | 1 | EAE   |        |
| 3  | 25 | 0.5   | KNO3          | lgK:9.76(0.02SD) | 6 | MGL   | 2392   |

## Reaction No. 30249



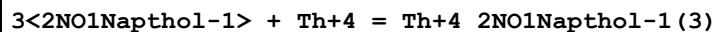
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:8.30(3SF) | 5 | CRV   | 2556   |

## Reaction No. 30250



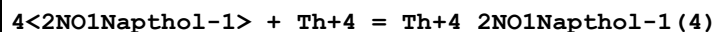
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:15.54(4SF) | 5 | CRV   | 2556   |

## Reaction No. 30251



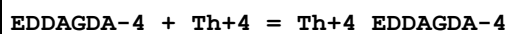
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:23.04(4SF) | 5 | CRV   | 2556   |

## Reaction No. 30252



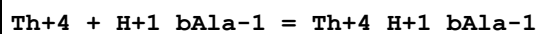
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:29.26(4SF) | 5 | CRV   | 2556   |

## Reaction No. 30639



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KNO3   | lgK:12.0(3SF) | 6 | MGL   | 1977   |

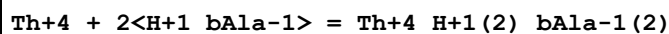
## Reaction No. 31271



| No | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:3.3(2SF) | 1 | EAE   |        |

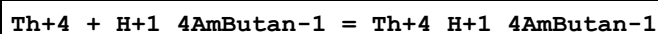
|   |    |   |         |                |   |     |      |
|---|----|---|---------|----------------|---|-----|------|
| 2 | 25 | 1 | Unknown | lgK:3.20 (3SF) | 5 | CRV | 2556 |
| 3 | 25 | 1 | Unknown | dH:1. (1SF)    | 4 | CRV | 2556 |

## Reaction No. 31272



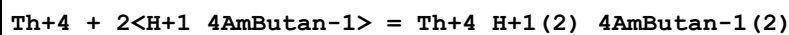
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.1 (2SF)  | 1 | EAE   |        |
| 2  | 25 | 1     | Unknown       | lgK:5.86 (3SF) | 5 | CRV   | 2556   |
| 3  | 25 | 1     | Unknown       | dH:2.4 (2SF)   | 4 | CRV   | 2556   |

## Reaction No. 31280



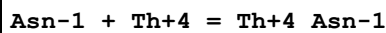
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 1     | Unknown | lgK:3.44 (3SF) | 5 | CRV   | 2556   |
| 2  | 25 | 1     | Unknown | dH:1.9 (2SF)   | 4 | CRV   | 2556   |

## Reaction No. 31281



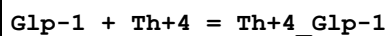
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 1     | Unknown | lgK:6.55 (3SF) | 5 | CRV   | 2556   |
| 2  | 25 | 1     | Unknown | dH:3.4 (2SF)   | 4 | CRV   | 2556   |

## Reaction No. 31321



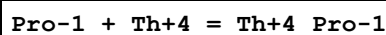
| No                                                      | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|---------------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 1                                                       | 20 | 0.5   | KNO3   | lgK:10.53 (4SF) | 0 | MIX   | 2261   |
| Note (1): Low wgt for internal consistency (c.f. Gly-1) |    |       |        |                 |   |       |        |
| 2                                                       | 25 | 0.1   | NaClO4 | lgK:8.28 (3SF)  | 3 | MGL   | 906    |

## Reaction No. 31884



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.5   | KNO3          | lgK:8.20 (3SF) | 5 | MIX   | 2261   |
| 2  | 25 | 0     | Inf. Dilution | lgK:8.0 (2SF)  | 1 | ESC   |        |

## Reaction No. 31944



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.5   | KNO3   | lgK:9.36(3SF)   | 5 | MIX   | 2261   |
| 2  | 25 | 0.5   | KNO3   | lgK:9.3(0.06SD) | 5 | MPT   | 906    |

| Reaction No. 31946                                      |    |       |        |                |   |       |        |
|---------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Glu-2 + Th+4 = Th+4_Glu-2                               |    |       |        |                |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 20 | 0.5   | KNO3   | lgK:11.40(4SF) | 0 | MIX   | 2261   |
| Note (1): Low wgt for internal consistency (c.f. Gly-1) |    |       |        |                |   |       |        |
| 2                                                       | 25 | 0.1   | NaClO4 | lgK:9.11(3SF)  | 3 | MGL   | 906    |

| Reaction No. 32063                              |    |       |               |              |   |       |        |
|-------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| 24DiNitrphenol-1 + Th+4 = Th+4_24DiNitrphenol-1 |    |       |               |              |   |       |        |
| No                                              | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                               | 21 | 0.1   | KCl           | lgK:3.2(2SF) | 4 | MGL   | 2261   |
| 2                                               | 25 | 0     | Inf. Dilution | lgK:3.4(2SF) | 1 | ESC   |        |

| Reaction No. 32064                              |    |       |               |               |   |       |        |
|-------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 26DiNitrphenol-1 + Th+4 = Th+4_26DiNitrphenol-1 |    |       |               |               |   |       |        |
| No                                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                               | 21 | 0.1   | KCl           | lgK:3.23(3SF) | 5 | MGL   | 2261   |
| 2                                               | 25 | 0     | Inf. Dilution | lgK:3.4(2SF)  | 1 | ESC   |        |

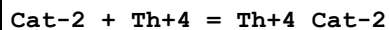
| Reaction No. 32079                        |    |       |               |              |   |       |        |
|-------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| 2NitrPhenol-1 + Th+4 = Th+4_2NitrPhenol-1 |    |       |               |              |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                         | 21 | 0.1   | KCl           | lgK:6.3(2SF) | 4 | MGL   | 2261   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:6.2(2SF) | 1 | ESC   |        |

| Reaction No. 32080                        |    |       |               |               |   |       |        |
|-------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 4NitrPhenol-1 + Th+4 = Th+4_4NitrPhenol-1 |    |       |               |               |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                         | 21 | 0.1   | KCl           | lgK:6.04(3SF) | 5 | MGL   | 2261   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:5.9(2SF)  | 1 | ESC   |        |

| Reaction No. 32149              |  |  |  |  |  |  |  |
|---------------------------------|--|--|--|--|--|--|--|
| Phenol-1 + Th+4 = Th+4_Phenol-1 |  |  |  |  |  |  |  |

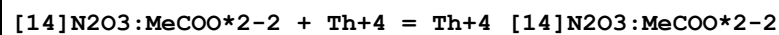
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:8.81(3SF) | 4 | EBU   | 6372   |
| 2  | 25 | 0.1   | KCl           | lgK:8.44(3SF) | 5 | MGL   | 2261   |

## Reaction No. 32150



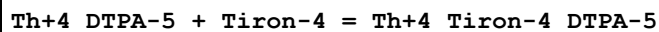
| No                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|--------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 1                                          | 25 | 0     | Inf. Dilution | lgK:24.24(4SF) | 0 | EBU   | 6372   |
| Note (1): Low wgt for internal consistency |    |       |               |                |   |       |        |
| 2                                          | 25 | 0.1   | KCl           | lgK:18.1(3SF)  | 5 | MGL   | 2261   |
| 3                                          | 30 | 0.1   | KNO3          | lgK:17.72(4SF) | 5 | MGL   | 3244   |

## Reaction No. 32891



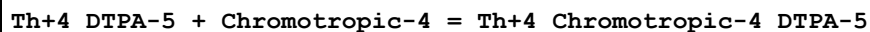
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | Me4NC1 | lgK:16.3(0.1SD) | 5 | MGL   | 1473   |

## Reaction No. 33227



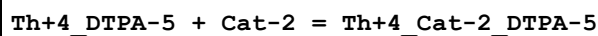
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:11.1(3SF)    | 1 | ESC   |        |
| 2  | 30 | 0.1   | KNO3          | lgK:10.8(0.04SD) | 5 | MGL   | 2986   |

## Reaction No. 33228



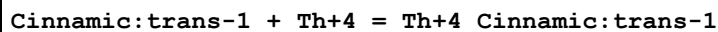
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:10.3(3SF)    | 1 | ESC   |        |
| 2  | 30 | 0.1   | KNO3          | lgK:10.0(0.06SD) | 5 | MGL   | 2986   |

## Reaction No. 33229



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:9.0(2SF)     | 1 | ESC   |        |
| 2  | 30 | 0.1   | KNO3          | lgK:8.83(0.02SD) | 5 | MGL   | 2986   |

## Reaction No. 33629



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:4.2 (2SF) | 5 | MDS   | 140    |

## Reaction No. 33630

Th+4\_Cinnamic:trans-1 + Cinnamic:trans-1 = Th+4\_Cinnamic:trans-1(2)

| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:3.8 (2SF) | 5 | MDS   | 140    |

## Reaction No. 33631

Th+4\_Cinnamic:trans-1(2) + Cinnamic:trans-1 = Th+4\_Cinnamic:trans-1(3)

| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:3.4 (2SF) | 5 | MDS   | 140    |

## Reaction No. 33632

Th+4\_Cinnamic:trans-1(3) + Cinnamic:trans-1 = Th+4\_Cinnamic:trans-1(4)

| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:3.0 (2SF) | 5 | MDS   | 140    |

## Reaction No. 33875

4&lt;Acetic-1&gt; + Th+4 = Th+4\_Acetic-1(4)

| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:13.28 (4SF) | 1 | EOR   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:16.95 (4SF) | 0 | EBU   | 6372   |
| 3  | 25 | 0     | Inf. Dilution | lgK:12.96 (4SF) | 2 | EDT   | 31309  |
| 4  | 25 | 0.3   | NaClO4        | lgK:10.28 (4SF) | 5 | MEP   | 31309  |
| 5  | 25 | 1     | NaClO4        | lgK:10.29 (4SF) | 5 | MGL   | 3099   |
| 6  | 25 | 1     | NaClO4        | dH:1. (0.2SD)   | 5 | MCL   | 3099   |

## Reaction No. 33876

5&lt;Acetic-1&gt; + Th+4 = Th+4\_Acetic-1(5)

| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:13.98 (4SF)    | 1 | EOR   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:19.44 (4SF)    | 0 | EBU   | 6372   |
| 3  | 25 | 0     | Inf. Dilution | lgK:14.39 (0.16SD) | 2 | EDT   | 31309  |
| 4  | 25 | 0.3   | NaClO4        | lgK:11.71 (4SF)    | 5 | MEP   | 31309  |
| 5  | 25 | 1     | NaClO4        | lgK:10.99 (4SF)    | 5 | MGL   | 3099   |
| 6  | 25 | 1     | NaClO4        | dH:0.89 (0.5SD)    | 5 | MCL   | 3099   |

| Reaction No. 34472                             |    |       |               |               |   |       |        |
|------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_2SHetol-1 + 2SHetol-1 = Th+4_2SHetol-1(2) |    |       |               |               |   |       |        |
| No                                             | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                              | 10 | 0.1   | NaClO4        | lgK:8.62(3SF) | 4 | MGL   | 3389   |
| 2                                              | 20 | 0.1   | NaClO4        | lgK:8.60(3SF) | 4 | MGL   | 3389   |
| 3                                              | 25 | 0     | Inf. Dilution | lgK:8.8(2SF)  | 1 | ESC   |        |
| 4                                              | 30 | 0.1   | NaClO4        | lgK:8.56(3SF) | 4 | MGL   | 3389   |

| Reaction No. 34473                                |    |       |               |               |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_2SHetol-1(2) + 2SHetol-1 = Th+4_2SHetol-1(3) |    |       |               |               |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 10 | 0.1   | NaClO4        | lgK:8.49(3SF) | 4 | MGL   | 3389   |
| 2                                                 | 20 | 0.1   | NaClO4        | lgK:8.46(3SF) | 4 | MGL   | 3389   |
| 3                                                 | 25 | 0     | Inf. Dilution | lgK:8.6(2SF)  | 1 | ESC   |        |
| 4                                                 | 30 | 0.1   | NaClO4        | lgK:8.41(3SF) | 4 | MGL   | 3389   |

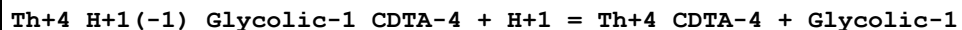
| Reaction No. 34474                                |    |       |               |               |   |       |        |
|---------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_2SHetol-1(3) + 2SHetol-1 = Th+4_2SHetol-1(4) |    |       |               |               |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 10 | 0.1   | NaClO4        | lgK:8.33(3SF) | 4 | MGL   | 3389   |
| 2                                                 | 20 | 0.1   | NaClO4        | lgK:8.25(3SF) | 4 | MGL   | 3389   |
| 3                                                 | 25 | 0     | Inf. Dilution | lgK:8.5(2SF)  | 1 | ESC   |        |
| 4                                                 | 30 | 0.1   | NaClO4        | lgK:8.30(3SF) | 4 | MGL   | 3389   |

| Reaction No. 34476                                              |    |       |               |                  |   |       |        |
|-----------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th+4_H+1(-1)_Glycolic-1_EDTA-4 + H+1 = Th+4_EDTA-4 + Glycolic-1 |    |       |               |                  |   |       |        |
| No                                                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                               | 25 | 0     | Inf. Dilution | lgK:4.8(2SF)     | 1 | ESC   |        |
| 2                                                               | 30 | 0.1   | KNO3          | lgK:4.62(0.03SD) | 5 | MGL   | 3393   |
| 3                                                               | 35 | 0.1   | KNO3          | lgK:4.64(0.02SD) | 5 | MGL   | 3393   |
| 4                                                               | 35 | 0.1   | KNO3          | dH:-1.71(3SF)    | 5 | MCL   | 3393   |

| Reaction No. 34477                                        |    |       |               |              |   |       |        |
|-----------------------------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Th+4_H+1(-1)_Malic-2_EDTA-4 + H+1 = Th+4_EDTA-4 + Malic-2 |    |       |               |              |   |       |        |
| No                                                        | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                         | 25 | 0     | Inf. Dilution | lgK:4.1(2SF) | 1 | ESC   |        |

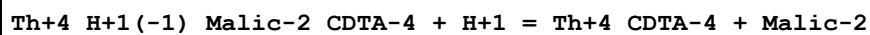
|   |    |     |      |                  |   |     |      |
|---|----|-----|------|------------------|---|-----|------|
| 2 | 30 | 0.1 | KNO3 | lgK:3.87(0.06SD) | 5 | MGL | 3393 |
| 3 | 35 | 0.1 | KNO3 | lgK:3.90(0.04SD) | 5 | MGL | 3393 |
| 4 | 35 | 0.1 | KNO3 | dH:-2.57(3SF)    | 5 | MCL | 3393 |

## Reaction No. 34478



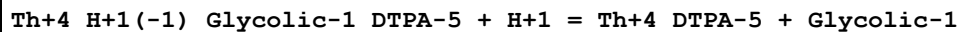
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:5.9(2SF)     | 1 | ESC   |        |
| 2  | 30 | 0.1   | KNO3          | lgK:5.70(0.05SD) | 5 | MGL   | 3393   |
| 3  | 35 | 0.1   | KNO3          | lgK:5.83(0.08SD) | 5 | MGL   | 3393   |
| 4  | 35 | 0.1   | KNO3          | dH:-11.12(4SF)   | 5 | MCL   | 3393   |

## Reaction No. 34479



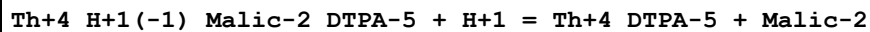
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:5.0(2SF)     | 1 | ESC   |        |
| 2  | 30 | 0.1   | KNO3          | lgK:4.72(0.04SD) | 5 | MGL   | 3393   |
| 3  | 35 | 0.1   | KNO3          | lgK:4.79(0.06SD) | 5 | MGL   | 3393   |
| 4  | 35 | 0.1   | KNO3          | dH:-5.99(3SF)    | 5 | MCL   | 3393   |

## Reaction No. 34480



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.6(2SF)     | 1 | ESC   |        |
| 2  | 30 | 0.1   | KNO3          | lgK:6.42(0.01SD) | 5 | MGL   | 3393   |
| 3  | 35 | 0.1   | KNO3          | lgK:6.48(0.03SD) | 5 | MGL   | 3393   |
| 4  | 35 | 0.1   | KNO3          | dH:-5.13(3SF)    | 5 | MCL   | 3393   |

## Reaction No. 34481



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.0(2SF)     | 1 | ESC   |        |
| 2  | 30 | 0.1   | KNO3          | lgK:5.89(0.02SD) | 5 | MGL   | 3393   |
| 3  | 35 | 0.1   | KNO3          | lgK:5.93(0.02SD) | 5 | MGL   | 3393   |
| 4  | 35 | 0.1   | KNO3          | dH:-3.42(3SF)    | 5 | MCL   | 3393   |

## Reaction No. 34482



| DTPA-5 + Th+4 + Lactic-1 = Th+4_Lactic-1_DTPA-5 |    |       |               |                 |   |       |        |
|-------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| No                                              | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                               | 25 | 0     | Inf. Dilution | lgK:6.3(2SF)    | 1 | ESC   |        |
| 2                                               | 30 | 0.1   | KNO3          | lgK:6.1(0.07SD) | 5 | MGL   | 3394   |

| Reaction No. 34483                                  |    |       |               |                 |   |       |        |
|-----------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| DTPA-5 + Th+4 + Mandelic-1 = Th+4_Mandelic-1_DTPA-5 |    |       |               |                 |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:6.1(2SF)    | 1 | ESC   |        |
| 2                                                   | 30 | 0.1   | KNO3          | lgK:5.9(0.09SD) | 5 | MGL   | 3394   |

| Reaction No. 34609                          |    |       |        |               |   |       |        |
|---------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| BenzHydroxam-1 + Th+4 = Th+4_BenzHydroxam-1 |    |       |        |               |   |       |        |
| No                                          | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 25 | 1     | HClO4  | lgK:9.60(3SF) | 5 | MGL   | 3463   |

| Reaction No. 34610                                |    |       |        |                |   |       |        |
|---------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<BenzHydroxam-1> + Th+4 = Th+4_BenzHydroxam-1(2) |    |       |        |                |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 1     | HClO4  | lgK:19.81(4SF) | 5 | MGL   | 3463   |

| Reaction No. 34611                                |    |       |        |                |   |       |        |
|---------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| 3<BenzHydroxam-1> + Th+4 = Th+4_BenzHydroxam-1(3) |    |       |        |                |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 1     | HClO4  | lgK:28.76(4SF) | 5 | MGL   | 3463   |

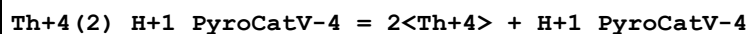
| Reaction No. 34850                                |    |       |         |                |   |       |        |
|---------------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Th+4_H+1(2)_PyroCatV-4 = Th+4 + H+1(2)_PyroCatV-4 |    |       |         |                |   |       |        |
| No                                                | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | Unknown | lgK:-9.83(3SF) | 5 | ENS   | 103    |

| Reaction No. 34851                                                  |    |       |         |                 |   |       |        |
|---------------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| Th+4_H+1(3)_PyroCatV-4(2) = Th+4_H+1(2)_PyroCatV-4 + H+1_PyroCatV-4 |    |       |         |                 |   |       |        |
| No                                                                  | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0.1   | Unknown | lgK:-15.38(4SF) | 5 | ENS   | 103    |

| Reaction No. 34852                                                     |  |  |  |  |  |  |  |
|------------------------------------------------------------------------|--|--|--|--|--|--|--|
| Th+4_H+1(4)_PyroCatV-4(3) = Th+4_H+1(3)_PyroCatV-4(2) + H+1_PyroCatV-4 |  |  |  |  |  |  |  |

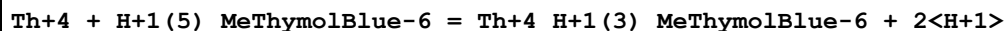
| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:-13.62 (4SF) | 5 | ENS   | 103    |

## Reaction No. 34853



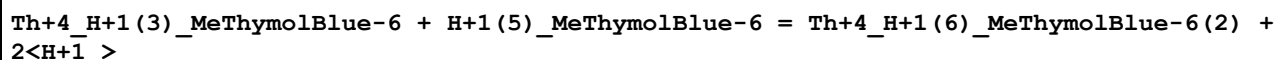
| No | t  | I-Str | Medium  | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:-20.04 (4SF) | 5 | ENS   | 103    |

## Reaction No. 34892



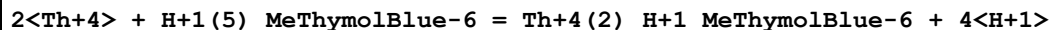
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:1.7 (0.05SD) | 5 | MGL   | 3674   |

## Reaction No. 34893



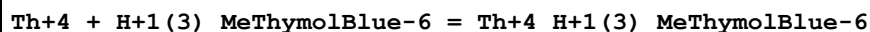
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:-1. (0.05SD) | 5 | MGL   | 3674   |

## Reaction No. 34894



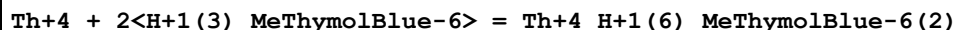
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:2.7 (0.05SD) | 5 | MGL   | 3674   |

## Reaction No. 34895



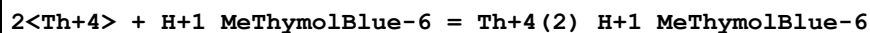
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:6.7 (0.2SD) | 5 | MGL   | 3674   |

## Reaction No. 34896



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:10.8 (0.2SD) | 5 | MGL   | 3674   |

## Reaction No. 34897



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:25. (0.3SD) | 5 | MGL   | 3674   |

| Reaction No. 35339                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| EDTA-4 + Th+4 + H+1 = Th+4_H+1_EDTA-4 |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:28.53(4SF) | 1 | EOR   |        |
| 2                                     | 25 | 0.1   | KNO3          | lgK:25.2(3SF)  | 5 | ENS   | 3800   |
| 3                                     | 25 | 0.1   | KNO3          | dH:-2.84(3SF)  | 5 | MTD   | 3800   |

| Reaction No. 35340                    |    |       |        |                 |   |       |        |
|---------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| H+1 + DTPA-5 + Th+4 = Th+4_H+1_DTPA-5 |    |       |        |                 |   |       |        |
| No                                    | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 25 | 0.1   | KNO3   | lgK:30.9(0.1SD) | 5 | ENS   | 3800   |
| 2                                     | 25 | 0.1   | KNO3   | dH:-2.94(3SF)   | 5 | MTD   | 3800   |

| Reaction No. 36442                              |    |       |               |                  |   |       |        |
|-------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<Citric-3> + Th+4 + H+1 = Th+4_H+1_Citric-3(2) |    |       |               |                  |   |       |        |
| No                                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0     | Inf. Dilution | lgK:27.14(4SF)   | 1 | EOR   |        |
| 2                                               | 25 | 0     | Inf. Dilution | lgK:27.21(4SF)   | 5 | MGL   | 3900   |
| 3                                               | 25 | 0.1   | NaCl          | lgK:23.6(0.04SD) | 5 | MGL   | 3900   |

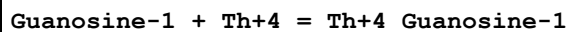
| Reaction No. 36443                                     |    |       |               |                  |   |       |        |
|--------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<Citric-3> + Th+4 - 2<H+1> = Th+4_H+1(-2)_Citric-3(2) |    |       |               |                  |   |       |        |
| No                                                     | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                      | 25 | 0     | Inf. Dilution | lgK:14.3(4SF)    | 1 | EOR   |        |
| 2                                                      | 25 | 0     | Inf. Dilution | lgK:14.67(4SF)   | 5 | MGL   | 3900   |
| 3                                                      | 25 | 0.1   | NaCl          | lgK:12.6(0.08SD) | 5 | MGL   | 3900   |

| Reaction No. 36444                    |    |       |               |                  |   |       |        |
|---------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 3<Citric-3> + Th+4 = Th+4_Citric-3(3) |    |       |               |                  |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:27.98(4SF)   | 1 | EOR   |        |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:28.00(4SF)   | 5 | MGL   | 3900   |
| 3                                     | 25 | 0.1   | NaCl          | lgK:26.1(0.07SD) | 5 | MGL   | 3900   |

| Reaction No. 36445                              |  |  |  |  |  |  |  |
|-------------------------------------------------|--|--|--|--|--|--|--|
| 3<Citric-3> + Th+4 + H+1 = Th+4_H+1_Citric-3(3) |  |  |  |  |  |  |  |

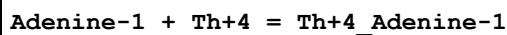
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:33.31(4SF)   | 1 | EOR   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:33.31(4SF)   | 5 | MGL   | 3900   |
| 3  | 25 | 0.1   | NaCl          | lgK:30.4(0.07SD) | 5 | MGL   | 3900   |

## Reaction No. 36851



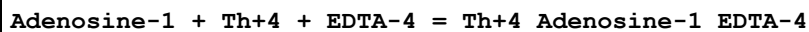
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:3.6(2SF)   | 1 | ESC   |        |
| 2  | 35 | 0.1   | KNO3          | lgK:3.4(0.2SD) | 5 | MGL   | 3855   |

## Reaction No. 36852



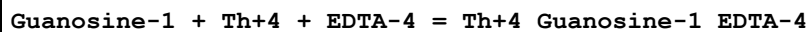
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:10.6(3SF)   | 1 | ESC   |        |
| 2  | 35 | 0.1   | KNO3          | lgK:10.3(0.1SD) | 5 | MGL   | 3855   |

## Reaction No. 36854



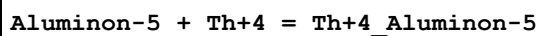
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.8(2SF)    | 1 | ESC   |        |
| 2  | 35 | 0.1   | KNO3          | lgK:2.6(0.08SD) | 5 | MGL   | 3855   |

## Reaction No. 36856



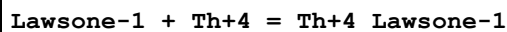
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:3.0(2SF)     | 1 | ESC   |        |
| 2  | 35 | 0.1   | KNO3          | lgK:2.81(0.02SD) | 5 | MGL   | 3855   |

## Reaction No. 37242



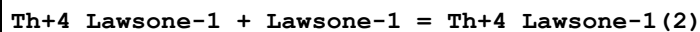
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:8.26(3SF) | 4 | MGL   | 906    |
| 2  | 25 |       |        | lgK:5.04(3SF) | 3 | MJM   | 140    |

## Reaction No. 37343



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:4.77(3SF) | 4 | MDS   | 140    |

## Reaction No. 37344



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:4.22(3SF) | 4 | MDS   | 140    |

## Reaction No. 37345



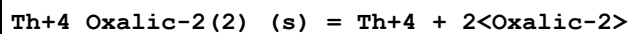
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:4.11(3SF) | 4 | MDS   | 140    |

## Reaction No. 37346



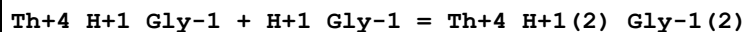
| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | Unknown | lgK:3.13(3SF) | 4 | MDS   | 140    |

## Reaction No. 37506



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-24.96(4SF) | 1 | EOR   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-24.96(4SF) | 4 | MSL   | 931    |
| 3  | 25 | 1     | NH4ClO4       | lgK:-21.38(4SF) | 5 | MSL   | 931    |

## Reaction No. 38155



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:1.66(0.02SD) | 5 | MGL   | 3841   |
| 2  | 25 | 1     | NaClO4 | dH:4.5(0.3SD)kJ  | 5 | MCL   | 3841   |

## Reaction No. 38156



| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 1     | NaClO4 | lgK:1.33(0.03SD) | 5 | MGL   | 3841   |
| 2  | 25 | 1     | NaClO4 | dH:2.3(0.4SD)kJ  | 5 | MCL   | 3841   |

## Reaction No. 38620

| DTPA-5 + Th+4 + 1OH2Naphthoic-2 = Th+4_1OH2Naphthoic-2_DTPA-5 |    |       |               |                 |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| No                                                            | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                             | 25 | 0     | Inf. Dilution | lgK:5.5(2SF)    | 1 | ESC   |        |
| 2                                                             | 30 | 0.1   | KNO3          | lgK:5.4(0.08SD) | 5 | MGL   | 3394   |

| Reaction No. 38621                                            |    |       |               |                 |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| DTPA-5 + Th+4 + 2OH3Naphthoic-2 = Th+4_2OH3Naphthoic-2_DTPA-5 |    |       |               |                 |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                             | 25 | 0     | Inf. Dilution | lgK:5.7(2SF)    | 1 | ESC   |        |
| 2                                                             | 30 | 0.1   | KNO3          | lgK:5.4(0.09SD) | 5 | MGL   | 3394   |

| Reaction No. 39500                         |    |       |               |               |   |       |        |
|--------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4 + 2<H+1_Gly-1> = Th+4_H+1(2)_Gly-1(2) |    |       |               |               |   |       |        |
| No                                         | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                          | 25 | 0     | Inf. Dilution | lgK:4.2(2SF)  | 1 | ESC   |        |
| 2                                          | 25 | 1     | Unknown       | lgK:4.21(3SF) | 6 | CRV   | 552    |
| 3                                          | 25 | 1     | Unknown       | dH:2.1(2SF)   | 5 | CRV   | 552    |

| Reaction No. 39529                                        |    |       |                          |               |   |       |        |
|-----------------------------------------------------------|----|-------|--------------------------|---------------|---|-------|--------|
| 4<DiBenzoylmethane-1> + Th+4 = Th+4_DiBenzoylmethane-1(4) |    |       |                          |               |   |       |        |
| No                                                        | t  | I-Str | Medium                   | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                         | 25 |       | Dioxan:Water<br>75:25Vol | lgK:53.3(3SF) | 5 | MGL   | 2360   |

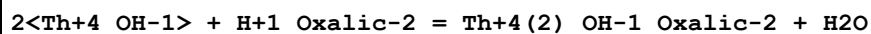
| Reaction No. 40179                          |    |       |                 |                |   |       |        |
|---------------------------------------------|----|-------|-----------------|----------------|---|-------|--------|
| Salicylic-2 + Th+4 = Th+4_Salicylic-2       |    |       |                 |                |   |       |        |
| No                                          | t  | I-Str | Medium          | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 20 | 0.1   | NaClO4          | lgK:15.45(4SF) | 4 | MGL   | 6788   |
| 2                                           | 25 | 0     | Inf. Dilution/m | lgK:4.25(3SF)  | 0 | EOD   | 6366   |
| Note (2): NDP problem (indicated in source) |    |       |                 |                |   |       |        |

| Reaction No. 40316                  |    |       |        |               |   |       |        |
|-------------------------------------|----|-------|--------|---------------|---|-------|--------|
| MeDiPhos-4 + Th+4 = Th+4_MeDiPhos-4 |    |       |        |               |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KCl    | lgK:23.9(3SF) | 5 | MGL   | 931    |

| Reaction No. 40317                        |  |  |  |  |  |  |  |
|-------------------------------------------|--|--|--|--|--|--|--|
| 2<MeDiPhos-4> + Th+4 = Th+4_MeDiPhos-4(2) |  |  |  |  |  |  |  |

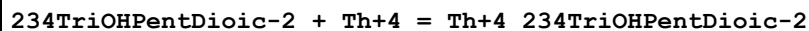
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:36.7 (3SF) | 5 | MGL   | 931    |

## Reaction No. 40403



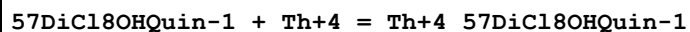
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0     | Unknown | lgK:22.9 (3SF) | 0 | MKN   | 931    |

## Reaction No. 40915



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 20 | 0.06  | Unknown       | lgK:3.70 (3SF) | 2 | MGL   | 931    |
| 2  | 23 | 0.3   | Unknown       | lgK:4.52 (3SF) | 2 | MIX   | 931    |
| 3  | 25 | 0     | Inf. Dilution | lgK:4.0 (2SF)  | 1 | ESC   |        |

## Reaction No. 41159



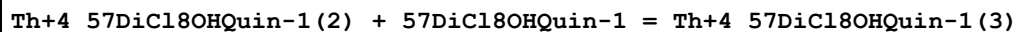
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:11.40 (4SF) | 5 | MDS   | 140    |

## Reaction No. 41160



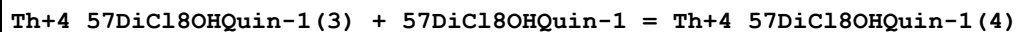
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:10.40 (4SF) | 5 | MDS   | 140    |

## Reaction No. 41161



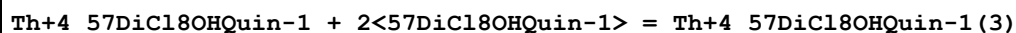
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:9.40 (3SF) | 5 | MDS   | 140    |

## Reaction No. 41162



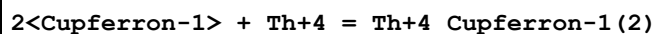
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:8.40 (3SF) | 5 | MDS   | 140    |
| 2  | 25 | 0.1   | NaClO4 | lgK:8.12 (3SF) | 5 | MDS   | 140    |

## Reaction No. 41163



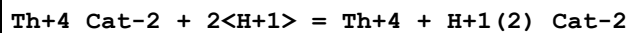
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:19.80 (4SF) | 5 | MDS   | 140    |

## Reaction No. 41941



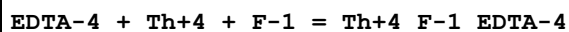
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:14.58 (4SF) | 5 | MDS   | 140    |

## Reaction No. 42249



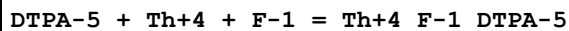
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:4.5 (2SF)   | 1 | ESC   |        |
| 2  | 25 |       |               | lgK:4.39 (3SF)  | 5 | MGL   | 906    |
| 3  | 30 | 0.1   | KNO3          | lgK:4.526 (4SF) | 5 | MGL   | 3244   |

## Reaction No. 42319



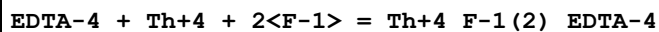
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 23 | 0.5   | KNO3          | lgK:4.83 (3SF) | 3 | MGL   | 3211   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.5 (2SF)  | 1 | EDH   |        |

## Reaction No. 42320



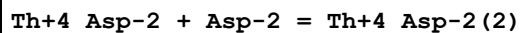
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 23 | 0.5   | KNO3          | lgK:3.81 (3SF) | 3 | MGL   | 3211   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9. (1SF)   | 1 | ESC   |        |

## Reaction No. 42321



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 23 | 0.5   | KNO3          | lgK:8.61 (3SF) | 3 | MGL   | 3211   |
| 2  | 25 | 0     | Inf. Dilution | lgK:8.8 (2SF)  | 1 | ESC   |        |

## Reaction No. 43353



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:8.7 (2SF) | 1 | ESC   |        |



|   |    |     |        |                |   |     |     |
|---|----|-----|--------|----------------|---|-----|-----|
| 2 | 25 | 0.1 | NaClO4 | lgK:8.57 (3SF) | 4 | MGL | 906 |
|---|----|-----|--------|----------------|---|-----|-----|

## Reaction No. 43354



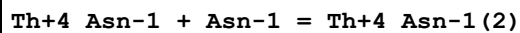
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:4.7 (2SF)  | 1 | ESC   |        |
| 2  | 25 | 0.1   | NaClO4        | lgK:4.55 (3SF) | 4 | MGL   | 906    |

## Reaction No. 43355



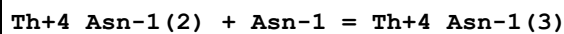
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:3.87 (3SF) | 4 | MGL   | 906    |

## Reaction No. 43418



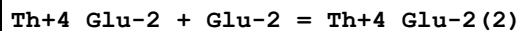
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:7.77 (3SF) | 4 | MGL   | 906    |

## Reaction No. 43419



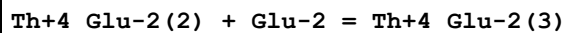
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:7.72 (3SF) | 4 | MGL   | 906    |

## Reaction No. 43625



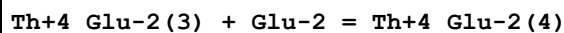
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:8.52 (3SF) | 4 | MGL   | 906    |

## Reaction No. 43626



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:4.18 (3SF) | 4 | MGL   | 906    |

## Reaction No. 43627



| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:3.62 (3SF) | 4 | MGL   | 906    |

| Reaction No. 43662                         |    |       |               |                |   |       |        |
|--------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4 + H+1(2)_Cat-2 = Th+4_H+1_Cat-2 + H+1 |    |       |               |                |   |       |        |
| No                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 25 | 0     | Inf. Dilution | lgK:-1.69(3SF) | 1 | ESC   |        |
| 2                                          | 25 |       |               | lgK:-1.69(3SF) | 0 | MGL   | 906    |

| Reaction No. 44607                |    |       |         |               |   |       |        |
|-----------------------------------|----|-------|---------|---------------|---|-------|--------|
| Azelaic-2 + Th+4 = Th+4_Azelaic-2 |    |       |         |               |   |       |        |
| No                                | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                 | 25 | 0.5   | Unknown | lgK:9.60(3SF) | 4 | MDS   | 906    |

| Reaction No. 44884                                     |    |       |               |               |   |       |        |
|--------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_EDTA-4 + H+1(2)_EDTA-4 = Th+4_H+1_EDTA-4(2) + H+1 |    |       |               |               |   |       |        |
| No                                                     | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                      | 25 | 0     | Inf. Dilution | lgK:-1.9(2SF) | 1 | ESC   |        |
| 2                                                      | 25 |       |               | lgK:-1.9(2SF) | 0 | MGL   | 906    |

| Reaction No. 45237                          |    |       |               |                 |   |       |        |
|---------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| AlizarinRedS-3 + Th+4 = Th+4_AlizarinRedS-3 |    |       |               |                 |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                           | 23 | 0.1   | Unknown       | lgK:11.52(4SF)w | 4 | MPH   | 906    |
| 2                                           | 25 | 0     | Inf. Dilution | lgK:11.7(3SF)   | 1 | ESC   |        |

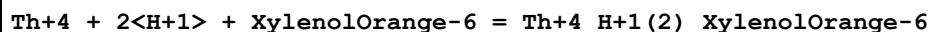
| Reaction No. 45238                                            |    |       |               |                |   |       |        |
|---------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_AlizarinRedS-3 + AlizarinRedS-3 = Th+4_AlizarinRedS-3(2) |    |       |               |                |   |       |        |
| No                                                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                             | 23 | 0.1   | Unknown       | lgK:6.63(3SF)w | 4 | MPH   | 906    |
| 2                                                             | 25 | 0     | Inf. Dilution | lgK:6.7(2SF)   | 1 | ESC   |        |

| Reaction No. 45631                                |    |       |        |               |   |       |        |
|---------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Aluminon-5 + Aluminon-5 = Th+4_Aluminon-5(2) |    |       |        |               |   |       |        |
| No                                                | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                 | 25 | 0.1   | NaClO4 | lgK:3.07(3SF) | 4 | MGL   | 906    |

| Reaction No. 45632                                   |   |       |        |             |   |       |        |
|------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Th+4_Aluminon-5(2) + Aluminon-5 = Th+4_Aluminon-5(3) |   |       |        |             |   |       |        |
| No                                                   | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |     |        |                |   |     |     |
|---|----|-----|--------|----------------|---|-----|-----|
| 1 | 25 | 0.1 | NaClO4 | lgK:2.80 (3SF) | 4 | MGL | 906 |
|---|----|-----|--------|----------------|---|-----|-----|

## Reaction No. 45689



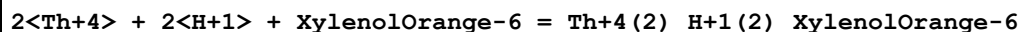
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:34.9 (0.08SD) | 5 | MSP   | 906    |

## Reaction No. 45690



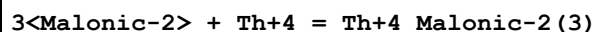
| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:64.5 (0.09SD) | 5 | MSP   | 906    |

## Reaction No. 45691



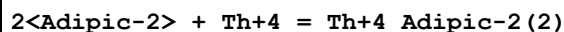
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:39.3 (0.1SD) | 5 | MSP   | 906    |

## Reaction No. 52424



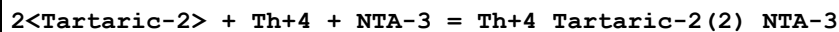
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:16.4 (3SF)   | 1 | ELC   |        |
| 2  | 25 | 1     | NaClO4        | lgK:16.279 (5SF) | 5 | MGL   | 4670   |
| 3  | 25 | 1     | NaClO4        | dH:1. (0.3SD)    | 4 | MCL   | 4670   |

## Reaction No. 52511



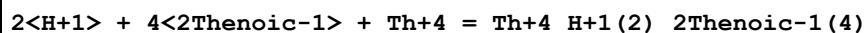
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15.6 (3SF) | 1 | ESC   |        |
| 2  | 40 | 0.1   | HClO4         | lgK:15.3 (3SF) | 0 | MEP   | 4506   |

## Reaction No. 52516



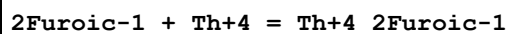
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:10.0 (3SF) | 1 | ESC   |        |
| 2  | 40 | 0.1   | NaClO4        | lgK:9.74 (3SF) | 3 | MEP   | 4507   |

## Reaction No. 52659



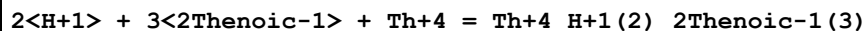
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:18.0(0.03SD) | 5 | MGL   | 5405   |

## Reaction No. 52660



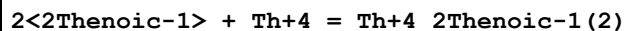
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:2.9(0.04SD) | 5 | MGL   | 5405   |

## Reaction No. 52661



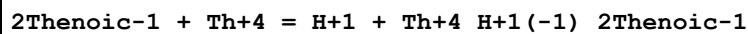
| No | t  | I-Str | Medium | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|--------|------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:14.2(0.05SD) | 5 | MGL   | 5405   |

## Reaction No. 52662



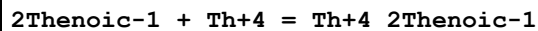
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:5.7(0.05SD) | 5 | MGL   | 5405   |

## Reaction No. 52663



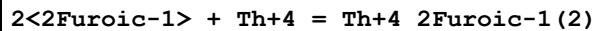
| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:0.36(2SF) | 4 | MGL   | 5405   |

## Reaction No. 52673



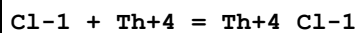
| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:3.0(0.07SD) | 5 | MGL   | 5405   |

## Reaction No. 52901



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:5.1(0.07SD) | 5 | MGL   | 5405   |

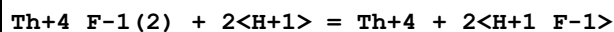
## Reaction No. 52997



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 23 | 4     | HClO4         | lgK:0.20(2SF) | 5 | MCE   | 931    |
| 2  | 25 | 0     | Inf. Dilution | lgK:1.38(3SF) | 2 | MDS   | 140    |

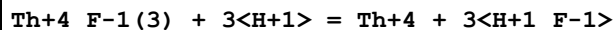
|    |    |     |                          |                    |   |     |       |
|----|----|-----|--------------------------|--------------------|---|-----|-------|
| 3  | 25 | 0   | Inf. Dilution            | lgK:1.57 (3SF)     | 3 | MAG | 5398  |
| 4  | 25 | 0   | Inf. Dilution            | lgK:1.22 (3SF)     | 0 | EBU | 6372  |
| 5  | 25 | 0   | Inf. Dilution            | lgK:1.10 (3SF)     | 2 | CRV | 9402  |
| 6  | 25 | 0   | Inf. Dilution            | lgK:1.39 (3SF)     | 2 | EMS | 12547 |
| 7  | 25 | 0   | Inf. Dilution            | lgK:1.7 (0.1SD)    | 5 | CRV | 25812 |
| 8  | 25 | 0   | Inf. Dilution            | lgK:1.700 (0.05SD) | 5 | CRV | 27322 |
| 9  | 25 | 0   | Inf. Dilution            | lgK:1.12 (3SF)     | 4 | CRV | 32224 |
| 10 | 25 | 0   | CHEMVAL2                 | lgK:1.38 (0.01SD)  | 0 | ENS | 5156  |
| 11 | 25 | 0   | CHEMVAL3                 | lgK:1.18 (0.01SD)  | 0 | ENS | 5156  |
| 12 | 25 | 0   | CHEMVAL4                 | lgK:1.18 (0.01SD)  | 0 | ENS | 5156  |
| 13 | 25 | 0.5 | H(ClO <sub>4</sub> )     | lgK:0.25 (2SF)     | 4 | MDS | 140   |
| 14 | 25 | 0.5 | NaClO <sub>4</sub>       | lgK:0.35 (2SF)     | 4 | MDS | 140   |
| 15 | 25 | 0.7 | NaClO <sub>4</sub>       | lgK:0.25 (2SF)     | 4 | MDS | 140   |
| 16 | 25 | 1   | NaClO <sub>4</sub>       | lgK:0.18 (2SF)     | 4 | MDS | 140   |
| 17 | 25 | 2   | H(Cl, ClO <sub>4</sub> ) | lgK:0.1 (0.05SD)   | 5 | MSE | 5584  |
| 18 | 25 | 2   | NaClO <sub>4</sub>       | lgK:0.08 (1SF)     | 4 | MDS | 140   |
| 19 | 25 | 4   | NaClO <sub>4</sub>       | lgK:0.23 (2SF)     | 4 | MDS | 140   |
| 20 | 25 | 4   | NaClO <sub>4</sub>       | lgK:0.11 (2SF)     | 4 | MDS | 366   |
| 21 | 25 | 6   | NaClO <sub>4</sub>       | lgK:0.32 (2SF)     | 4 | MDS | 140   |
| 22 | 25 | 0   | Inf. Dilution            | dH:0.0 (1SF)       | 3 | MCL | 5398  |
| 23 | 25 | 0   | Inf. Dilution            | dH:-0.20 (2SF)     | 3 | CRV | 9402  |
| 24 | 25 | 0   | Inf. Dilution            | dH:0.87 (2SF) kJ   | 4 | CRV | 32222 |

## Reaction No. 53163



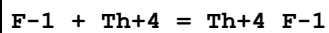
| No | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------------------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO <sub>4</sub> | lgK:-7.46 (3SF) | 5 | MDS   | 366    |
| 2  | 25 | 3     | NaClO <sub>4</sub> | lgK:-7.26 (3SF) | 5 | MIS   | 5398   |

## Reaction No. 53164



| No | t  | I-Str | Medium             | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------------------|----------------|---|-------|--------|
| 1  | 25 | 3     | NaClO <sub>4</sub> | lgK:-8.9 (2SF) | 5 | MIS   | 5398   |

## Reaction No. 53165



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|                   |    |      |               |                      |   |     |       |
|-------------------|----|------|---------------|----------------------|---|-----|-------|
| 1                 | 5  | 0    | Inf. Dilution | lgK:8.46 (3SF)       | 4 | MIS | 295   |
| 2                 | 5  | 0.01 | NH4NO3        | lgK:8.11 (3SF)       | 5 | MIS | 295   |
| 3                 | 25 | 0    | Inf. Dilution | lgK:8.65 (3SF)       | 3 | MSL | 140   |
| 4                 | 25 | 0    | Inf. Dilution | lgK:8.44 (3SF)       | 4 | MIS | 295   |
| 5                 | 25 | 0    | Inf. Dilution | lgK:7.94 (3SF)       | 0 | EBU | 6372  |
| 6                 | 25 | 0    | Inf. Dilution | lgK:8.00 (3SF)       | 3 | CRV | 9402  |
| 7                 | 25 | 0    | Inf. Dilution | lgK:8.87 (0.15SD)    | 2 | CRV | 25812 |
| 8                 | 25 | 0    | Inf. Dilution | lgK:8.870 (0.08SD)   | 2 | CRV | 27322 |
| 9                 | 25 | 0    | Inf. Dilution | lgK:7.64 (3SF)       | 2 | CRV | 32222 |
| 10                | 25 | 0    | Inf. Dilution | lgK:8.16 (3SF)       | 4 | CRV | 32224 |
| 11                | 25 | 0    | CHEMVAL2      | lgK:8.44 (0.01SD)    | 0 | ENS | 5156  |
| 12                | 25 | 0    | CHEMVAL3      | lgK:8.44 (0.01SD)    | 0 | ENS | 5156  |
| 13                | 25 | 0    | CHEMVAL4      | lgK:8.44 (0.01SD)    | 4 | ENS | 5156  |
| 14                | 25 | 0.01 | NH4NO3        | lgK:8.08 (3SF)       | 5 | MIS | 295   |
| 15                | 25 | 0.5  | KNO3          | lgK:7.6 (0.04SD)     | 0 | MIS | 2261  |
| 16                | 25 | 1    | Unknown       | lgK:7.46 (3SF)       | 5 | CRV | 2556  |
| 17                | 25 | 2    | HClO4         | lgK:7.56 (3SF)       | 5 | CRV | 2556  |
| 18                | 25 | 3    | NaClO4        | lgK:7.80 (3SF)       | 5 | CRV | 2556  |
| 19                | 25 | 4    | HClO4         | lgK:8.21 (3SF)       | 5 | CRV | 2556  |
| 20                | 25 |      |               | lgK:5.9 (2SF)        | 0 | MSL | 140   |
| 21                | 25 |      |               | lgK:6.0 (2SF)        | 0 | MSP | 931   |
| 22                | 45 | 0    | Inf. Dilution | lgK:8.32 (3SF)       | 4 | MIS | 295   |
| 23                | 45 | 0.01 | NH4NO3        | lgK:7.95 (3SF)       | 2 | MIS | 295   |
| 24                | 25 | 0    | Inf. Dilution | dG:-11.39 (4SF)      | 3 | EOD | 32225 |
| Note (24): rxn 75 |    |      |               |                      |   |     |       |
| 25                | 25 | 0    | Inf. Dilution | dH:-1.2 (2SF)        | 2 | MTD | 295   |
| 26                | 25 | 0    | Inf. Dilution | dH:-1.0 (5.SD)       | 2 | EVK | 437   |
| 27                | 25 | 0    | Inf. Dilution | dH:-1.40 (3SF)       | 0 | CRV | 9402  |
| 28                | 25 | 0    | Inf. Dilution | dH:-0.400 (1.0SD) kJ | 4 | CRV | 27322 |
| 29                | 25 | 0    | Inf. Dilution | dH:-2.01 (3SF) kJ    | 4 | CRV | 32222 |
| 30                | 25 | 0    | Inf. Dilution | dH:-1.2 (2SF)        | 2 | EOD | 32225 |
| Note (30): rxn 79 |    |      |               |                      |   |     |       |
| 31                | 25 | 0.5  | NaClO4        | dH:1.6 (0.03SD) kJ   | 3 | MCL | 5280  |
| 32                | 25 | 1    | NaClO4        | dH:0.717 (3SF)       | 0 | MCL | 437   |
| 33                | 25 | 1    | NaClO4        | dH:0.4 (5.SD)        | 2 | EVK | 437   |
| 34                | 25 | 1    | NaClO4        | dH:0.55 (2SF)        | 3 | MTD | 437   |

|    |    |   |        |                |   |     |     |
|----|----|---|--------|----------------|---|-----|-----|
| 35 | 25 | 1 | NaClO4 | dH:0.263 (3SF) | 3 | MTD | 437 |
| 36 | 25 | 4 | HClO4  | dH:-0.6 (1SF)  | 2 | CRV | 552 |

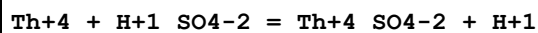
| Reaction No. 53166            |    |       |               |                |   |       |        |
|-------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_F-1 + F-1 = Th+4_F-1 (2) |    |       |               |                |   |       |        |
| No                            | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 5  | 0     | Inf. Dilution | lgK:6.55 (3SF) | 4 | MIS   | 295    |
| 2                             | 5  | 0.01  | NH4NO3        | lgK:6.29 (3SF) | 5 | MIS   | 295    |
| 3                             | 25 | 0     | Inf. Dilution | lgK:6.62 (3SF) | 4 | MIS   | 295    |
| 4                             | 25 | 0.01  | NH4NO3        | lgK:6.36 (3SF) | 5 | MIS   | 295    |
| 5                             | 45 | 0     | Inf. Dilution | lgK:6.48 (3SF) | 4 | MIS   | 295    |
| 6                             | 45 | 0.01  | NH4NO3        | lgK:6.20 (3SF) | 5 | MIS   | 295    |
| 7                             | 25 | 0     | Inf. Dilution | dG:-8.98 (3SF) | 3 | EOD   | 32225  |
| Note (7): rxn 87              |    |       |               |                |   |       |        |
| 8                             | 25 | 0     | Inf. Dilution | dH:-0.8 (1SF)  | 3 | MTD   | 295    |
| 9                             | 25 | 0     | Inf. Dilution | dH:-0.8 (5.SD) | 4 | EVK   | 437    |

| Reaction No. 53167                |    |       |               |                |   |       |        |
|-----------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_F-1 (2) + F-1 = Th+4_F-1 (3) |    |       |               |                |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 5  | 0     | Inf. Dilution | lgK:4.81 (3SF) | 4 | MIS   | 295    |
| 2                                 | 5  | 0.01  | NH4NO3        | lgK:4.64 (3SF) | 5 | MIS   | 295    |
| 3                                 | 25 | 0     | Inf. Dilution | lgK:4.75 (3SF) | 4 | MIS   | 295    |
| 4                                 | 25 | 0.01  | NH4NO3        | lgK:4.57 (3SF) | 5 | MIS   | 295    |
| 5                                 | 45 | 0     | Inf. Dilution | lgK:4.73 (3SF) | 4 | MIS   | 295    |
| 6                                 | 45 | 0.01  | NH4NO3        | lgK:4.55 (3SF) | 5 | MIS   | 295    |
| 7                                 | 25 | 0     | Inf. Dilution | dG:-6.46 (3SF) | 3 | EOD   | 32225  |
| Note (7): rxn 91                  |    |       |               |                |   |       |        |
| 8                                 | 25 | 0     | Inf. Dilution | dH:-0.8 (1SF)  | 3 | MTD   | 295    |
| 9                                 | 25 | 0     | Inf. Dilution | dH:-0.8 (5.SD) | 4 | EVK   | 437    |

| Reaction No. 53168                |    |       |               |                |   |       |        |
|-----------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_F-1 (3) + F-1 = Th+4_F-1 (4) |    |       |               |                |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 5  | 0     | Inf. Dilution | lgK:3.41 (3SF) | 4 | MIS   | 295    |
| 2                                 | 5  | 0.01  | NH4NO3        | lgK:3.33 (3SF) | 5 | MIS   | 295    |
| 3                                 | 25 | 0     | Inf. Dilution | lgK:3.36 (3SF) | 4 | MIS   | 295    |

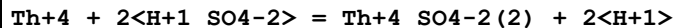
|   |    |      |               |                |   |     |     |
|---|----|------|---------------|----------------|---|-----|-----|
| 4 | 25 | 0.01 | NH4NO3        | lgK:3.28 (3SF) | 5 | MIS | 295 |
| 5 | 45 | 0    | Inf. Dilution | lgK:3.80 (3SF) | 3 | MIS | 295 |
| 6 | 45 | 0.01 | NH4NO3        | lgK:3.71 (3SF) | 3 | MIS | 295 |
| 7 | 25 | 0    | Inf. Dilution | dH:-0.9 (1SF)  | 3 | MTD | 295 |
| 8 | 25 | 0    | Inf. Dilution | dH:-0.9 (5.SD) | 4 | EVK | 437 |

## Reaction No. 53309



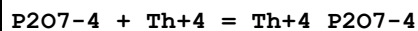
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 23 | 1.7   | NaClO4 | lgK:2.3 (2SF)  | 4 | MDS   | 140    |
| 2  | 25 | 1     | NaClO4 | lgK:2.22 (3SF) | 5 | MIX   | 365    |
| 3  | 25 | 2     | NaClO4 | lgK:2.21 (3SF) | 4 | MIX   | 365    |
| 4  | 25 | 2     | NaClO4 | lgK:2.20 (3SF) | 5 | MDS   | 366    |
| 5  | 10 | 2     | HClO4  | lgR:2.34 (3SF) | 5 | MDS   | 5398   |
| 6  | 25 | 2     | HClO4  | lgR:2.26 (3SF) | 5 | MDS   | 5398   |
| 7  | 40 | 2     | HClO4  | lgR:2.24 (3SF) | 5 | MDS   | 5398   |
| 8  | 25 | 2     | HClO4  | dH:-0.54 (2SF) | 4 | MCL   | 140    |

## Reaction No. 53310



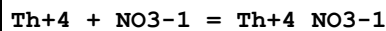
| No          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|-------------|----|-------|---------------|----------------|---|-------|--------|
| 1           | 25 | 0     | Inf. Dilution | lgK:5.4 (2SF)  | 1 | ELC   |        |
| Note (1): n |    |       |               |                |   |       |        |
| 2           | 25 | 2     | NaClO4        | lgK:3.45 (3SF) | 5 | MDS   | 366    |
| 3           | 10 | 2     | HClO4         | lgR:3.59 (3SF) | 5 | MDS   | 5398   |
| 4           | 25 | 2     | HClO4         | lgR:3.57 (3SF) | 5 | MDS   | 5398   |
| 5           | 40 | 2     | HClO4         | lgR:3.51 (3SF) | 5 | MDS   | 5398   |

## Reaction No. 53420



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 25 | 0.1   | NaClO4 | lgK:18.05 (4SF) | 5 | MSL   | 5398   |

## Reaction No. 53660

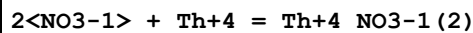


| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 2     | HClO4  | lgK:0.78 (2SF) | 3 | MDS   | 140    |



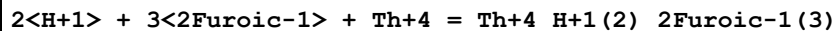
|                    |    |      |               |                   |   |     |       |
|--------------------|----|------|---------------|-------------------|---|-----|-------|
| 2                  | 23 | 4    | HClO4         | lgK:0.55 (2SF)    | 4 | MCE | 931   |
| 3                  | 25 | 0    | Inf. Dilution | lgK:1.75 (3SF)    | 0 | EBU | 6372  |
| 4                  | 25 | 0    | Inf. Dilution | lgK:1.90 (3SF)    | 0 | EMS | 12547 |
| 5                  | 25 | 0    | Inf. Dilution | lgK:1.3 (2SF)     | 5 | CRV | 25812 |
| 6                  | 25 | 0    | Inf. Dilution | lgK:1.300 (0.1SD) | 5 | CRV | 27322 |
| 7                  | 25 | 0    | Inf. Dilution | lgK:0.80 (0.20SD) | 3 | CRV | 32222 |
| 8                  | 25 | 0    | CHEMVAL4      | lgK:1.52 (0.01SD) | 0 | ENS | 5156  |
| 9                  | 25 | 0.5  | HClO4         | lgK:0.67 (2SF)    | 3 | MDS | 140   |
| 10                 | 25 | 1    | NaClO4        | lgK:-0.66 (2SF)   | 5 | MCL | 24917 |
| 11                 | 25 | 2    | HClO4         | lgK:1.22 (3SF)    | 0 | MCE | 931   |
| 12                 | 25 | 5.97 | NaClO4        | lgK:0.45 (2SF)    | 4 | MDS | 366   |
| 13                 | 25 |      |               | lgK:0.1 (1SF)     | 0 | MRM | 5398  |
| 14                 | 25 | 0    | Inf. Dilution | dG:-1.8 (2SF)     | 3 | EOD | 32225 |
| Note (14): rxn 230 |    |      |               |                   |   |     |       |
| 15                 | 25 | 1    | NaClO4        | dH:5.2 (2.7MD) kJ | 5 | MCL | 24917 |

## Reaction No. 53661



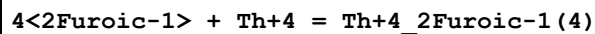
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 23 | 4     | HClO4         | lgK:0.32 (2SF)    | 4 | MCE   | 931    |
| 2  | 25 | 0     | Inf. Dilution | lgK:2.78 (3SF)    | 0 | EBU   | 6372   |
| 3  | 25 | 0     | Inf. Dilution | lgK:2.300 (0.2SD) | 5 | CRV   | 27322  |
| 4  | 25 | 0     | CHEMVAL4      | lgK:2.03 (0.01SD) | 2 | ENS   | 5156   |
| 5  | 25 | 2     | HClO4         | lgK:1.53 (3SF)    | 0 | MCE   | 931    |
| 6  | 25 | 5.97  | NaClO4        | lgK:0.15 (2SF)    | 5 | MDS   | 366    |
| 7  | 25 |       |               | lgK:0.8 (1SF)     | 0 | MRM   | 5398   |

## Reaction No. 53735



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:12.8 (0.06SD) | 5 | MGL   | 5405   |

## Reaction No. 53736



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:10.1 (0.03SD) | 5 | MGL   | 5405   |

| Reaction No. 53737                                                                      |    |       |        |                   |   |       |        |
|-----------------------------------------------------------------------------------------|----|-------|--------|-------------------|---|-------|--------|
| $2\text{<H+1>} + 4\text{<2Furoic-1>} + \text{Th+4} = \text{Th+4\_H+1(2)\_2Furoic-1(4)}$ |    |       |        |                   |   |       |        |
| No                                                                                      | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                       | 25 | 0.5   | NaClO4 | lgK:15.1 (0.07SD) | 5 | MGL   | 5405   |

| Reaction No. 53813                                |    |       |          |                |   |       |        |
|---------------------------------------------------|----|-------|----------|----------------|---|-------|--------|
| $\text{NO2-1} + \text{Th+4} = \text{Th+4\_NO2-1}$ |    |       |          |                |   |       |        |
| No                                                | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                 | 23 |       | DMF      | lgK:2.11 (3SF) | 4 | MPL   | 931    |
| 2                                                 | 23 |       | Methanol | lgK:2.23 (3SF) | 4 | MPL   | 931    |

| Reaction No. 53814                                      |    |       |          |                |   |       |        |
|---------------------------------------------------------|----|-------|----------|----------------|---|-------|--------|
| $2\text{<NO2-1>} + \text{Th+4} = \text{Th+4\_NO2-1(2)}$ |    |       |          |                |   |       |        |
| No                                                      | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 23 |       | DMF      | lgK:4.03 (3SF) | 4 | MPL   | 931    |
| 2                                                       | 23 |       | Methanol | lgK:4.17 (3SF) | 4 | MPL   | 931    |

| Reaction No. 53815                                      |    |       |          |                |   |       |        |
|---------------------------------------------------------|----|-------|----------|----------------|---|-------|--------|
| $3\text{<NO2-1>} + \text{Th+4} = \text{Th+4\_NO2-1(3)}$ |    |       |          |                |   |       |        |
| No                                                      | t  | I-Str | Medium   | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 23 |       | DMF      | lgK:5.80 (3SF) | 4 | MPL   | 931    |
| 2                                                       | 23 |       | Methanol | lgK:5.80 (3SF) | 4 | MPL   | 931    |

| Reaction No. 53816                                      |    |       |        |                |   |       |        |
|---------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $4\text{<NO2-1>} + \text{Th+4} = \text{Th+4\_NO2-1(4)}$ |    |       |        |                |   |       |        |
| No                                                      | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                       | 23 |       | DMF    | lgK:7.65 (3SF) | 3 | MPL   | 931    |

| Reaction No. 53838                                      |    |       |               |                   |   |       |        |
|---------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $3\text{<NO3-1>} + \text{Th+4} = \text{Th+4\_NO3-1(3)}$ |    |       |               |                   |   |       |        |
| No                                                      | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                       | 23 | 4     | HClO4         | lgK:-0.30 (2SF)   | 3 | MCE   | 931    |
| 2                                                       | 25 | 0     | Inf. Dilution | lgK:3.33 (3SF)    | 0 | EBU   | 6372   |
| 3                                                       | 25 | 0     | CHEMVAL4      | lgK:3.41 (0.01SD) | 0 | ENS   | 5156   |
| 4                                                       | 25 | 2     | HClO4         | lgK:1.1 (2SF)     | 3 | MCE   | 931    |

| Reaction No. 53839 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| 4<NO3-1> + Th+4 = Th+4_NO3-1(4) |    |       |               |                  |   |       |        |
|---------------------------------|----|-------|---------------|------------------|---|-------|--------|
| No                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                               | 23 | 4     | HClO4         | lgK:-0.72(2SF)   | 4 | MCE   | 931    |
| 2                               | 25 | 0     | Inf. Dilution | lgK:3.52(3SF)    | 4 | EBU   | 6372   |
| 3                               | 25 | 0     | CHEMVAL4      | lgK:3.42(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 53895                |    |       |               |                |   |       |        |
|-----------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4 + H+1_PO4-3 = Th+4_H+1_PO4-3 |    |       |               |                |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:12.37(4SF) | 0 | EBU   | 6372   |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:12.(2SF)   | 3 | EOD   | 32233  |
| 3                                 | 25 | 0.35  | HClO4         | lgK:10.8(3SF)  | 4 | MSL   | 931    |

| Reaction No. 53896                         |    |       |               |                 |   |       |        |
|--------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Th+4 + 2<H+1_PO4-3> = Th+4_H+1(2)_PO4-3(2) |    |       |               |                 |   |       |        |
| No                                         | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                          | 25 | 0     | Inf. Dilution | lgK:23.14(4SF)  | 0 | EBU   | 6372   |
| 2                                          | 25 | 0     | Inf. Dilution | lgK:21.5(3.5SD) | 3 | EOD   | 32233  |
| 3                                          | 25 | 0.35  | HClO4         | lgK:22.8(3SF)   | 4 | MSL   | 931    |

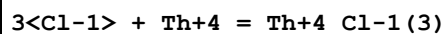
| Reaction No. 53897                         |    |       |               |                 |   |       |        |
|--------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Th+4 + 3<H+1_PO4-3> = Th+4_H+1(3)_PO4-3(3) |    |       |               |                 |   |       |        |
| No                                         | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                          | 25 | 0     | Inf. Dilution | lgK:32.56(4SF)  | 0 | EBU   | 6372   |
| 2                                          | 25 | 0     | Inf. Dilution | lgK:28.3(0.3SD) | 3 | EOD   | 32233  |
| 3                                          | 25 | 0.35  | HClO4         | lgK:31.3(3SF)   | 4 | MSL   | 931    |

| Reaction No. 54057                    |    |       |               |               |   |       |        |
|---------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_SO4-2(2) + SO4-2 = Th+4_SO4-2(3) |    |       |               |               |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:0.76(2SF) | 4 | MDS   | 140    |
| 2                                     | 25 | 2     | NaClO4        | lgK:1.08(3SF) | 4 | MDS   | 365    |

| Reaction No. 54175            |    |       |        |                |   |       |        |
|-------------------------------|----|-------|--------|----------------|---|-------|--------|
| 2<Cl-1> + Th+4 = Th+4_Cl-1(2) |    |       |        |                |   |       |        |
| No                            | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                             | 23 | 4     | HClO4  | lgK:-0.80(2SF) | 0 | MCE   | 931    |

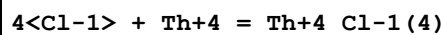
|    |    |   |               |                   |   |     |       |
|----|----|---|---------------|-------------------|---|-----|-------|
| 2  | 25 | 0 | Inf. Dilution | lgK:1.8 (2SF)     | 1 | ELC |       |
| 3  | 25 | 0 | Inf. Dilution | lgK:2.13 (3SF)    | 0 | EBU | 6372  |
| 4  | 25 | 0 | Inf. Dilution | lgK:0.80 (2SF)    | 0 | CRV | 9402  |
| 5  | 25 | 0 | Inf. Dilution | lgK:0.84 (2SF)    | 0 | CRV | 32224 |
| 6  | 25 | 0 | CHEMVAL2      | lgK:0.76 (0.01SD) | 0 | ENS | 5156  |
| 7  | 25 | 0 | CHEMVAL3      | lgK:0.92 (0.01SD) | 0 | ENS | 5156  |
| 8  | 25 | 0 | CHEMVAL4      | lgK:0.92 (0.01SD) | 0 | ENS | 5156  |
| 9  | 25 | 2 | H(Cl,C1O4)    | lgK:-0.2 (0.1SD)  | 0 | MSE | 5584  |
| 10 | 25 | 4 | NaClO4        | lgK:-0.90 (2SF)   | 2 | MDS | 366   |
| 11 | 25 | 0 | Inf. Dilution | dH:20.90 (4SF)    | 2 | CRV | 9402  |

## Reaction No. 54176



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 23 | 4     | HClO4         | lgK:-0.85 (2SF)   | 0 | MCE   | 931    |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.5 (2SF)     | 1 | ELC   |        |
| 3  | 25 | 0     | Inf. Dilution | lgK:2.86 (3SF)    | 0 | EBU   | 6372   |
| 4  | 25 | 0     | Inf. Dilution | lgK:1.65 (3SF)    | 0 | CRV   | 9402   |
| 5  | 25 | 0     | Inf. Dilution | lgK:1.67 (3SF)    | 0 | CRV   | 32224  |
| 6  | 25 | 0     | CHEMVAL2      | lgK:1.67 (0.01SD) | 0 | ENS   | 5156   |
| 7  | 25 | 0     | CHEMVAL3      | lgK:1.71 (0.01SD) | 0 | ENS   | 5156   |
| 8  | 25 | 0     | CHEMVAL4      | lgK:1.71 (0.01SD) | 0 | ENS   | 5156   |
| 9  | 25 | 4     | NaClO4        | lgK:-1.43 (3SF)   | 5 | MDS   | 366    |
| 10 | 25 | 0     | Inf. Dilution | dH:15.75 (4SF)    | 3 | CRV   | 9402   |

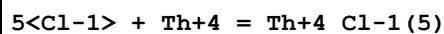
## Reaction No. 54177



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 23 | 4     | HClO4         | lgK:-1.46 (3SF)   | 3 | MCE   | 931    |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.45 (3SF)    | 0 | EBU   | 6372   |
| 3  | 25 | 0     | Inf. Dilution | lgK:1.26 (3SF)    | 3 | CRV   | 9402   |
| 4  | 25 | 0     | Inf. Dilution | lgK:1.21 (3SF)    | 4 | CRV   | 32222  |
| 5  | 25 | 0     | Inf. Dilution | lgK:1.25 (3SF)    | 4 | CRV   | 32224  |
| 6  | 25 | 0     | CHEMVAL2      | lgK:1.26 (0.01SD) | 0 | ENS   | 5156   |
| 7  | 25 | 0     | CHEMVAL3      | lgK:0.94 (0.01SD) | 0 | ENS   | 5156   |
| 8  | 25 | 0     | CHEMVAL4      | lgK:0.94 (0.01SD) | 4 | ENS   | 5156   |

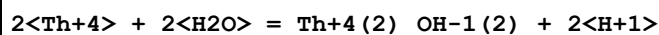
|    |    |   |                    |                 |   |     |      |
|----|----|---|--------------------|-----------------|---|-----|------|
| 9  | 25 | 4 | NaClO <sub>4</sub> | lgK:-1.85 (3SF) | 5 | MDS | 366  |
| 10 | 25 | 0 | Inf. Dilution      | dH:12.30 (4SF)  | 3 | CRV | 9402 |

## Reaction No. 54178



| No | t  | I-Str | Medium            | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-------------------|-----------------|---|-------|--------|
| 1  | 23 | 4     | HClO <sub>4</sub> | lgK:-2.46 (3SF) | 5 | MCE   | 931    |
| 2  | 25 | 0     | Inf. Dilution     | lgK:3.94 (3SF)  | 0 | EBU   | 6372   |

## Reaction No. 55342



| No               | t  | I-Str | Medium                | Value (Dev)         | W | Tech. | Ref. # |
|------------------|----|-------|-----------------------|---------------------|---|-------|--------|
| 1                | 0  | 1     | NaClO <sub>4</sub> /m | lgK:-5.60 (0.01SD)  | 3 | CRV   | 27322  |
| Note (1): p. 487 |    |       |                       |                     |   |       |        |
| 2                | 25 | 0     | Inf. Dilution         | lgK:-6.10 (3SF)     | 3 | CRV   | 9402   |
| 3                | 25 | 0     | Inf. Dilution         | lgK:-5.900 (0.25SD) | 5 | CRV   | 27322  |
| 4                | 25 | 0     | Inf. Dilution         | lgK:-5.7 (2SF)      | 4 | CRV   | 32222  |
| 5                | 25 | 0     | Inf. Dilution         | lgK:-6.13 (3SF)     | 4 | CRV   | 32224  |
| 6                | 25 | 0     | CHEMVAL2              | lgK:-6.10 (0.01SD)  | 0 | ENS   | 5156   |
| 7                | 25 | 0     | CHEMVAL3              | lgK:-5.41 (0.01SD)  | 0 | ENS   | 5156   |
| 8                | 25 | 0     | CHEMVAL4              | lgK:-5.41 (0.01SD)  | 3 | ENS   | 5156   |
| 9                | 25 | 0.5   | NaCl                  | lgK:-4.90 (0.16MD)  | 5 | MPT   | 4882   |
| 10               | 25 | 1     | LiCl                  | lgK:-5.1 (0.04SD)   | 2 | MGL   | 2261   |
| 11               | 25 | 1     | NaCl                  | lgK:-4.88 (0.03MD)  | 5 | MPT   | 4882   |
| 12               | 25 | 1     | NaClO <sub>4</sub> /m | lgK:-4.61 (0.01SD)  | 3 | CRV   | 27322  |

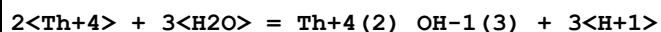
## Note (12): p. 487

|    |    |     |                       |                     |   |     |       |
|----|----|-----|-----------------------|---------------------|---|-----|-------|
| 13 | 25 | 1.5 | NaCl                  | lgK:-4.95 (0.15MD)  | 5 | MPT | 4882  |
| 14 | 25 | 2   | NaCl                  | lgK:-5.02 (0.16MD)  | 5 | MPT | 4882  |
| 15 | 25 | 2.5 | (Mg) Cl               | lgK:-4.9 (0.04SD)   | 5 | MGL | 2261  |
| 16 | 25 | 2.5 | NaCl                  | lgK:-5.06 (0.12MD)  | 5 | MPT | 4882  |
| 17 | 25 | 3   | KCl                   | lgK:-5.04 (0.11MD)  | 4 | MPT | 4882  |
| 18 | 25 | 3   | LiCl                  | lgK:-5.23 (0.15MD)  | 3 | MPT | 4882  |
| 19 | 25 | 3   | NaCl                  | lgK:-5.08 (0.2MD)   | 5 | MPT | 4882  |
| 20 | 25 | 3   | NaClO <sub>4</sub>    | lgK:-4.7 (0.04SD)   | 0 | MGL | 2261  |
| 21 | 95 | 1   | NaClO <sub>4</sub> /m | lgK:-2.55 (0.015SD) | 3 | CRV | 27322 |

## Note (21): p. 487

|    |    |     |               |                     |   |     |       |
|----|----|-----|---------------|---------------------|---|-----|-------|
| 22 | 25 | 0   | Inf. Dilution | dH:14.80 (4SF)      | 0 | CRV | 9402  |
| 23 | 25 | 0   | Inf. Dilution | dH:58.3 (2.9SD) kJ  | 5 | CRV | 27322 |
| 24 | 25 | 0.5 | NaCl          | dH:86.7 (0.1SD) kJ  | 0 | MCL | 4882  |
| 25 | 25 | 1   | NaCl          | dH:87.0 (0.1SD) kJ  | 0 | MCL | 4882  |
| 26 | 25 | 1.5 | NaCl          | dH:98.3 (0.1SD) kJ  | 0 | MCL | 4882  |
| 27 | 25 | 2   | NaCl          | dH:109.8 (0.1SD) kJ | 0 | MCL | 4882  |
| 28 | 25 | 2.5 | NaCl          | dH:110.3 (0.1SD) kJ | 0 | MCL | 4882  |
| 29 | 25 | 3   | KCl           | dH:101.9 (0.1SD) kJ | 0 | MCL | 4882  |
| 30 | 25 | 3   | LiCl          | dH:133.6 (0.1SD) kJ | 0 | MCL | 4882  |
| 31 | 25 | 3   | NaCl          | dH:119.0 (0.1SD) kJ | 0 | MCL | 4882  |

## Reaction No. 55343



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-6.8 (0.1SD)   | 5 | CRV   | 27322  |
| 2  | 25 | 0     | CHEMVAL2      | lgK:-9.55 (0.01SD) | 0 | ENS   | 5156   |
| 3  | 25 | 0     | CHEMVAL3      | lgK:-8.44 (0.01SD) | 0 | ENS   | 5156   |
| 4  | 25 | 0     | CHEMVAL4      | lgK:-8.44 (0.01SD) | 0 | ENS   | 5156   |
| 5  | 25 | 0.5   | NaCl          | lgK:-7.72 (0.05MD) | 5 | MPT   | 4882   |
| 6  | 25 | 1     | LiCl          | lgK:-7.9 (0.04SD)  | 5 | MGL   | 2261   |
| 7  | 25 | 1     | NaCl          | lgK:-7.93 (0.13MD) | 5 | MPT   | 4882   |
| 8  | 25 | 1.5   | NaCl          | lgK:-8.01 (0.10MD) | 5 | MPT   | 4882   |
| 9  | 25 | 2     | NaCl          | lgK:-8.11 (0.12MD) | 3 | MPT   | 4882   |
| 10 | 25 | 2.5   | (Mg) Cl       | lgK:-8.4 (0.04SD)  | 0 | MGL   | 2261   |
| 11 | 25 | 2.5   | NaCl          | lgK:-8.12 (0.15MD) | 5 | MPT   | 4882   |
| 12 | 25 | 3     | KCl           | lgK:-8.16 (0.17MD) | 3 | MPT   | 4882   |
| 13 | 25 | 3     | LiCl          | lgK:-8.28 (0.12MD) | 5 | MPT   | 4882   |
| 14 | 25 | 3     | NaCl          | lgK:-8.23 (0.18MD) | 5 | MPT   | 4882   |
| 15 | 25 | 0.5   | NaCl          | dH:90.7 (0.1SD) kJ | 0 | MCL   | 4882   |
| 16 | 25 | 1     | NaCl          | dH:76.8 (0.1SD) kJ | 0 | MCL   | 4882   |
| 17 | 25 | 1.5   | NaCl          | dH:60.1 (0.1SD) kJ | 0 | MCL   | 4882   |
| 18 | 25 | 2     | NaCl          | dH:45.3 (0.1SD) kJ | 0 | MCL   | 4882   |
| 19 | 25 | 2.5   | NaCl          | dH:33.3 (0.1SD) kJ | 5 | MCL   | 4882   |
| 20 | 25 | 3     | KCl           | dH:38.1 (0.1SD) kJ | 1 | MCL   | 4882   |
| 21 | 25 | 3     | LiCl          | dH:23.1 (0.1SD) kJ | 1 | MCL   | 4882   |
| 22 | 25 | 3     | NaCl          | dH:21.2 (0.1SD) kJ | 1 | MCL   | 4882   |

| Reaction No. 55701                                                                                        |    |       |               |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $4\text{Th(s)} + 4\text{O}_2\text{(g)} + 4\text{H}_2\text{(g)} = \text{Th}_4\text{OH-1(8)} + 8\text{e-1}$ |    |       |               |                 |   |       |        |
| No                                                                                                        | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0     | Inf. Dilution | lgK:805.2(4SF)  | 1 | EOR   |        |
| 2                                                                                                         | 25 | 1     | NaClO4        | dG:-1100.(5.SD) | 3 | ENS   | 3006   |
| 3                                                                                                         | 25 | 1     | NaClO4        | dH:-1220.(2.SD) | 3 | ENS   | 3006   |

| Reaction No. 55702                                                                                     |    |       |               |                 |   |       |        |
|--------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Th(s)} + \text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{Th}_4\text{OH-1(2)} + 2\text{e-1}$ |    |       |               |                 |   |       |        |
| No                                                                                                     | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                      | 25 | 0     | Inf. Dilution | lgK:200.0(4SF)  | 1 | EOR   |        |
| 2                                                                                                      | 25 | 1     | NaClO4        | dG:-271.(1.SD)  | 4 | ENS   | 3006   |
| 3                                                                                                      | 25 | 1     | NaClO4        | dH:-307.(0.5SD) | 5 | ENS   | 3006   |

| Reaction No. 55703                                                                                      |    |       |               |                |   |       |        |
|---------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $2\text{Th(s)} + \text{O}_2\text{(g)} + \text{H}_2\text{(g)} = \text{Th}_4\text{OH-1(2)} + 6\text{e-1}$ |    |       |               |                |   |       |        |
| No                                                                                                      | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                       | 25 | 0     | Inf. Dilution | lgK:324.2(4SF) | 1 | EOR   |        |
| 2                                                                                                       | 25 | 1     | NaClO4        | dG:-440.(3.SD) | 3 | ENS   | 3006   |
| 3                                                                                                       | 25 | 1     | NaClO4        | dH:-489.(1.SD) | 3 | ENS   | 3006   |

| Reaction No. 55704                                                                                        |    |       |               |                 |   |       |        |
|-----------------------------------------------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Th(s)} + 0.5\text{O}_2\text{(g)} + 0.5\text{H}_2\text{(g)} = \text{Th}_4\text{OH-1} + 3\text{e-1}$ |    |       |               |                 |   |       |        |
| No                                                                                                        | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                         | 25 | 0     | Inf. Dilution | lgK:162.6(4SF)  | 1 | ELC   |        |
| 2                                                                                                         | 25 | 1     | NaClO4        | dG:-220.(1.SD)  | 0 | ENS   | 3006   |
| 3                                                                                                         | 25 | 1     | NaClO4        | dH:-246.(0.5SD) | 2 | ENS   | 3006   |

| Reaction No. 55705                                             |     |       |               |                |   |       |        |
|----------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| $\text{O}_2\text{(g)} + \text{Th(s)} = \text{ThO}_2\text{(s)}$ |     |       |               |                |   |       |        |
| No                                                             | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                              | 25  | 0     | Inf. Dilution | lgK:204.8(4SF) | 4 | ENS   | 6422   |
| 2                                                              | 25  | 0     | Inf. Dilution | lgK:204.8(4SF) | 4 | ENS   | 6494   |
| 3                                                              | 27  | 0     | Inf. Dilution | lgK:203.4(4SF) | 0 | ENS   | 6422   |
| 4                                                              | 27  | 0     | Inf. Dilution | lgK:203.5(4SF) | 0 | ENS   | 6494   |
| 5                                                              | 127 | 0     | Inf. Dilution | lgK:150.1(4SF) | 4 | ENS   | 6422   |

|    |     |   |               |                        |   |     |       |
|----|-----|---|---------------|------------------------|---|-----|-------|
| 6  | 127 | 0 | Inf. Dilution | lgK:150.1(4SF)         | 4 | ENS | 6494  |
| 7  | 227 | 0 | Inf. Dilution | lgK:118.1(4SF)         | 3 | ENS | 6422  |
| 8  | 227 | 0 | Inf. Dilution | lgK:118.1(4SF)         | 3 | ENS | 6494  |
| 9  | 327 | 0 | Inf. Dilution | lgK:96.76(4SF)         | 0 | ENS | 6422  |
| 10 | 327 | 0 | Inf. Dilution | lgK:96.85(4SF)         | 0 | ENS | 6494  |
| 11 | 25  | 0 | Inf. Dilution | dG:-279.4(0.1SD)       | 4 | ENS | 802   |
| 12 | 25  | 0 | Inf. Dilution | dG:-279.3(4SF)         | 4 | EFE | 6422  |
| 13 | 25  | 0 | Inf. Dilution | dG:-279.4(4SF)         | 4 | EFE | 6494  |
| 14 | 25  | 0 | Inf. Dilution | dG:-1169.(4SF) kJ      | 6 | CRV | 8746  |
| 15 | 25  | 0 | Inf. Dilution | dG:-1169.2(3.504SD) kJ | 6 | CRV | 14923 |
| 16 | 25  | 0 | Inf. Dilution | dG:-279.35(5SF)        | 3 | CRV | 31291 |
| 17 | 25  | 0 | Inf. Dilution | dH:-293.(0.8SD)        | 5 | CRV | 5605  |
| 18 | 25  | 0 | Inf. Dilution | dH:-293.1(4SF)         | 4 | MTD | 6422  |
| 19 | 25  | 0 | Inf. Dilution | dH:-293.1(4SF)         | 4 | MTD | 6494  |
| 20 | 25  | 0 | Inf. Dilution | dH:-1226.4(3.5MD) kJ   | 6 | CRV | 8746  |
| 21 | 25  | 0 | Inf. Dilution | dH:-1226.4(3.500SD) kJ | 6 | CRV | 14923 |
| 22 | 127 | 0 | Inf. Dilution | dH:-292.9(4SF)         | 4 | MTD | 6422  |
| 23 | 127 | 0 | Inf. Dilution | dH:-292.9(4SF)         | 4 | MTD | 6494  |
| 24 | 227 | 0 | Inf. Dilution | dH:-292.7(4SF)         | 3 | MTD | 6422  |
| 25 | 227 | 0 | Inf. Dilution | dH:-292.6(4SF)         | 3 | MTD | 6494  |
| 26 | 327 | 0 | Inf. Dilution | dH:-292.5(4SF)         | 0 | MTD | 6422  |
| 27 | 327 | 0 | Inf. Dilution | dH:-292.4(4SF)         | 0 | MTD | 6494  |

## Reaction No. 55706

Th(s) = Th(g)

| No | t  | I-Str | Medium        | Value (Dev)           | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:133.(0.5SD)        | 0 | ENS   | 3006   |
| 2  | 25 | 0     | Inf. Dilution | dG:560.7(4SF) kJ      | 6 | CRV   | 8746   |
| 3  | 25 | 0     | Inf. Dilution | dG:560.75(6.002SD) kJ | 6 | CRV   | 14923  |
| 4  | 25 | 0     | Inf. Dilution | dH:143.(1.SD)         | 5 | CRV   | 5605   |
| 5  | 25 | 0     | Inf. Dilution | dH:602.(6.MD) kJ      | 6 | CRV   | 8746   |
| 6  | 25 | 0     | Inf. Dilution | dH:602.00(6.000SD) kJ | 6 | CRV   | 14923  |
| 7  | 25 | 0     | Inf. Dilution | dH:597.(3SF) kJ       | 3 | CRV   | 22524  |

## Reaction No. 55754

3&lt;Th(s)&gt; + 2&lt;N2(g)&gt; = Th3N4(s)



| No | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|-----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:212.5 (4SF)   | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:211.1 (4SF)   | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:153.9 (4SF)   | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:119.6 (4SF)   | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:96.75 (4SF)   | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-289.9 (0.1SD) | 4 | ENS   | 802    |
| 7  | 25  | 0     | Inf. Dilution | dG:-289.9 (4SF)   | 4 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-314.3 (0.1SD) | 4 | ENS   | 802    |
| 9  | 25  | 0     | Inf. Dilution | dH:-314.3 (4SF)   | 4 | MTD   | 6422   |
| 10 | 127 | 0     | Inf. Dilution | dH:-314.0 (4SF)   | 4 | MTD   | 6422   |
| 11 | 227 | 0     | Inf. Dilution | dH:-313.5 (4SF)   | 4 | MTD   | 6422   |
| 12 | 327 | 0     | Inf. Dilution | dH:-313.1 (4SF)   | 4 | MTD   | 6422   |

## Reaction No. 55755



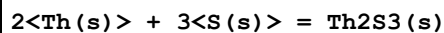
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:118. (0.5SD) | 4 | ENS   | 802    |

## Reaction No. 55756



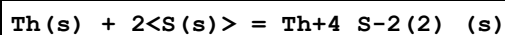
| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:63.67 (4SF)  | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:63.25 (4SF)  | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:46.23 (4SF)  | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:36.03 (4SF)  | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:29.24 (4SF)  | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-86.9 (0.1SD) | 4 | ENS   | 802    |
| 7  | 25  | 0     | Inf. Dilution | dG:-86.86 (4SF)  | 4 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-93.5 (0.1SD) | 4 | ENS   | 802    |
| 9  | 25  | 0     | Inf. Dilution | dH:-93.5 (3SF)   | 4 | MTD   | 6422   |
| 10 | 127 | 0     | Inf. Dilution | dH:-93.4 (3SF)   | 4 | MTD   | 6422   |
| 11 | 227 | 0     | Inf. Dilution | dH:-93.3 (3SF)   | 4 | MTD   | 6422   |
| 12 | 327 | 0     | Inf. Dilution | dH:-93.1 (3SF)   | 4 | MTD   | 6422   |

## Reaction No. 55757



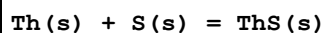
| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:188.7(4SF)   | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:187.5(4SF)   | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:140.3(4SF)   | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:111.8(4SF)   | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:92.66(4SF)   | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-257.4(0.1SD) | 4 | ENS   | 802    |
| 7  | 25  | 0     | Inf. Dilution | dG:-257.4(4SF)   | 4 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-259.0(0.1SD) | 4 | ENS   | 802    |
| 9  | 25  | 0     | Inf. Dilution | dH:-259.1(4SF)   | 4 | MTD   | 6422   |
| 10 | 127 | 0     | Inf. Dilution | dH:-260.7(4SF)   | 4 | MTD   | 6422   |
| 11 | 227 | 0     | Inf. Dilution | dH:-261.9(4SF)   | 4 | MTD   | 6422   |
| 12 | 327 | 0     | Inf. Dilution | dH:-262.8(4SF)   | 4 | MTD   | 6422   |

## Reaction No. 55758



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:108.6(4SF)   | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:107.9(4SF)   | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:80.62(4SF)   | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:64.12(4SF)   | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:53.06(4SF)   | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-148.2(0.1SD) | 4 | ENS   | 802    |
| 7  | 25  | 0     | Inf. Dilution | dG:-148.1(4SF)   | 4 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-149.7(0.1SD) | 4 | ENS   | 802    |
| 9  | 25  | 0     | Inf. Dilution | dH:-149.6(4SF)   | 4 | MTD   | 6422   |
| 10 | 127 | 0     | Inf. Dilution | dH:-150.7(4SF)   | 4 | MTD   | 6422   |
| 11 | 227 | 0     | Inf. Dilution | dH:-151.4(4SF)   | 4 | MTD   | 6422   |
| 12 | 327 | 0     | Inf. Dilution | dH:-152.0(4SF)   | 4 | MTD   | 6422   |

## Reaction No. 55759



| No | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-----|-------|---------------|----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:68.46(4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:68.03(4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:50.80(4SF) | 4 | ENS   | 6422   |

|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 4  | 227 | 0 | Inf. Dilution | lgK:40.39(4SF)  | 4 | ENS | 6422 |
| 5  | 327 | 0 | Inf. Dilution | lgK:33.42(4SF)  | 4 | ENS | 6422 |
| 6  | 25  | 0 | Inf. Dilution | dG:-93.4(0.1SD) | 4 | ENS | 802  |
| 7  | 25  | 0 | Inf. Dilution | dG:-93.39(4SF)  | 4 | EFE | 6422 |
| 8  | 25  | 0 | Inf. Dilution | dH:-94.5(0.1SD) | 4 | ENS | 802  |
| 9  | 25  | 0 | Inf. Dilution | dH:-94.5(3SF)   | 4 | MTD | 6422 |
| 10 | 127 | 0 | Inf. Dilution | dH:-95.1(3SF)   | 4 | MTD | 6422 |
| 11 | 227 | 0 | Inf. Dilution | dH:-95.5(3SF)   | 4 | MTD | 6422 |
| 12 | 327 | 0 | Inf. Dilution | dH:-95.9(3SF)   | 4 | MTD | 6422 |

| Reaction No. 55760                 |    |       |               |                  |   |       |        |
|------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th(s) + 2<I2(s)> = Th+4_I-1(4)_(g) |    |       |               |                  |   |       |        |
| No                                 | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                  | 25 | 0     | Inf. Dilution | dG:-123.1(0.1SD) | 4 | ENS   | 802    |
| 2                                  | 25 | 0     | Inf. Dilution | dH:-110.1(0.1SD) | 4 | ENS   | 802    |

| Reaction No. 55761                                                  |    |       |               |                  |   |       |        |
|---------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th(s) + 2<Br2(l)> + 7<H2(g)> + 3.5<O2(g)> = Th+4_Br-1(4)_H2O(7)_(s) |    |       |               |                  |   |       |        |
| No                                                                  | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0     | Inf. Dilution | dH:-757.2(0.1SD) | 4 | ENS   | 802    |

| Reaction No. 55762                   |    |       |               |                  |   |       |        |
|--------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th(s) + 2<Br2(l)> = Th+4_Br-1(4)_(g) |    |       |               |                  |   |       |        |
| No                                   | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                    | 25 | 0     | Inf. Dilution | dG:-187.5(0.1SD) | 4 | ENS   | 802    |
| 2                                    | 25 | 0     | Inf. Dilution | dH:-182.3(0.1SD) | 4 | ENS   | 802    |

| Reaction No. 55763                                                  |    |       |               |                  |   |       |        |
|---------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th(s) + 2<Cl2(g)> + 7<H2(g)> + 3.5<O2(g)> = Th+4_Cl-1(4)_H2O(7)_(s) |    |       |               |                  |   |       |        |
| No                                                                  | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0     | Inf. Dilution | dH:-804.4(0.1SD) | 4 | ENS   | 802    |

| Reaction No. 55764                                                |    |       |               |                  |   |       |        |
|-------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th(s) + 2<Cl2(g)> + 4<H2(g)> + 2<O2(g)> = Th+4_Cl-1(4)_H2O(4)_(s) |    |       |               |                  |   |       |        |
| No                                                                | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                 | 25 | 0     | Inf. Dilution | dH:-587.8(0.1SD) | 4 | ENS   | 802    |

| Reaction No. 55765                                                                                                                               |    |       |               |                  |   |       |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{Th(s)} + 2\text{Cl}_2(\text{g}) + 2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) = \text{Th}_4\text{Cl}_{10}\text{H}_2\text{O}_2(\text{s})$ |    |       |               |                  |   |       |        |
| No                                                                                                                                               | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                                                                                                | 25 | 0     | Inf. Dilution | dH:-436.0(0.1SD) | 4 | ENS   | 802    |

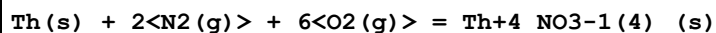
| Reaction No. 55766                                                          |    |       |               |                  |   |       |        |
|-----------------------------------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| $\text{Th(s)} + 2\text{F}_2(\text{g}) = \text{Th}_4\text{F}_{10}(\text{g})$ |    |       |               |                  |   |       |        |
| No                                                                          | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                                                           | 25 | 0     | Inf. Dilution | dG:-409.7(0.1SD) | 4 | ENS   | 802    |
| 2                                                                           | 25 | 0     | Inf. Dilution | dH:-418.0(0.1SD) | 4 | ENS   | 802    |

| Reaction No. 55767                                             |     |       |               |                 |   |       |        |
|----------------------------------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| $\text{Th(s)} + \text{H}_2(\text{g}) = \text{ThH}_2(\text{s})$ |     |       |               |                 |   |       |        |
| No                                                             | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                              | 25  | 0     | Inf. Dilution | lgK:17.51(4SF)  | 4 | ENS   | 6422   |
| 2                                                              | 27  | 0     | Inf. Dilution | lgK:17.36(4SF)  | 4 | ENS   | 6422   |
| 3                                                              | 127 | 0     | Inf. Dilution | lgK:11.24(4SF)  | 4 | ENS   | 6422   |
| 4                                                              | 227 | 0     | Inf. Dilution | lgK:7.522(4SF)  | 4 | ENS   | 6422   |
| 5                                                              | 327 | 0     | Inf. Dilution | lgK:5.017(4SF)  | 4 | ENS   | 6422   |
| 6                                                              | 25  | 0     | Inf. Dilution | dG:-23.9(0.1SD) | 4 | ENS   | 802    |
| 7                                                              | 25  | 0     | Inf. Dilution | dG:-23.89(4SF)  | 4 | EFE   | 6422   |
| 8                                                              | 25  | 0     | Inf. Dilution | dH:-33.4(0.1SD) | 4 | ENS   | 802    |
| 9                                                              | 25  | 0     | Inf. Dilution | dH:-33.4(3SF)   | 4 | MTD   | 6422   |
| 10                                                             | 127 | 0     | Inf. Dilution | dH:-33.8(3SF)   | 4 | MTD   | 6422   |
| 11                                                             | 227 | 0     | Inf. Dilution | dH:-34.2(3SF)   | 4 | MTD   | 6422   |
| 12                                                             | 327 | 0     | Inf. Dilution | dH:-34.5(3SF)   | 4 | MTD   | 6422   |

| Reaction No. 55768                                              |     |       |               |                 |   |       |        |
|-----------------------------------------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| $\text{Th(s)} + 0.5\text{O}_2(\text{g}) = \text{ThO}(\text{g})$ |     |       |               |                 |   |       |        |
| No                                                              | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                               | 25  | 0     | Inf. Dilution | lgK:8.791(4SF)  | 4 | ENS   | 6422   |
| 2                                                               | 27  | 0     | Inf. Dilution | lgK:8.764(4SF)  | 4 | ENS   | 6422   |
| 3                                                               | 127 | 0     | Inf. Dilution | lgK:7.649(4SF)  | 4 | ENS   | 6422   |
| 4                                                               | 227 | 0     | Inf. Dilution | lgK:6.952(4SF)  | 4 | ENS   | 6422   |
| 5                                                               | 327 | 0     | Inf. Dilution | lgK:6.467(4SF)  | 4 | ENS   | 6422   |
| 6                                                               | 25  | 0     | Inf. Dilution | dG:-12.0(0.1SD) | 4 | ENS   | 802    |
| 7                                                               | 25  | 0     | Inf. Dilution | dG:-11.99(4SF)  | 4 | EFE   | 6422   |

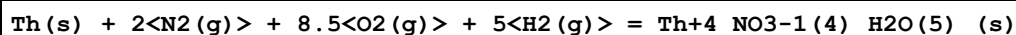
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 8  | 25  | 0 | Inf. Dilution | dH:-6.0 (0.1SD) | 4 | ENS | 802  |
| 9  | 25  | 0 | Inf. Dilution | dH:-6.0 (2SF)   | 4 | MTD | 6422 |
| 10 | 127 | 0 | Inf. Dilution | dH:-6.3 (2SF)   | 4 | MTD | 6422 |
| 11 | 227 | 0 | Inf. Dilution | dH:-6.5 (2SF)   | 4 | MTD | 6422 |
| 12 | 327 | 0 | Inf. Dilution | dH:-6.8 (2SF)   | 4 | MTD | 6422 |

## Reaction No. 55789



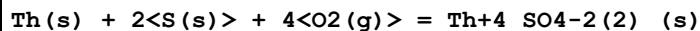
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dH:-345. (0.5SD) | 4 | ENS   | 802    |

## Reaction No. 55826



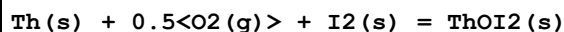
| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | dG:-556. (1.SD)     | 4 | ENS   | 3006   |
| 2  | 25 | 0     | Inf. Dilution | dG:-2324.9 (5SF) kJ | 5 | CRV   | 7065   |
| 3  | 25 | 0     | Inf. Dilution | dG:-2322.6 (5SF) kJ | 7 | MSP   | 31759  |
| 4  | 25 | 0     | Inf. Dilution | dH:-719. (1.SD)     | 4 | ENS   | 3006   |

## Reaction No. 55844



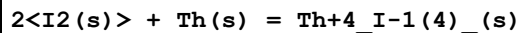
| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:404.8 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:402.0 (4SF) | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:291.3 (4SF) | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:224.7 (4SF) | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:180.3 (4SF) | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-550. (5.SD) | 4 | ENS   | 3006   |
| 7  | 25  | 0     | Inf. Dilution | dG:-552.2 (4SF) | 4 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-610. (5.SD) | 4 | ENS   | 3006   |
| 9  | 25  | 0     | Inf. Dilution | dH:-607.7 (4SF) | 4 | MTD   | 6422   |
| 10 | 127 | 0     | Inf. Dilution | dH:-609.0 (4SF) | 4 | MTD   | 6422   |
| 11 | 227 | 0     | Inf. Dilution | dH:-609.5 (4SF) | 4 | MTD   | 6422   |
| 12 | 327 | 0     | Inf. Dilution | dH:-609.4 (4SF) | 4 | MTD   | 6422   |

## Reaction No. 55845



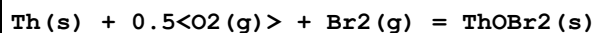
| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:169.4 (4SF)  | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:168.3 (4SF)  | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:124.7 (4SF)  | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:97.72 (4SF)  | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:79.28 (4SF)  | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-229. (0.7SD) | 4 | ENS   | 3006   |
| 7  | 25  | 0     | Inf. Dilution | dG:-231.1 (4SF)  | 4 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-238. (0.5SD) | 4 | ENS   | 3006   |
| 9  | 25  | 0     | Inf. Dilution | dH:-239.2 (4SF)  | 4 | MTD   | 6422   |
| 10 | 127 | 0     | Inf. Dilution | dH:-243.1 (4SF)  | 4 | MTD   | 6422   |
| 11 | 227 | 0     | Inf. Dilution | dH:-253.3 (4SF)  | 4 | MTD   | 6422   |
| 12 | 327 | 0     | Inf. Dilution | dH:-252.8 (4SF)  | 4 | MTD   | 6422   |

## Reaction No. 55871



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:114.9 (4SF)  | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:114.2 (4SF)  | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:85.05 (4SF)  | 2 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:65.96 (4SF)  | 1 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:52.33 (4SF)  | 0 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-157. (0.5SD) | 4 | ENS   | 3006   |
| 7  | 25  | 0     | Inf. Dilution | dG:-156.7 (4SF)  | 4 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dG:-158.40 (5SF) | 3 | CRV   | 31291  |
| 9  | 25  | 0     | Inf. Dilution | dH:-160. (0.5SD) | 4 | ENS   | 3006   |
| 10 | 25  | 0     | Inf. Dilution | dH:-158.9 (4SF)  | 4 | MTD   | 6422   |
| 11 | 25  | 0     | Inf. Dilution | dH:-160.57 (5SF) | 3 | CRV   | 31291  |
| 12 | 127 | 0     | Inf. Dilution | dH:-166.8 (4SF)  | 1 | MTD   | 6422   |
| 13 | 227 | 0     | Inf. Dilution | dH:-187.4 (4SF)  | 0 | MTD   | 6422   |
| 14 | 327 | 0     | Inf. Dilution | dH:-186.7 (4SF)  | 0 | MTD   | 6422   |

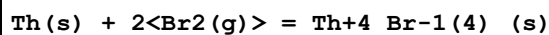
## Reaction No. 55872



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:198.8 (4SF) | 4 | ENS   | 6422   |

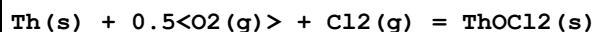
|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 2  | 27  | 0 | Inf. Dilution | lgK:197.5 (4SF) | 0 | ENS | 6422 |
| 3  | 127 | 0 | Inf. Dilution | lgK:145.0 (4SF) | 4 | ENS | 6422 |
| 4  | 227 | 0 | Inf. Dilution | lgK:113.3 (4SF) | 4 | ENS | 6422 |
| 5  | 327 | 0 | Inf. Dilution | lgK:92.12 (4SF) | 4 | ENS | 6422 |
| 6  | 25  | 0 | Inf. Dilution | dG:-260. (3.SD) | 0 | ENS | 3006 |
| 7  | 25  | 0 | Inf. Dilution | dG:-271.2 (4SF) | 4 | EFE | 6422 |
| 8  | 25  | 0 | Inf. Dilution | dH:-270. (3.SD) | 0 | ENS | 3006 |
| 9  | 25  | 0 | Inf. Dilution | dH:-283.8 (4SF) | 4 | MTD | 6422 |
| 10 | 127 | 0 | Inf. Dilution | dH:-290.8 (4SF) | 4 | MTD | 6422 |
| 11 | 227 | 0 | Inf. Dilution | dH:-290.3 (4SF) | 4 | MTD | 6422 |
| 12 | 327 | 0 | Inf. Dilution | dH:-289.9 (4SF) | 4 | MTD | 6422 |

## Reaction No. 55873



| No | t   | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|-----|-------|---------------|------------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:162.0 (4SF)  | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:161.0 (4SF)  | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:117.3 (4SF)  | 0 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:90.61 (4SF)  | 0 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:72.85 (4SF)  | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-222. (0.5SD) | 2 | ENS   | 802    |
| 7  | 25  | 0     | Inf. Dilution | dG:-221.0 (4SF)  | 4 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-231. (0.5SD) | 4 | ENS   | 802    |
| 9  | 25  | 0     | Inf. Dilution | dH:-230.8 (4SF)  | 4 | MTD   | 6422   |
| 10 | 127 | 0     | Inf. Dilution | dH:-244.9 (4SF)  | 0 | MTD   | 6422   |
| 11 | 227 | 0     | Inf. Dilution | dH:-244.2 (4SF)  | 0 | MTD   | 6422   |
| 12 | 327 | 0     | Inf. Dilution | dH:-243.5 (4SF)  | 4 | MTD   | 6422   |

## Reaction No. 55919



| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:202.5 (4SF) | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:201.2 (4SF) | 0 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:147.6 (4SF) | 2 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:115.5 (4SF) | 1 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:94.10 (4SF) | 0 | ENS   | 6422   |

|    |     |   |               |                 |   |     |      |
|----|-----|---|---------------|-----------------|---|-----|------|
| 6  | 25  | 0 | Inf. Dilution | dG:-278.(2.SD)  | 3 | ENS | 3006 |
| 7  | 25  | 0 | Inf. Dilution | dG:-276.3(4SF)  | 3 | EFE | 6422 |
| 8  | 25  | 0 | Inf. Dilution | dH:-295.(0.5SD) | 2 | ENS | 802  |
| 9  | 25  | 0 | Inf. Dilution | dH:-294.5(4SF)  | 2 | MTD | 6422 |
| 10 | 127 | 0 | Inf. Dilution | dH:-294.1(4SF)  | 1 | MTD | 6422 |
| 11 | 227 | 0 | Inf. Dilution | dH:-293.6(4SF)  | 0 | MTD | 6422 |
| 12 | 327 | 0 | Inf. Dilution | dH:-293.2(4SF)  | 0 | MTD | 6422 |

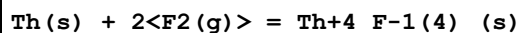
| Reaction No. 55932                   |     |       |               |                   |   |       |        |
|--------------------------------------|-----|-------|---------------|-------------------|---|-------|--------|
| 2<Cl2(g)> + Th(s) = Th+4_Cl-1(4)_(s) |     |       |               |                   |   |       |        |
| No                                   | t   | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                    | 25  | 0     | Inf. Dilution | lgK:191.8(4SF)    | 4 | ENS   | 6422   |
| 2                                    | 27  | 0     | Inf. Dilution | lgK:190.5(4SF)    | 0 | ENS   | 6422   |
| 3                                    | 127 | 0     | Inf. Dilution | lgK:138.9(4SF)    | 4 | ENS   | 6422   |
| 4                                    | 227 | 0     | Inf. Dilution | lgK:108.0(4SF)    | 3 | ENS   | 6422   |
| 5                                    | 327 | 0     | Inf. Dilution | lgK:87.43(4SF)    | 0 | ENS   | 6422   |
| 6                                    | 25  | 0     | Inf. Dilution | dG:-262.(0.5SD)   | 4 | ENS   | 3006   |
| 7                                    | 25  | 0     | Inf. Dilution | dG:-1094.5(5SF)kJ | 3 | CRV   | 6330   |
| 8                                    | 25  | 0     | Inf. Dilution | dG:-261.6(4SF)    | 4 | EFE   | 6422   |
| 9                                    | 25  | 0     | Inf. Dilution | dH:-283.(0.4SD)   | 4 | ENS   | 3006   |
| 10                                   | 25  | 0     | Inf. Dilution | dH:-1186.6(5SF)kJ | 3 | CRV   | 6330   |
| 11                                   | 25  | 0     | Inf. Dilution | dH:-283.6(4SF)    | 4 | MTD   | 6422   |
| 12                                   | 127 | 0     | Inf. Dilution | dH:-283.0(4SF)    | 4 | MTD   | 6422   |
| 13                                   | 227 | 0     | Inf. Dilution | dH:-282.4(4SF)    | 3 | MTD   | 6422   |
| 14                                   | 327 | 0     | Inf. Dilution | dH:-281.7(4SF)    | 0 | MTD   | 6422   |

| Reaction No. 55934                    |     |       |               |                |   |       |        |
|---------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| Th(s) + 0.5<O2(g)> + F2(g) = ThOF2(s) |     |       |               |                |   |       |        |
| No                                    | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25  | 0     | Inf. Dilution | lgK:278.5(4SF) | 4 | ENS   | 6422   |
| 2                                     | 27  | 0     | Inf. Dilution | lgK:276.7(4SF) | 4 | ENS   | 6422   |
| 3                                     | 127 | 0     | Inf. Dilution | lgK:204.2(4SF) | 4 | ENS   | 6422   |
| 4                                     | 227 | 0     | Inf. Dilution | lgK:160.8(4SF) | 4 | ENS   | 6422   |
| 5                                     | 327 | 0     | Inf. Dilution | lgK:131.8(4SF) | 4 | ENS   | 6422   |
| 6                                     | 25  | 0     | Inf. Dilution | dG:-380.(2.SD) | 4 | ENS   | 3006   |
| 7                                     | 25  | 0     | Inf. Dilution | dG:-379.9(4SF) | 4 | EFE   | 6422   |



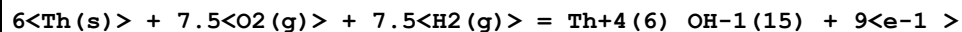
|    |     |   |               |                |   |     |      |
|----|-----|---|---------------|----------------|---|-----|------|
| 8  | 25  | 0 | Inf. Dilution | dH:-399.(2.SD) | 4 | ENS | 3006 |
| 9  | 25  | 0 | Inf. Dilution | dH:-398.0(4SF) | 4 | MTD | 6422 |
| 10 | 127 | 0 | Inf. Dilution | dH:-397.6(4SF) | 4 | MTD | 6422 |
| 11 | 227 | 0 | Inf. Dilution | dH:-397.3(4SF) | 4 | MTD | 6422 |
| 12 | 327 | 0 | Inf. Dilution | dH:-396.9(4SF) | 4 | MTD | 6422 |

## Reaction No. 55935



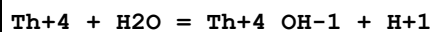
| No | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|-----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25  | 0     | Inf. Dilution | lgK:351.0(4SF)  | 4 | ENS   | 6422   |
| 2  | 27  | 0     | Inf. Dilution | lgK:348.7(4SF)  | 4 | ENS   | 6422   |
| 3  | 127 | 0     | Inf. Dilution | lgK:257.5(4SF)  | 4 | ENS   | 6422   |
| 4  | 227 | 0     | Inf. Dilution | lgK:202.8(4SF)  | 4 | ENS   | 6422   |
| 5  | 327 | 0     | Inf. Dilution | lgK:166.3(4SF)  | 4 | ENS   | 6422   |
| 6  | 25  | 0     | Inf. Dilution | dG:-478.(0.5SD) | 4 | ENS   | 3006   |
| 7  | 25  | 0     | Inf. Dilution | dG:-478.9(4SF)  | 4 | EFE   | 6422   |
| 8  | 25  | 0     | Inf. Dilution | dH:-500.(0.5SD) | 4 | ENS   | 802    |
| 9  | 25  | 0     | Inf. Dilution | dH:-501.4(4SF)  | 4 | MTD   | 6422   |
| 10 | 127 | 0     | Inf. Dilution | dH:-500.9(4SF)  | 4 | MTD   | 6422   |
| 11 | 227 | 0     | Inf. Dilution | dH:-500.3(4SF)  | 4 | MTD   | 6422   |
| 12 | 327 | 0     | Inf. Dilution | dH:-499.8(4SF)  | 4 | MTD   | 6422   |

## Reaction No. 55982



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:1327.3(5SF) | 1 | EOR   |        |
| 2  | 25 | 1     | NaClO4        | dG:-1810.(8.SD) | 3 | ENS   | 3006   |
| 3  | 25 | 1     | NaClO4        | dH:-2020.(4.SD) | 3 | ENS   | 3006   |

## Reaction No. 56006



| No               | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|------------------|----|-------|---------------|-------------------|---|-------|--------|
| 1                | 0  | 1     | NaClO4/m      | lgK:-4.32(0.01SD) | 3 | CRV   | 27322  |
| Note (1): p. 487 |    |       |               |                   |   |       |        |
| 2                | 25 | 0     | Inf. Dilution | lgK:-2.69(3SF)    | 0 | EBU   | 6372   |
| 3                | 25 | 0     | Inf. Dilution | lgK:-3.20(3SF)    | 0 | CRV   | 6478   |
| 4                | 25 | 0     | Inf. Dilution | lgK:-3.20(3SF)    | 0 | CRV   | 9402   |

|                   |    |     |               |                       |   |     |       |
|-------------------|----|-----|---------------|-----------------------|---|-----|-------|
| 5                 | 25 | 0   | Inf. Dilution | lgK:-2.500 (0.25SD)   | 5 | CRV | 27322 |
| 6                 | 25 | 0   | Inf. Dilution | lgK:-2.20 (0.20SD)    | 4 | CRV | 32222 |
| 7                 | 25 | 0   | Inf. Dilution | lgK:-3.21 (3SF)       | 2 | CRV | 32224 |
| 8                 | 25 | 0   | CHEMVAL2      | lgK:-3.20 (0.01SD)    | 0 | ENS | 5156  |
| 9                 | 25 | 0   | CHEMVAL3      | lgK:-2.34 (0.01SD)    | 0 | ENS | 5156  |
| 10                | 25 | 0   | CHEMVAL4      | lgK:-2.34 (0.01SD)    | 4 | ENS | 5156  |
| 11                | 25 | 0.1 | KNO3          | lgK:-2.98 (0.007SD)   | 6 | MPT | 5586  |
| 12                | 25 | 0.7 | Seawater      | lgK:-3.74 (3SF)       | 0 | ENS | 6496  |
| 13                | 25 | 1   | NaClO4/m      | lgK:-4.15 (0.02SD)    | 0 | CRV | 27322 |
| Note (13): p. 487 |    |     |               |                       |   |     |       |
| 14                | 95 | 1   | NaClO4/m      | lgK:-2.29 (0.01SD)    | 3 | CRV | 27322 |
| Note (14): p. 487 |    |     |               |                       |   |     |       |
| 15                | 25 | 0   | Inf. Dilution | dH:11.9 (3SF)         | 0 | CRV | 6478  |
| 16                | 25 | 0   | Inf. Dilution | dH:11.2 (2.SD)        | 2 | CRV | 6478  |
| 17                | 25 | 0   | Inf. Dilution | dH:5.95 (3SF)         | 0 | CRV | 9402  |
| 18                | 25 | 0   | Inf. Dilution | dH:44.200 (3.2SD) kJ  | 5 | CRV | 27322 |
| 19                | 25 | 0   | Inf. Dilution | dH:53.67 (10.49SD) kJ | 3 | CRV | 32222 |
| 20                | 25 | 0.7 | Seawater      | dH:11.2 (3SF)         | 0 | ENS | 6496  |

| Reaction No. 56007                     |    |       |               |                     |   |       |        |
|----------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Th+4 + 2<H2O> = Th+4_OH-1 (2) + 2<H+1> |    |       |               |                     |   |       |        |
| No                                     | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                      | 0  | 1     | NaClO4/m      | lgK:-8.48 (0.015SD) | 3 | CRV   | 27322  |
| 2                                      | 25 | 0     | Inf. Dilution | lgK:-5.91 (3SF)     | 0 | EBU   | 6372   |
| 3                                      | 25 | 0     | Inf. Dilution | lgK:-6.95 (3SF)     | 0 | CRV   | 9402   |
| 4                                      | 25 | 0     | Inf. Dilution | lgK:-6.2 (0.25SD)   | 3 | CRV   | 27322  |
| 5                                      | 25 | 0     | Inf. Dilution | lgK:-6.00 (0.60SD)  | 4 | CRV   | 32222  |
| 6                                      | 25 | 0     | Inf. Dilution | lgK:-6.93 (3SF)     | 3 | CRV   | 32224  |
| 7                                      | 25 | 0     | CHEMVAL2      | lgK:-6.90 (0.01SD)  | 0 | ENS   | 5156   |
| 8                                      | 25 | 0     | CHEMVAL3      | lgK:-6.36 (0.01SD)  | 0 | ENS   | 5156   |
| 9                                      | 25 | 0     | CHEMVAL4      | lgK:-6.36 (0.01SD)  | 0 | ENS   | 5156   |
| 10                                     | 25 | 0.7   | Seawater      | lgK:-8.02 (3SF)     | 0 | ENS   | 6496   |
| 11                                     | 25 | 1     | NaClO4/m      | lgK:-7.70 (0.015SD) | 2 | CRV   | 27322  |
| Note (11): p. 487                      |    |       |               |                     |   |       |        |
| 12                                     | 95 | 1     | NaClO4/m      | lgK:-4.50 (0.005SD) | 2 | CRV   | 27322  |
| Note (12): p. 487                      |    |       |               |                     |   |       |        |

|    |    |     |               |                       |   |     |       |
|----|----|-----|---------------|-----------------------|---|-----|-------|
| 13 | 25 | 0   | Inf. Dilution | dH:13.80 (4SF)        | 0 | CRV | 9402  |
| 14 | 25 | 0   | Inf. Dilution | dH:85.700 (20.7SD) kJ | 3 | CRV | 27322 |
| 15 | 25 | 0   | Inf. Dilution | dH:90.56 (10.98SD) kJ | 3 | CRV | 32222 |
| 16 | 25 | 0.7 | Seawater      | dH:21.9 (3SF)         | 0 | ENS | 6496  |

| Reaction No. 56008                     |    |       |               |                        |   |       |        |
|----------------------------------------|----|-------|---------------|------------------------|---|-------|--------|
| Th+4 + 3<H2O> = Th+4_OH-1 (3) + 3<H+1> |    |       |               |                        |   |       |        |
| No                                     | t  | I-Str | Medium        | Value (Dev)            | W | Tech. | Ref. # |
| 1                                      | 25 | 0     | Inf. Dilution | lgK:-9.53 (3SF)        | 0 | EBU   | 6372   |
| 2                                      | 25 | 0     | Inf. Dilution | lgK:-11.70 (4SF)       | 3 | CRV   | 9402   |
| 3                                      | 25 | 0     | Inf. Dilution | lgK:-11.00 (1.00SD)    | 4 | CRV   | 32222  |
| 4                                      | 25 | 0     | Inf. Dilution | lgK:-11.68 (4SF)       | 2 | CRV   | 32224  |
| Note (4): source indicates uncertainty |    |       |               |                        |   |       |        |
| 5                                      | 25 | 0     | CHEMVAL2      | lgK:-11.70 (0.01SD)    | 0 | ENS   | 5156   |
| 6                                      | 25 | 0     | CHEMVAL3      | lgK:-11.70 (0.01SD)    | 0 | ENS   | 5156   |
| 7                                      | 25 | 0     | CHEMVAL4      | lgK:-11.70 (0.01SD)    | 4 | ENS   | 5156   |
| 8                                      | 25 | 0.7   | Seawater      | lgK:-13.08 (4SF)       | 0 | ENS   | 6496   |
| 9                                      | 25 | 0     | Inf. Dilution | dH:20.40 (4SF)         | 0 | CRV   | 9402   |
| 10                                     | 25 | 0     | Inf. Dilution | dH:121.52 (11.89SD) kJ | 3 | CRV   | 32222  |
| 11                                     | 25 | 0.7   | Seawater      | dH:30.4 (3SF)          | 0 | ENS   | 6496   |

| Reaction No. 56009                     |    |       |               |                     |   |       |        |
|----------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| Th+4 + 4<H2O> = Th+4_OH-1 (4) + 4<H+1> |    |       |               |                     |   |       |        |
| No                                     | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                      | 25 | 0     | Inf. Dilution | lgK:-13.51 (4SF)    | 0 | EBU   | 6372   |
| 2                                      | 25 | 0     | Inf. Dilution | lgK:-19.7 (3SF)     | 0 | EOD   | 8120   |
| 3                                      | 25 | 0     | Inf. Dilution | lgK:-15.90 (4SF)    | 0 | CRV   | 9402   |
| 4                                      | 25 | 0     | Inf. Dilution | lgK:-17.4 (0.35SD)  | 5 | CRV   | 27322  |
| 5                                      | 25 | 0     | Inf. Dilution | lgK:-17.50 (0.50SD) | 4 | CRV   | 32222  |
| 6                                      | 25 | 0     | Inf. Dilution | lgK:-15.8 (3SF)     | 0 | CRV   | 32224  |
| 7                                      | 25 | 0     | CHEMVAL2      | lgK:-15.90 (0.01SD) | 0 | ENS   | 5156   |
| 8                                      | 25 | 0     | CHEMVAL3      | lgK:-15.90 (0.01SD) | 0 | ENS   | 5156   |
| 9                                      | 25 | 0     | CHEMVAL4      | lgK:-15.90 (0.01SD) | 0 | ENS   | 5156   |
| 10                                     | 25 | 0.7   | Seawater      | lgK:-17.28 (4SF)    | 0 | ENS   | 6496   |
| 11                                     | 25 | 3     | NaClO4        | lgK:-15.7 (0.04SD)  | 5 | MGL   | 2261   |
| 12                                     | 25 | 0     | Inf. Dilution | dH:24.70 (4SF)      | 0 | CRV   | 9402   |

|    |    |     |               |                        |   |     |       |
|----|----|-----|---------------|------------------------|---|-----|-------|
| 13 | 25 | 0   | Inf. Dilution | dH:153.74 (10.81SD) kJ | 3 | CRV | 32222 |
| 14 | 25 | 0.7 | Seawater      | dH:34.4 (3SF)          | 0 | ENS | 6496  |

| Reaction No. 56010                          |    |       |               |                        |   |       |        |
|---------------------------------------------|----|-------|---------------|------------------------|---|-------|--------|
| 4<Th+4> + 8<H2O> = Th+4(4)_OH-1(8) + 8<H+1> |    |       |               |                        |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)            | W | Tech. | Ref. # |
| 1                                           | 0  | 1     | NaClO4/m      | lgK:-22.79 (0.01SD)    | 3 | CRV   | 27322  |
| Note (1): p. 487                            |    |       |               |                        |   |       |        |
| 2                                           | 25 | 0     | Inf. Dilution | lgK:-21.10 (4SF)       | 0 | CRV   | 9402   |
| 3                                           | 25 | 0     | Inf. Dilution | lgK:-20.400 (0.2SD)    | 5 | CRV   | 27322  |
| 4                                           | 25 | 0     | Inf. Dilution | lgK:-20.4 (3SF)        | 4 | CRV   | 32222  |
| 5                                           | 25 | 0     | Inf. Dilution | lgK:-21.06 (4SF)       | 4 | CRV   | 32224  |
| 6                                           | 25 | 0     | CHEMVAL2      | lgK:-21.10 (0.01SD)    | 0 | ENS   | 5156   |
| 7                                           | 25 | 0     | CHEMVAL3      | lgK:-20.08 (0.01SD)    | 0 | ENS   | 5156   |
| 8                                           | 25 | 0     | CHEMVAL4      | lgK:-20.10 (0.01SD)    | 0 | ENS   | 5156   |
| 9                                           | 25 | 1     | NaClO4/m      | lgK:-19.01 (0.01SD)    | 3 | CRV   | 27322  |
| Note (9): p. 487                            |    |       |               |                        |   |       |        |
| 10                                          | 25 | 3     | NaClO4        | lgK:-19.2 (0.04SD)     | 4 | MGL   | 2261   |
| 11                                          | 95 | 1     | NaClO4/m      | lgK:-10.49 (0.015SD)   | 3 | CRV   | 27322  |
| Note (11): p. 487                           |    |       |               |                        |   |       |        |
| 12                                          | 25 | 0     | Inf. Dilution | dH:57.80 (4SF)         | 3 | CRV   | 9402   |
| 13                                          | 25 | 0     | Inf. Dilution | dH:243.000 (10.7SD) kJ | 5 | CRV   | 27322  |

| Reaction No. 56011                             |    |       |          |                     |   |       |        |
|------------------------------------------------|----|-------|----------|---------------------|---|-------|--------|
| 6<Th+4> + 14<H2O> = Th+4(6)_OH-1(14) + 14<H+1> |    |       |          |                     |   |       |        |
| No                                             | t  | I-Str | Medium   | Value (Dev)         | W | Tech. | Ref. # |
| 1                                              | 25 | 0     | CHEMVAL2 | lgK:-35.23 (0.01SD) | 0 | ENS   | 5156   |
| 2                                              | 25 | 0     | CHEMVAL3 | lgK:-38.83 (0.01SD) | 0 | ENS   | 5156   |
| 3                                              | 25 | 0     | CHEMVAL4 | lgK:-38.80 (0.01SD) | 4 | ENS   | 5156   |
| 4                                              | 25 | 3     | NaClO4   | lgK:-33.8 (0.04SD)  | 5 | MGL   | 2261   |

| Reaction No. 56012                             |    |       |               |                     |   |       |        |
|------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| 6<Th+4> + 15<H2O> = Th+4(6)_OH-1(15) + 15<H+1> |    |       |               |                     |   |       |        |
| No                                             | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                              | 0  | 1     | NaClO4/m      | lgK:-43.84 (0.01SD) | 3 | CRV   | 27322  |
| Note (1): p. 487                               |    |       |               |                     |   |       |        |
| 2                                              | 25 | 0     | Inf. Dilution | lgK:-36.80 (0.01SD) | 3 | ENS   | 5156   |

|                   |    |     |               |                     |   |     |       |
|-------------------|----|-----|---------------|---------------------|---|-----|-------|
| 3                 | 25 | 0   | Inf. Dilution | lgK:-39.82 (4SF)    | 0 | EBU | 6372  |
| 4                 | 25 | 0   | Inf. Dilution | lgK:-36.70 (4SF)    | 3 | CRV | 9402  |
| 5                 | 25 | 0   | Inf. Dilution | lgK:-36.64 (4SF)    | 4 | CRV | 32224 |
| 6                 | 25 | 0   | CHEMVAL2      | lgK:-36.80 (0.01SD) | 0 | ENS | 5156  |
| 7                 | 25 | 0   | CHEMVAL3      | lgK:-43.84 (0.01SD) | 0 | ENS | 5156  |
| 8                 | 25 | 0   | CHEMVAL4      | lgK:-43.80 (0.01SD) | 0 | ENS | 5156  |
| 9                 | 25 | 0.1 | KNO3          | lgK:-34.4 (0.03SD)  | 0 | MPT | 5586  |
| 10                | 25 | 1   | NaClO4/m      | lgK:-36.76 (0.01SD) | 2 | CRV | 27322 |
| Note (10): p. 487 |    |     |               |                     |   |     |       |
| 11                | 95 | 1   | NaClO4/m      | lgK:-20.63 (0.01SD) | 3 | CRV | 27322 |
| Note (11): p. 487 |    |     |               |                     |   |     |       |
| 12                | 25 | 0   | Inf. Dilution | dH:108.40 (5SF)     | 2 | CRV | 9402  |

| Reaction No. 56013        |    |       |               |                    |   |       |        |
|---------------------------|----|-------|---------------|--------------------|---|-------|--------|
| CO3-2 + Th+4 = Th+4_CO3-2 |    |       |               |                    |   |       |        |
| No                        | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                         | 25 | 0     | Inf. Dilution | lgK:11.31 (4SF)    | 0 | EBU   | 6372   |
| 2                         | 25 | 0     | Inf. Dilution | lgK:11.00 (4SF)    | 4 | CRV   | 9402   |
| 3                         | 25 | 0     | CHEMVAL2      | lgK:11.03 (0.01SD) | 0 | ENS   | 5156   |
| 4                         | 25 | 0     | CHEMVAL3      | lgK:11.03 (0.01SD) | 0 | ENS   | 5156   |
| 5                         | 25 | 0     | CHEMVAL4      | lgK:11.00 (0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56014                  |    |       |          |                    |   |       |        |
|-------------------------------------|----|-------|----------|--------------------|---|-------|--------|
| Th+4 + H+1 + PO4-3 = Th+4_H+1_PO4-3 |    |       |          |                    |   |       |        |
| No                                  | t  | I-Str | Medium   | Value (Dev)        | W | Tech. | Ref. # |
| 1                                   | 25 | 0     | CHEMVAL2 | lgK:25.25 (0.01SD) | 0 | ENS   | 5156   |
| 2                                   | 25 | 0     | CHEMVAL3 | lgK:25.25 (0.01SD) | 0 | ENS   | 5156   |
| 3                                   | 25 | 0     | CHEMVAL4 | lgK:25.30 (0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56015                                |    |       |          |                    |   |       |        |
|---------------------------------------------------|----|-------|----------|--------------------|---|-------|--------|
| Th+4 + 2<H+1> + 2<PO4-3> = Th+4_H+1 (2)_PO4-3 (2) |    |       |          |                    |   |       |        |
| No                                                | t  | I-Str | Medium   | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | CHEMVAL2 | lgK:47.49 (0.01SD) | 0 | ENS   | 5156   |
| 2                                                 | 25 | 0     | CHEMVAL3 | lgK:47.49 (0.01SD) | 0 | ENS   | 5156   |
| 3                                                 | 25 | 0     | CHEMVAL4 | lgK:50.70 (0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56016 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Th+4 + 3<H+1> + 3<PO4-3> = Th+4_H+1(3)_PO4-3(3) |    |       |          |                   |   |       |        |
|-------------------------------------------------|----|-------|----------|-------------------|---|-------|--------|
| No                                              | t  | I-Str | Medium   | Value (Dev)       | W | Tech. | Ref. # |
| 1                                               | 25 | 0     | CHEMVAL2 | lgK:68.30(0.01SD) | 0 | ENS   | 5156   |
| 2                                               | 25 | 0     | CHEMVAL3 | lgK:68.30(0.01SD) | 0 | ENS   | 5156   |
| 3                                               | 25 | 0     | CHEMVAL4 | lgK:71.60(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56017                        |    |       |               |                   |   |       |        |
|-------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Th+4 + 2<H+1> + PO4-3 = Th+4_H+1(2)_PO4-3 |    |       |               |                   |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | Inf. Dilution | lgK:24.2(0.03SD)  | 4 | ENS   | 5458   |
| 2                                         | 25 | 0     | Inf. Dilution | lgK:24.16(4SF)    | 4 | CRV   | 32224  |
| 3                                         | 25 | 0     | CHEMVAL2      | lgK:24.07(0.01SD) | 0 | ENS   | 5156   |
| 4                                         | 25 | 0     | CHEMVAL3      | lgK:24.07(0.01SD) | 0 | ENS   | 5156   |
| 5                                         | 25 | 0     | CHEMVAL4      | lgK:24.80(0.01SD) | 4 | ENS   | 5156   |

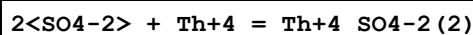
| Reaction No. 56018                              |    |       |               |                   |   |       |        |
|-------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Th+4 + 4<H+1> + 2<PO4-3> = Th+4_H+1(4)_PO4-3(2) |    |       |               |                   |   |       |        |
| No                                              | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                               | 25 | 0     | Inf. Dilution | lgK:46.5(3SF)     | 1 | ELC   |        |
| 2                                               | 25 | 0     | Inf. Dilution | lgK:47.95(4SF)    | 0 | CRV   | 32224  |
| 3                                               | 25 | 0     | CHEMVAL2      | lgK:47.98(0.01SD) | 0 | ENS   | 5156   |
| 4                                               | 25 | 0     | CHEMVAL3      | lgK:47.98(0.01SD) | 0 | ENS   | 5156   |
| 5                                               | 25 | 0     | CHEMVAL4      | lgK:48.90(0.01SD) | 0 | ENS   | 5156   |

| Reaction No. 56019                        |    |       |          |                   |   |       |        |
|-------------------------------------------|----|-------|----------|-------------------|---|-------|--------|
| Th+4 + 3<H+1> + PO4-3 = Th+4_H+1(3)_PO4-3 |    |       |          |                   |   |       |        |
| No                                        | t  | I-Str | Medium   | Value (Dev)       | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | CHEMVAL2 | lgK:23.60(0.01SD) | 0 | ENS   | 5156   |
| 2                                         | 25 | 0     | CHEMVAL3 | lgK:23.60(0.01SD) | 0 | ENS   | 5156   |
| 3                                         | 25 | 0     | CHEMVAL4 | lgK:23.70(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56020        |    |       |               |                  |   |       |        |
|---------------------------|----|-------|---------------|------------------|---|-------|--------|
| SO4-2 + Th+4 = Th+4_SO4-2 |    |       |               |                  |   |       |        |
| No                        | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                         | 25 | 0     | Inf. Dilution | lgK:5.45(0.02SD) | 0 | ENS   | 5458   |
| 2                         | 25 | 0     | Inf. Dilution | lgK:4.78(3SF)    | 0 | EBU   | 6372   |
| 3                         | 25 | 0     | Inf. Dilution | lgK:5.45(3SF)    | 0 | CRV   | 9402   |

|    |    |   |               |                   |   |     |       |
|----|----|---|---------------|-------------------|---|-----|-------|
| 4  | 25 | 0 | Inf. Dilution | lgK:6.17(0.32SD)  | 5 | CRV | 25812 |
| 5  | 25 | 0 | Inf. Dilution | lgK:6.170(0.16SD) | 5 | CRV | 27322 |
| 6  | 25 | 0 | Inf. Dilution | lgK:6.17(0.32SD)  | 3 | EOD | 31756 |
| 7  | 25 | 0 | Inf. Dilution | lgK:7.9(2SF)      | 0 | CRV | 32222 |
| 8  | 25 | 0 | Inf. Dilution | lgK:5.45(3SF)     | 3 | CRV | 32224 |
| 9  | 25 | 0 | CHEMVAL2      | lgK:5.45(0.01SD)  | 0 | ENS | 5156  |
| 10 | 25 | 0 | CHEMVAL3      | lgK:5.45(0.01SD)  | 0 | ENS | 5156  |
| 11 | 25 | 0 | CHEMVAL4      | lgK:5.86(0.01SD)  | 0 | ENS | 5156  |
| 12 | 25 | 2 | NaClO4        | lgK:3.28(3SF)     | 4 | MDS | 140   |
| 13 | 25 | 2 | NaClO4        | lgK:3.32(3SF)     | 4 | MDS | 140   |
| 14 | 25 | 2 | Unknown       | lgK:3.22(0.01SD)  | 6 | CRV | 552   |
| 15 | 25 | 0 | Inf. Dilution | dH:3.70(3SF)      | 0 | CRV | 9402  |
| 16 | 25 | 2 | Unknown       | dH:5.0(2SF)       | 4 | CRV | 2556  |

## Reaction No. 56021



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:9.7(0.03SD)    | 4 | ENS   | 5458   |
| 2  | 25 | 0     | Inf. Dilution | lgK:8.19(3SF)      | 0 | EBU   | 6372   |
| 3  | 25 | 0     | Inf. Dilution | lgK:11.59(4SF)     | 0 | EOD   | 8120   |
| 4  | 25 | 0     | Inf. Dilution | lgK:9.75(3SF)      | 3 | CRV   | 9402   |
| 5  | 25 | 0     | Inf. Dilution | lgK:9.69(0.14SD)   | 5 | CRV   | 27322  |
| 6  | 25 | 0     | Inf. Dilution | lgK:9.69(0.27SD)   | 3 | EOD   | 31756  |
| 7  | 25 | 0     | Inf. Dilution | lgK:11.8(3SF)      | 0 | CRV   | 32222  |
| 8  | 25 | 0     | Inf. Dilution | lgK:9.72(3SF)      | 4 | CRV   | 32224  |
| 9  | 25 | 0     | CHEMVAL2      | lgK:9.73(0.01SD)   | 0 | ENS   | 5156   |
| 10 | 25 | 0     | CHEMVAL3      | lgK:9.73(0.01SD)   | 0 | ENS   | 5156   |
| 11 | 25 | 0     | CHEMVAL4      | lgK:8.69(0.01SD)   | 0 | ENS   | 5156   |
| 12 | 25 | 2     | NaClO4        | lgK:5.61(3SF)      | 5 | MDS   | 366    |
| 13 | 25 | 2     | NaClO4        | lgK:5.70(3SF)      | 2 | MDS   | 8136   |
| 14 | 25 | 2     | Unknown       | lgK:5.53(3SF)      | 3 | CRV   | 2556   |
| 15 | 25 | 0     | Inf. Dilution | dH:7.50(3SF)       | 0 | CRV   | 9402   |
| 16 | 25 | 0     | Inf. Dilution | dH:40.38(0.5SD) kJ | 5 | CRV   | 27322  |
| 17 | 25 | 2     | Unknown       | dH:9.6(2SF)        | 0 | CRV   | 2556   |

## Reaction No. 56022

| 3<SO4-2> + Th+4 = Th+4_SO4-2 (3) |    |       |               |                     |   |       |        |
|----------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| No                               | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                | 25 | 0     | Inf. Dilution | lgK:10.50 (0.02SD)  | 4 | ENS   | 5458   |
| 2                                | 25 | 0     | Inf. Dilution | lgK:10.48 (4SF)     | 2 | EBU   | 6372   |
| 3                                | 25 | 0     | Inf. Dilution | lgK:12.42 (4SF)     | 0 | EOD   | 8120   |
| 4                                | 25 | 0     | Inf. Dilution | lgK:10.50 (4SF)     | 3 | CRV   | 9402   |
| 5                                | 25 | 0     | Inf. Dilution | lgK:10.748 (0.04SD) | 4 | CRV   | 27322  |
| 6                                | 25 | 0     | Inf. Dilution | lgK:10.75 (0.08SD)  | 3 | EOD   | 31756  |
| 7                                | 25 | 0     | Inf. Dilution | lgK:12.4 (3SF)      | 0 | CRV   | 32222  |
| 8                                | 25 | 0     | Inf. Dilution | lgK:10.47 (4SF)     | 4 | CRV   | 32224  |
| 9                                | 25 | 0     | CHEMVAL2      | lgK:10.50 (0.01SD)  | 0 | ENS   | 5156   |
| 10                               | 25 | 0     | CHEMVAL3      | lgK:10.50 (0.01SD)  | 0 | ENS   | 5156   |
| 11                               | 25 | 0     | CHEMVAL4      | lgK:10.50 (0.01SD)  | 2 | ENS   | 5156   |
| 12                               | 25 | 0     | Inf. Dilution | dH:11.70 (4SF)      | 3 | CRV   | 9402   |

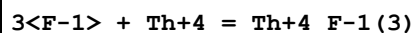
| Reaction No. 56023                                   |    |       |               |                   |   |       |        |
|------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 4<SO4-2> + Th+4 = Th+4_SO4-2 (4)                     |    |       |               |                   |   |       |        |
| Note: Consider with caution. Reversal of stepwise K. |    |       |               |                   |   |       |        |
| No                                                   | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                    | 25 | 0     | Inf. Dilution | lgK:8.4 (0.03SD)  | 4 | ENS   | 5458   |
| 2                                                    | 25 | 0     | Inf. Dilution | lgK:11.75 (4SF)   | 0 | EBU   | 6372   |
| 3                                                    | 25 | 0     | Inf. Dilution | lgK:8.50 (3SF)    | 3 | CRV   | 9402   |
| 4                                                    | 25 | 0     | Inf. Dilution | lgK:8.44 (3SF)    | 2 | EOD   | 31756  |
| Note (4): Deprecated                                 |    |       |               |                   |   |       |        |
| 5                                                    | 25 | 0     | Inf. Dilution | lgK:8.44 (3SF)    | 4 | CRV   | 32224  |
| 6                                                    | 25 | 0     | CHEMVAL2      | lgK:8.48 (0.01SD) | 0 | ENS   | 5156   |
| 7                                                    | 25 | 0     | CHEMVAL3      | lgK:8.48 (0.01SD) | 0 | ENS   | 5156   |
| 8                                                    | 25 | 0     | CHEMVAL4      | lgK:8.48 (0.01SD) | 2 | ENS   | 5156   |
| 9                                                    | 25 | 0     | Inf. Dilution | dH:13.10 (4SF)    | 3 | CRV   | 9402   |

| Reaction No. 56024           |    |       |               |                  |   |       |        |
|------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<F-1> + Th+4 = Th+4_F-1 (2) |    |       |               |                  |   |       |        |
| No                           | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                            | 25 | 0     | Inf. Dilution | lgK:15.1 (0.2SD) | 2 | CRV   | 552    |
| 2                            | 25 | 0     | Inf. Dilution | lgK:14.61 (4SF)  | 0 | EBU   | 6372   |
| 3                            | 25 | 0     | Inf. Dilution | lgK:14.20 (4SF)  | 0 | CRV   | 9402   |



|    |    |     |               |                      |   |     |       |
|----|----|-----|---------------|----------------------|---|-----|-------|
| 4  | 25 | 0   | Inf. Dilution | lgK:15.630 (0.12SD)  | 4 | CRV | 27322 |
| 5  | 25 | 0   | Inf. Dilution | lgK:13.36 (4SF)      | 1 | CRV | 32222 |
| 6  | 25 | 0   | Inf. Dilution | lgK:14.50 (4SF)      | 2 | CRV | 32224 |
| 7  | 25 | 0   | CHEMVAL2      | lgK:15.08 (0.01SD)   | 0 | ENS | 5156  |
| 8  | 25 | 0   | CHEMVAL3      | lgK:15.06 (0.01SD)   | 0 | ENS | 5156  |
| 9  | 25 | 0   | CHEMVAL4      | lgK:15.10 (0.01SD)   | 0 | ENS | 5156  |
| 10 | 25 | 0.5 | Unknown       | lgK:13.44 (4SF)      | 5 | CRV | 2556  |
| 11 | 25 | 3   | NaClO4        | lgK:13.80 (4SF)      | 2 | CRV | 2556  |
| 12 | 25 | 4   | HClO4         | lgK:14.72 (4SF)      | 2 | CRV | 2556  |
| 13 | 25 |     |               | lgK:8.7 (2SF)        | 0 | MSL | 140   |
| 14 | 25 | 0   | Inf. Dilution | dH:-2.10 (3SF)       | 3 | CRV | 9402  |
| 15 | 25 | 0   | Inf. Dilution | dH:-3.300 (0.2SD) kJ | 5 | CRV | 27322 |
| 16 | 25 | 0   | Inf. Dilution | dH:-2.25 (3SF) kJ    | 4 | CRV | 32222 |
| 17 | 25 | 0.5 | NaClO4        | dH:4.3 (0.07SD) kJ   | 0 | MCL | 5280  |
| 18 | 25 | 4   | HClO4         | dH:-0.8 (1SF)        | 0 | CRV | 552   |

## Reaction No. 56025



| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:19.8 (3SF)      | 2 | CRV   | 552    |
| 2  | 25 | 0     | Inf. Dilution | lgK:20.15 (4SF)     | 0 | EBU   | 6372   |
| 3  | 25 | 0     | Inf. Dilution | lgK:18.89 (4SF)     | 0 | EOD   | 8120   |
| 4  | 25 | 0     | Inf. Dilution | lgK:18.90 (4SF)     | 0 | CRV   | 9402   |
| 5  | 25 | 0     | Inf. Dilution | lgK:20.670 (0.08SD) | 4 | CRV   | 27322  |
| 6  | 25 | 0     | Inf. Dilution | lgK:17.54 (4SF)     | 0 | CRV   | 32222  |
| 7  | 25 | 0     | Inf. Dilution | lgK:19.27 (4SF)     | 0 | CRV   | 32224  |
| 8  | 25 | 0     | CHEMVAL2      | lgK:19.80 (0.01SD)  | 0 | ENS   | 5156   |
| 9  | 25 | 0     | CHEMVAL3      | lgK:19.81 (0.01SD)  | 0 | ENS   | 5156   |
| 10 | 25 | 0     | CHEMVAL4      | lgK:19.80 (0.01SD)  | 0 | ENS   | 5156   |
| 11 | 25 | 0.5   | Unknown       | lgK:17.9 (3SF)      | 0 | CRV   | 2556   |
| 12 | 25 | 3     | NaClO4        | lgK:18.8 (3SF)      | 5 | CRV   | 2556   |
| 13 | 25 | 0     | Inf. Dilution | dH:-3.00 (3SF)      | 0 | CRV   | 9402   |
| 14 | 25 | 0     | Inf. Dilution | dH:-3.23 (3SF) kJ   | 1 | CRV   | 32222  |
| 15 | 25 | 0.5   | NaClO4        | dH:7.8 (0.9SD) kJ   | 0 | MCL   | 5280   |

## Reaction No. 56026

| 4<F-1> + Th+4 = Th+4_F-1(4) |    |       |               |                   |   |       |        |
|-----------------------------|----|-------|---------------|-------------------|---|-------|--------|
| No                          | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:23.2(3SF)     | 1 | ELC   |        |
| 2                           | 25 | 0     | Inf. Dilution | lgK:23.2(3SF)     | 0 | CRV   | 552    |
| 3                           | 25 | 0     | Inf. Dilution | lgK:24.59(4SF)    | 0 | EBU   | 6372   |
| 4                           | 25 | 0     | Inf. Dilution | lgK:22.33(4SF)    | 0 | EOD   | 8120   |
| 5                           | 25 | 0     | Inf. Dilution | lgK:22.30(4SF)    | 0 | CRV   | 9402   |
| 6                           | 25 | 0     | Inf. Dilution | lgK:25.58(0.09SD) | 0 | CRV   | 27322  |
| 7                           | 25 | 0     | Inf. Dilution | lgK:20.41(4SF)    | 0 | CRV   | 32222  |
| 8                           | 25 | 0     | Inf. Dilution | lgK:22.81(4SF)    | 4 | CRV   | 32224  |
| 9                           | 25 | 0     | CHEMVAL2      | lgK:23.20(0.01SD) | 0 | ENS   | 5156   |
| 10                          | 25 | 0     | CHEMVAL3      | lgK:23.17(0.01SD) | 0 | ENS   | 5156   |
| 11                          | 25 | 0     | CHEMVAL4      | lgK:23.20(0.01SD) | 0 | ENS   | 5156   |
| 12                          | 25 | 0     | Inf. Dilution | dH:-3.80(3SF)     | 2 | CRV   | 9402   |
| 13                          | 25 | 0     | Inf. Dilution | dH:-4.46(3SF) kJ  | 1 | CRV   | 32222  |

| Reaction No. 56141                   |    |       |          |                    |   |       |        |
|--------------------------------------|----|-------|----------|--------------------|---|-------|--------|
| Th+4_SO4-2(2)_ (s) = Th+4 + 2<SO4-2> |    |       |          |                    |   |       |        |
| No                                   | t  | I-Str | Medium   | Value (Dev)        | W | Tech. | Ref. # |
| 1                                    | 25 | 0     | CHEMVAL2 | lgK:-20.35(0.01SD) | 4 | ENS   | 5156   |
| 2                                    | 25 | 0     | CHEMVAL3 | lgK:-20.35(0.01SD) | 4 | ENS   | 5156   |
| 3                                    | 25 | 0     | CHEMVAL4 | lgK:-20.30(0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 56142                                                 |    |       |               |                    |   |       |        |
|--------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Th+4_F-1(4)_ (s) = Th+4 + 4<F-1>                                   |    |       |               |                    |   |       |        |
| Note: Inconsistent with [08RFN_27322] which has the solid hydrated |    |       |               |                    |   |       |        |
| No                                                                 | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                  | 25 | 0     | Inf. Dilution | lgK:-27.4(3SF)     | 0 | ENS   | 5648   |
| 2                                                                  | 25 | 0     | Inf. Dilution | lgK:-30.66(4SF)    | 4 | CRV   | 32224  |
| 3                                                                  | 25 | 0     | CHEMVAL2      | lgK:-30.17(0.01SD) | 0 | ENS   | 5156   |
| 4                                                                  | 25 | 0     | CHEMVAL3      | lgK:-30.17(0.01SD) | 0 | ENS   | 5156   |
| 5                                                                  | 25 | 0     | CHEMVAL4      | lgK:-31.00(0.01SD) | 2 | ENS   | 5156   |
| 6                                                                  | 25 | 3     | NaClO4        | lgK:-28.3(3SF)     | 2 | CRV   | 2556   |

| Reaction No. 56271                         |   |       |        |             |   |       |        |
|--------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Th+4_OH-1(4)_ (s) + 4<H+1> = Th+4 + 4<H2O> |   |       |        |             |   |       |        |
| No                                         | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |   |               |                    |   |     |       |
|---|----|---|---------------|--------------------|---|-----|-------|
| 1 | 25 | 0 | Inf. Dilution | lgK:11.8 (3SF)     | 1 | ELC |       |
| 2 | 25 | 0 | Inf. Dilution | lgK:8.68 (3SF)     | 0 | EBU | 6372  |
| 3 | 25 | 0 | Inf. Dilution | lgK:-9.45 (3SF)    | 0 | CRV | 32224 |
| 4 | 25 | 0 | CHEMVAL2      | lgK:12.60 (0.01SD) | 0 | ENS | 5156  |
| 5 | 25 | 0 | CHEMVAL3      | lgK:12.60 (0.01SD) | 0 | ENS | 5156  |
| 6 | 25 | 0 | CHEMVAL4      | lgK:12.60 (0.01SD) | 0 | ENS | 5156  |

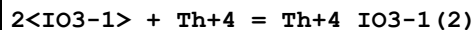
| Reaction No. 56272                                                 |    |       |               |                   |   |       |        |
|--------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| ThO2(s) + 4<H+1> = Th+4 + 2<H2O>                                   |    |       |               |                   |   |       |        |
| No                                                                 | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                  | 25 | 0     | Inf. Dilution | lgK:1.8 (2SF)     | 1 | ELC   |        |
| 2                                                                  | 25 | 0     | Inf. Dilution | lgK:6.3 (1.SD)    | 0 | CRV   | 6478   |
| 3                                                                  | 25 | 0     | Inf. Dilution | lgK:1.80 (1.30SD) | 4 | CRV   | 32222  |
| Note (3): sign changed                                             |    |       |               |                   |   |       |        |
| 4                                                                  | 25 | 0     | Inf. Dilution | lgK:6.27 (3SF)    | 0 | CRV   | 32224  |
| Note (4): seems to be confusion over ThO2(s), not being thorianite |    |       |               |                   |   |       |        |
| 5                                                                  | 25 | 0     | Inf. Dilution | lgK:1.76 (3SF)    | 4 | CRV   | 32224  |
| 6                                                                  | 25 | 0     | CHEMVAL2      | lgK:6.30 (0.01SD) | 0 | ENS   | 5156   |
| 7                                                                  | 25 | 0     | CHEMVAL3      | lgK:6.30 (0.01SD) | 0 | ENS   | 5156   |
| 8                                                                  | 25 | 0     | CHEMVAL4      | lgK:6.30 (0.01SD) | 0 | ENS   | 5156   |
| 9                                                                  | 25 | 0     | Inf. Dilution | dH:-33.4 (1.8SD)  | 0 | CRV   | 6478   |

| Reaction No. 56273                                                  |    |       |          |                     |   |       |        |
|---------------------------------------------------------------------|----|-------|----------|---------------------|---|-------|--------|
| Th+4_H+1(2)_PO4-3(2)_H2O(4)_(s) = Th+4 + 2<H+1> + 2<PO4-3> + 4<H2O> |    |       |          |                     |   |       |        |
| No                                                                  | t  | I-Str | Medium   | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                                   | 25 | 0     | CHEMVAL2 | lgK:-51.56 (0.01SD) | 4 | ENS   | 5156   |
| 2                                                                   | 25 | 0     | CHEMVAL3 | lgK:-51.56 (0.01SD) | 4 | ENS   | 5156   |
| 3                                                                   | 25 | 0     | CHEMVAL4 | lgK:-51.60 (0.01SD) | 4 | ENS   | 5156   |

| Reaction No. 57116        |    |       |               |                   |   |       |        |
|---------------------------|----|-------|---------------|-------------------|---|-------|--------|
| IO3-1 + Th+4 = Th+4_IO3-1 |    |       |               |                   |   |       |        |
| No                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                         | 25 | 0     | Inf. Dilution | lgK:3.07 (3SF)    | 0 | EBU   | 6372   |
| 2                         | 25 | 0     | Inf. Dilution | lgK:4.14 (0.05SD) | 5 | CRV   | 27322  |
| 3                         | 25 | 0.5   | HClO4         | lgK:2.68 (3SF)    | 4 | MDS   | 140    |
| 4                         | 25 | 0.5   | Li+1 salt     | lgK:2.9 (0.1SD)   | 3 | MSL   | 5188   |
| 5                         | 25 | 1     | NaClO4        | lgK:2.5 (0.05SD)  | 5 | MSE   | 5193   |

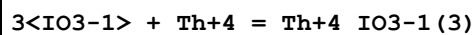
|   |    |     |                    |                 |   |     |      |
|---|----|-----|--------------------|-----------------|---|-----|------|
| 6 | 25 | 1   | NaClO <sub>4</sub> | dG:-3.4(0.05SD) | 5 | MSL | 5193 |
| 7 | 25 | 0.5 | Unknown            | dH:0.8(1SF)     | 4 | CRV | 552  |
| 8 | 25 | 1   | NaClO <sub>4</sub> | dH:1.6(0.1SD)   | 3 | MCL | 5193 |

## Reaction No. 57117



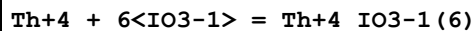
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:5.48(3SF)    | 0 | EBU   | 6372   |
| 2  | 25 | 0     | Inf. Dilution | lgK:6.97(0.06SD) | 5 | CRV   | 27322  |
| 3  | 25 | 0.5   | Li+1 salt     | lgK:4.8(0.1SD)   | 4 | MSL   | 5188   |
| 4  | 25 | 0.5   | Unknown       | dH:1.0(2SF)      | 4 | CRV   | 552    |

## Reaction No. 57118



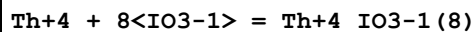
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.50(3SF)     | 0 | EBU   | 6372   |
| 2  | 25 | 0     | Inf. Dilution | lgK:9.87(0.055SD) | 5 | CRV   | 27322  |
| 3  | 25 | 0.5   | Li+1 salt     | lgK:7.2(0.1SD)    | 4 | MSL   | 5188   |

## Reaction No. 57119



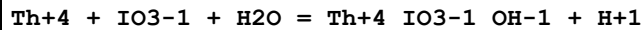
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:11.96(4SF)  | 0 | EBU   | 6372   |
| 2  | 25 | 0.5   | Li+1 salt     | lgK:11.0(0.1SD) | 3 | MSL   | 5188   |

## Reaction No. 57120



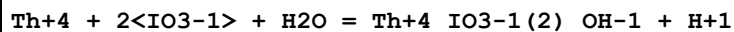
| No | t  | I-Str | Medium    | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|-----------|-----------------|---|-------|--------|
| 1  | 25 | 0.5   | Li+1 salt | lgK:13.0(0.1SD) | 4 | MSL   | 5188   |

## Reaction No. 57121



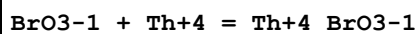
| No | t  | I-Str | Medium    | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|-----------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | Li+1 salt | lgK:0.4(0.1SD) | 4 | MSL   | 5188   |

## Reaction No. 57122



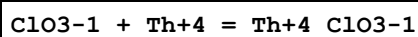
| No | t  | I-Str | Medium    | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|-----------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | Li+1 salt | lgK:2.4(0.1SD) | 4 | MSL   | 5188   |

## Reaction No. 57158



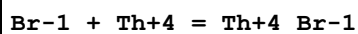
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:1.900(0.05SD) | 5 | CRV   | 27322  |
| 2  | 25 | 0.5   | H(ClO4)       | lgK:0.81(2SF)     | 4 | MDS   | 140    |
| 3  | 25 | 1     | NaClO4        | lgK:0.6(0.06SD)   | 5 | MSE   | 5193   |
| 4  | 25 | 1     | NaClO4        | dG:-0.9(0.08SD)   | 5 | MSE   | 5193   |
| 5  | 25 | 1     | NaClO4        | dH:0.6(0.1SD)     | 4 | MSL   | 5193   |

## Reaction No. 57160



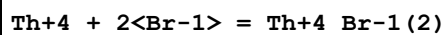
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:1.550(0.07SD) | 5 | CRV   | 27322  |
| 2  | 25 | 0.5   | H(ClO4)       | lgK:0.26(2SF)     | 4 | MDS   | 140    |
| 3  | 25 | 1     | NaClO4        | lgK:0.1(0.08SD)   | 5 | MSE   | 5193   |
| 4  | 25 | 1     | NaClO4        | dG:-0.2(0.1SD)    | 5 | MSE   | 5193   |
| 5  | 25 | 1     | NaClO4        | dH:0.6(0.2SD)     | 4 | MCL   | 5193   |

## Reaction No. 57393



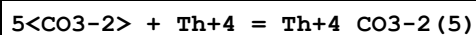
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:0.74(2SF)     | 0 | EBU   | 6372   |
| 2  | 25 | 0     | Inf. Dilution | lgK:1.38(0.07SD)  | 5 | CRV   | 27322  |
| 3  | 25 | 2     | HClO4         | lgK:-0.13(2SF)    | 6 | CRV   | 552    |
| 4  | 25 | 2     | Unknown       | lgK:-0.10(0.02SD) | 6 | MSE   | 5259   |

## Reaction No. 57394



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:1.18(3SF)    | 3 | EBU   | 6372   |
| 2  | 25 | 2     | Unknown       | lgK:-0.6(0.07SD) | 6 | MSE   | 5259   |

## Reaction No. 57902



| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 20 | 1     | Unknown       | lgK:26.2 (0.1SD)    | 0 | MDS   | 2261   |
| 2  | 20 | 2.5   | Unknown       | lgK:26.3 (0.1SD)    | 0 | MDS   | 2261   |
| 3  | 25 | 0     | Inf. Dilution | lgK:32.7 (3SF)      | 1 | ELC   |        |
| 4  | 25 | 0     | Inf. Dilution | lgK:49.24 (4SF)     | 0 | EBU   | 6372   |
| 5  | 25 | 0     | Inf. Dilution | lgK:31.000 (0.35SD) | 4 | CRV   | 27322  |
| 6  | 25 | 0     | Inf. Dilution | lgK:31.0 (3SF)      | 2 | EOD   | 29848  |
| 7  | 25 | 0     | Inf. Dilution | lgK:28.60 (2.00SD)  | 0 | CRV   | 32222  |

| Reaction No. 57903      |       |       |               |                   |   |       |        |
|-------------------------|-------|-------|---------------|-------------------|---|-------|--------|
| Th+4 + OH-1 = Th+4_OH-1 |       |       |               |                   |   |       |        |
| No                      | t     | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                       | 25    | 0     | Inf. Dilution | lgK:11.6 (3SF)    | 1 | ELC   |        |
| 2                       | 25    | 0     | Inf. Dilution | lgK:10.8 (3SF)    | 0 | CRV   | 2556   |
| 3                       | 25    | 0     | Inf. Dilution | lgK:11.2 (3SF)    | 3 | EOD   | 16981  |
| 4                       | 25    | 0     | Inf. Dilution | lgK:12.2 (3SF)    | 2 | EOD   | 29848  |
| 5                       | 25    | 0.1   | Unknown       | lgK:10.0 (0.2SD)  | 0 | CRV   | 2556   |
| 6                       | 25    | 0.5   | Unknown       | lgK:9.6 (0.1SD)   | 0 | CRV   | 2556   |
| 7                       | 25    | 1     | Unknown       | lgK:9.6 (0.1SD)   | 0 | CRV   | 2556   |
| 8                       | 25    | 3     | NaCl          | lgK:9.0 (2SF)     | 0 | CRV   | 2556   |
| 9                       | 25    | 3     | NaClO4        | lgK:-4.1 (0.04SD) | 0 | MGL   | 2261   |
| 10                      | 0:100 | 1     | Unknown       | dH:-8. (1SF)      | 0 | CRV   | 2556   |

| Reaction No. 57904                               |    |       |               |                    |   |       |        |
|--------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| 4<Th+4> + 12<H2O> = Th+4 (4)_OH-1 (12) + 12<H+1> |    |       |               |                    |   |       |        |
| No                                               | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                | 25 | 0     | Inf. Dilution | lgK:-33.62 (4SF)   | 0 | EBU   | 6372   |
| 2                                                | 25 | 0     | Inf. Dilution | lgK:-26.6 (0.1SD)  | 4 | CRV   | 27322  |
| 3                                                | 25 | 0     | Inf. Dilution | lgK:-26.7 (3SF)    | 4 | CRV   | 32222  |
| 4                                                | 25 | 0.1   | KNO3          | lgK:-30.6 (0.03SD) | 4 | MPT   | 5586   |

| Reaction No. 57905                |    |       |               |                   |   |       |        |
|-----------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| ThO2 (s) + 2<H2O> = Th+4_OH-1 (4) |    |       |               |                   |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:-15.6 (3SF)   | 1 | ELC   |        |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:-8.5 (2SF)    | 0 | CRV   | 32222  |
| 3                                 | 25 | 0     | NaClO4        | lgK:-6.7 (0.04SD) | 0 | MSL   | 2261   |

| Reaction No. 57909 |    |       |               |                 |   |       |        |
|--------------------|----|-------|---------------|-----------------|---|-------|--------|
| Th+3 = Th+4 + e-1  |    |       |               |                 |   |       |        |
| No                 | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                  | 25 | 0     | Inf. Dilution | lgK:59.2(0.1SD) | 3 | ENS   | 5458   |
| 2                  | 25 | 0     | Inf. Dilution | lgK:59.17(4SF)  | 4 | CRV   | 32224  |
| 3                  | 25 | 0     | Inf. Dilution | E:3.8(2SF)      | 0 | CRV   | 22551  |

| Reaction No. 57910                             |    |       |               |                  |   |       |        |
|------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th(s) + 2<O2(g)> + 2<H2(g)> = Th+4_OH-1(4)_(s) |    |       |               |                  |   |       |        |
| No                                             | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                              | 25 | 0     | Inf. Dilution | dG:-379.0(0.1SD) | 5 | ENS   | 5585   |
| 2                                              | 25 | 0     | Inf. Dilution | dH:-421.5(0.1SD) | 5 | ENS   | 5585   |

| Reaction No. 57912                        |    |       |               |                 |   |       |        |
|-------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Th+4(3)_PO4-3(4)_(s) = 3<Th+4> + 4<PO4-3> |    |       |               |                 |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | Inf. Dilution | lgK:-100.(1.SD) | 5 | MSL   | 5587   |

| Reaction No. 58392                 |    |       |        |                |   |       |        |
|------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4_NO3-1 + NO3-1 = Th+4_NO3-1(2) |    |       |        |                |   |       |        |
| No                                 | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                  | 20 | 2     | HClO4  | lgK:0.33(2SF)  | 4 | MDS   | 140    |
| 2                                  | 25 | 5.97  | NaClO4 | lgK:-0.30(2SF) | 4 | MDS   | 140    |

| Reaction No. 58393                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_NO3-1(2) + NO3-1 = Th+4_NO3-1(3) |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 20 | 2     | HClO4         | lgK:-0.11(2SF) | 4 | MDS   | 140    |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:3.2(2SF)   | 1 | ELC   |        |

| Reaction No. 58394                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_NO3-1(3) + NO3-1 = Th+4_NO3-1(4) |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 20 | 2     | HClO4         | lgK:-0.26(2SF) | 4 | MDS   | 140    |
| 2                                     | 25 | 0     | Inf. Dilution | lgK:-2.8(2SF)  | 1 | ELC   |        |
| 3                                     | 25 |       |               | lgK:-0.22(2SF) | 0 | MAX   | 140    |

| Reaction No. 58395                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_NO3-1(4) + NO3-1 = Th+4_NO3-1(5) |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:-0.8(1SF)  | 1 | ESC   |        |
| 2                                     | 25 |       |               | lgK:-0.80(2SF) | 0 | MAX   | 140    |

| Reaction No. 58396                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_NO3-1(5) + NO3-1 = Th+4_NO3-1(6) |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:-0.9(1SF)  | 1 | ESC   |        |
| 2                                     | 25 |       |               | lgK:-0.90(2SF) | 0 | MAX   | 140    |

| Reaction No. 58421                    |    |       |               |              |   |       |        |
|---------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Th+4_P2O7-4 + P2O7-4 = Th+4_P2O7-4(2) |    |       |               |              |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:5.3(2SF) | 1 | ESC   |        |
| 2                                     | 25 |       |               | lgK:5.3(2SF) | 3 | MCN   | 140    |

| Reaction No. 58507                      |    |       |               |              |   |       |        |
|-----------------------------------------|----|-------|---------------|--------------|---|-------|--------|
| Th+4 + H+1_Adipic-2 = Th+4_H+1_Adipic-2 |    |       |               |              |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
| 1                                       | 25 | 0     | Inf. Dilution | lgK:5.5(2SF) | 1 | ESC   |        |
| 2                                       | 40 | 0.1   | HClO4         | lgK:5.2(2SF) | 3 | MEP   | 4506   |

| Reaction No. 58514                           |    |       |         |               |   |       |        |
|----------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Th+4_SO4-2 + H+1_SO4-2 = Th+4_SO4-2(2) + H+1 |    |       |         |               |   |       |        |
| No                                           | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                            | 23 | 1.7   | NaClO4  | lgK:1.1(2SF)  | 4 | MDS   | 140    |
| 2                                            | 25 | 2     | NaClO4  | lgK:1.25(3SF) | 5 | MDS   | 140    |
| 3                                            | 25 | 2     | Unknown | lgK:1.34(3SF) | 4 | MCE   | 140    |
| 4                                            | 25 | 2     | HClO4   | dH:-0.89(2SF) | 4 | MCL   | 140    |

| Reaction No. 58515                 |    |       |        |               |   |       |        |
|------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_SO4-2 + SO4-2 = Th+4_SO4-2(2) |    |       |        |               |   |       |        |
| No                                 | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                  | 25 | 2     | NaClO4 | lgK:2.33(3SF) | 4 | MDS   | 140    |
| 2                                  | 25 | 2     | NaClO4 | lgK:2.38(3SF) | 4 | MDS   | 140    |



| Reaction No. 58516                                                             |    |       |        |              |   |       |        |
|--------------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| $\text{Th+4} + 2\text{<H+1\_SO4-2>} = \text{Th+4\_H+1\_SO4-2(2)} + \text{H+1}$ |    |       |        |              |   |       |        |
| No                                                                             | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                              | 25 | 2     | NaClO4 | lgK:2.9(2SF) | 3 | MDS   | 366    |

| Reaction No. 58517                                                                       |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Th+4\_SO4-2(3)} + \text{SO4-2} = \text{Th+4\_SO4-2(4)}$                           |    |       |               |                |   |       |        |
| Note: Consider with caution. See $4\text{<SO4-2>} + \text{Th+4} = \text{Th+4\_SO4-2(4)}$ |    |       |               |                |   |       |        |
| No                                                                                       | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                        | 25 | 0     | Inf. Dilution | lgK:-2.02(3SF) | 2 | MDS   | 140    |

| Reaction No. 58581                                       |    |       |        |                |   |       |        |
|----------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Th+4\_Cl-1} + \text{Cl-1} = \text{Th+4\_Cl-1(2)}$ |    |       |        |                |   |       |        |
| No                                                       | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                        | 25 | 2     | NaClO4 | lgK:-1.08(3SF) | 4 | MDS   | 140    |
| 2                                                        | 25 | 4     | NaClO4 | lgK:-1.03(3SF) | 4 | MDS   | 140    |
| 3                                                        | 25 | 4     | NaClO4 | lgK:-1.08(3SF) | 4 | MDS   | 140    |
| 4                                                        | 25 | 6     | NaClO4 | lgK:-0.58(2SF) | 4 | MDS   | 140    |

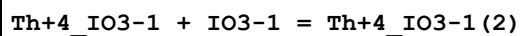
| Reaction No. 58582                                          |    |       |        |                |   |       |        |
|-------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| $\text{Th+4\_Cl-1(2)} + \text{Cl-1} = \text{Th+4\_Cl-1(3)}$ |    |       |        |                |   |       |        |
| No                                                          | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                           | 25 | 2     | NaClO4 | lgK:0.30(2SF)  | 4 | MDS   | 140    |
| 2                                                           | 25 | 4     | NaClO4 | lgK:-0.15(2SF) | 4 | MDS   | 140    |
| 3                                                           | 25 | 4     | NaClO4 | lgK:-0.51(2SF) | 4 | MDS   | 140    |
| 4                                                           | 25 | 6     | NaClO4 | lgK:-0.20(2SF) | 4 | MDS   | 140    |

| Reaction No. 58583                                          |    |       |               |                |   |       |        |
|-------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Th+4\_Cl-1(3)} + \text{Cl-1} = \text{Th+4\_Cl-1(4)}$ |    |       |               |                |   |       |        |
| No                                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                           | 25 | 0     | Inf. Dilution | lgK:-2.47(3SF) | 1 | ELC   |        |
| 2                                                           | 25 | 4     | NaClO4        | lgK:-0.42(2SF) | 4 | MDS   | 140    |
| 3                                                           | 25 | 4     | NaClO4        | lgK:-0.74(2SF) | 4 | MDS   | 140    |

| Reaction No. 58803                                             |  |  |  |  |  |  |  |
|----------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{Th+4\_BrO3-1} + \text{BrO3-1} = \text{Th+4\_BrO3-1(2)}$ |  |  |  |  |  |  |  |

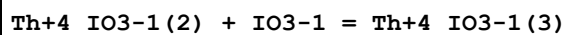
| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | H(ClO4) | lgK:0.10 (2SF) | 4 | MDS   | 140    |

## Reaction No. 58842



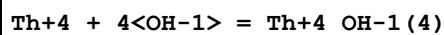
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | HClO4  | lgK:1.91 (3SF) | 4 | MDS   | 140    |

## Reaction No. 58843



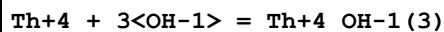
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | HClO4  | lgK:2.36 (3SF) | 4 | MDS   | 140    |

## Reaction No. 59321



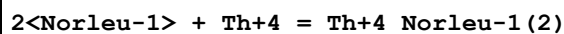
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:38.6 (3SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:39.1 (3SF) | 0 | EOD   | 29848  |

## Reaction No. 59322



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:30.3 (3SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:32.7 (3SF) | 0 | EOD   | 29848  |

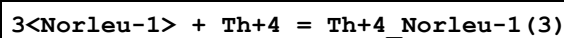
## Reaction No. 60024



Note:

| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO4 | lgK:8.52 (3SF) | 0 | MGL   | 2392   |
| 2  | 30 | 0.1   | NaClO4 | lgK:8.57 (3SF) | 0 | MGL   | 2392   |
| 3  | 40 | 0.1   | NaClO4 | lgK:8.62 (3SF) | 0 | MGL   | 2392   |

## Reaction No. 60025



| No | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|--------|-----------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO4 | lgK:16.72 (4SF) | 0 | MGL   | 2392   |

|   |       |     |                    |                 |   |     |      |
|---|-------|-----|--------------------|-----------------|---|-----|------|
| 2 | 25    | 0   | Inf. Dilution      | lgK:16.9 (3SF)  | 1 | ESC |      |
| 3 | 30    | 0.1 | NaClO <sub>4</sub> | lgK:16.80 (4SF) | 0 | MGL | 2392 |
| 4 | 40    | 0.1 | NaClO <sub>4</sub> | lgK:16.89 (4SF) | 0 | MGL | 2392 |
| 5 | 20:40 | 0.1 | NaClO <sub>4</sub> | dH:-10.4 (3SF)  | 3 | MTD | 2392 |

| Reaction No. 60026                     |    |       |                    |                 |   |       |        |
|----------------------------------------|----|-------|--------------------|-----------------|---|-------|--------|
| 4<Norleu-1> + Th+4 = Th+4_Norleu-1 (4) |    |       |                    |                 |   |       |        |
| No                                     | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
| 1                                      | 20 | 0.1   | NaClO <sub>4</sub> | lgK:21.71 (4SF) | 0 | MGL   | 2392   |
| 2                                      | 25 | 0     | Inf. Dilution      | lgK:22.3 (3SF)  | 1 | ESC   |        |
| 3                                      | 30 | 0.1   | NaClO <sub>4</sub> | lgK:22.29 (4SF) | 0 | MGL   | 2392   |
| 4                                      | 40 | 0.1   | NaClO <sub>4</sub> | lgK:24.88 (4SF) | 0 | MGL   | 2392   |

| Reaction No. 60178                                      |    |       |                    |                 |   |       |        |
|---------------------------------------------------------|----|-------|--------------------|-----------------|---|-------|--------|
| 2<Th+4_EDTA-4> + 2<OH-1> = Th+4 (2)_OH-1 (2)_EDTA-4 (2) |    |       |                    |                 |   |       |        |
| No                                                      | t  | I-Str | Medium             | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                       | 20 | 0.1   | NaClO <sub>4</sub> | lgK:17.97 (4SF) | 5 | MRX   | 903    |
| 2                                                       | 25 | 0     | Inf. Dilution      | lgK:17.9 (3SF)  | 1 | ESC   |        |

| Reaction No. 60433             |        |       |               |                |   |       |        |
|--------------------------------|--------|-------|---------------|----------------|---|-------|--------|
| Th+4 + 2<OH-1> = Th+4_OH-1 (2) |        |       |               |                |   |       |        |
| No                             | t      | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                              | 25     | 0     | Inf. Dilution | lgK:21.4 (3SF) | 1 | ELC   |        |
| 2                              | 25     | 0     | Inf. Dilution | lgK:21.1 (3SF) | 0 | CRV   | 2556   |
| 3                              | 25     | 0     | Inf. Dilution | lgK:21.8 (3SF) | 3 | EOD   | 16981  |
| 4                              | 25     | 0     | Inf. Dilution | lgK:23.0 (3SF) | 0 | EOD   | 29848  |
| 5                              | 25     | 1     | Unknown       | lgK:19.9 (3SF) | 2 | CRV   | 2556   |
| 6                              | 10:100 | 1     | Unknown       | dH:-13. (2SF)  | 2 | CRV   | 2556   |

| Reaction No. 60434                    |       |       |               |                 |   |       |        |
|---------------------------------------|-------|-------|---------------|-----------------|---|-------|--------|
| 2<Th+4> + 2<OH-1> = Th+4 (2)_OH-1 (2) |       |       |               |                 |   |       |        |
| No                                    | t     | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                     | 25    | 0     | Inf. Dilution | lgK:21.9 (3SF)  | 3 | CRV   | 2556   |
| 2                                     | 25    | 1     | Unknown       | lgK:22.97 (4SF) | 5 | CRV   | 2556   |
| 3                                     | 25    | 3     | NaCl          | lgK:23.58 (4SF) | 5 | CRV   | 2556   |
| 4                                     | 0:100 | 1     | Unknown       | dH:-12.0 (3SF)  | 5 | CRV   | 2556   |

| Reaction No. 60435                                               |    |       |               |               |   |       |        |
|------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $2<\text{Th}+4> + 3<\text{OH}-1> = \text{Th}+4(2)\text{OH}-1(3)$ |    |       |               |               |   |       |        |
| No                                                               | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                | 25 | 0     | Inf. Dilution | lgK:35.(2SF)  | 0 | ELC   |        |
| 2                                                                | 25 | 3     | NaCl          | lgK:33.8(3SF) | 3 | CRV   | 2556   |

| Reaction No. 60436                                               |    |       |        |               |   |       |        |
|------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $2<\text{Th}+4> + 5<\text{OH}-1> = \text{Th}+4(2)\text{OH}-1(5)$ |    |       |        |               |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                | 25 | 3     | NaCl   | lgK:53.7(3SF) | 4 | CRV   | 2556   |

| Reaction No. 60437                                         |    |       |        |               |   |       |        |
|------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $3<\text{Th}+4> + \text{OH}-1 = \text{Th}+4(3)\text{OH}-1$ |    |       |        |               |   |       |        |
| No                                                         | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                          | 25 | 3     | NaCl   | lgK:12.7(3SF) | 4 | CRV   | 2556   |

| Reaction No. 60438                                               |    |       |        |               |   |       |        |
|------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| $3<\text{Th}+4> + 3<\text{OH}-1> = \text{Th}+4(3)\text{OH}-1(3)$ |    |       |        |               |   |       |        |
| No                                                               | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                | 25 | 3     | NaCl   | lgK:35.7(3SF) | 4 | CRV   | 2556   |

| Reaction No. 60439                                               |       |       |               |               |   |       |        |
|------------------------------------------------------------------|-------|-------|---------------|---------------|---|-------|--------|
| $4<\text{Th}+4> + 8<\text{OH}-1> = \text{Th}+4(4)\text{OH}-1(8)$ |       |       |               |               |   |       |        |
| No                                                               | t     | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                | 25    | 0     | Inf. Dilution | lgK:90.9(3SF) | 1 | ELC   |        |
| 2                                                                | 25    | 0     | Inf. Dilution | lgK:90.9(3SF) | 0 | CRV   | 2556   |
| 3                                                                | 25    | 1     | Unknown       | lgK:91.2(3SF) | 0 | CRV   | 2556   |
| 4                                                                | 0:100 | 1     | Unknown       | dH:-51.(2SF)  | 2 | CRV   | 2556   |

| Reaction No. 60440                                                 |    |       |               |                |   |       |        |
|--------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $6<\text{Th}+4> + 14<\text{OH}-1> = \text{Th}+4(6)\text{OH}-1(14)$ |    |       |               |                |   |       |        |
| No                                                                 | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                  | 25 | 0     | Inf. Dilution | lgK:157.2(4SF) | 1 | EOR   |        |
| 2                                                                  | 25 | 3     | NaCl          | lgK:162.1(4SF) | 3 | CRV   | 2556   |

| Reaction No. 60441                                                 |  |  |  |  |  |  |  |
|--------------------------------------------------------------------|--|--|--|--|--|--|--|
| $6<\text{Th}+4> + 15<\text{OH}-1> = \text{Th}+4(6)\text{OH}-1(15)$ |  |  |  |  |  |  |  |

| No | t     | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|-------|-------|---------------|----------------|---|-------|--------|
| 1  | 25    | 0     | Inf. Dilution | lgK:173.3(4SF) | 1 | ELC   |        |
| 2  | 25    | 0     | Inf. Dilution | lgK:173.2(4SF) | 0 | CRV   | 2556   |
| 3  | 25    | 1     | Unknown       | lgK:169.8(4SF) | 0 | CRV   | 2556   |
| 4  | 0:100 | 1     | Unknown       | dH:-96.(2SF)   | 2 | CRV   | 2556   |

| Reaction No. 60442                |    |       |               |                |   |       |        |
|-----------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_OH-1(4)_(s) = Th+4 + 4<OH-1> |    |       |               |                |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 22 | 0     | Inf. Dilution | lgK:-44.7(3SF) | 0 | CRV   | 2556   |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:-44.2(3SF) | 1 | ELC   |        |
| 3                                 | 25 | 0     | Inf. Dilution | lgK:-44.7(3SF) | 3 | EOD   | 16981  |
| 4                                 | 25 | 0     | Inf. Dilution | lgK:-48.6(3SF) | 0 | CRV   | 22551  |

| Reaction No. 60443                |    |       |               |                |   |       |        |
|-----------------------------------|----|-------|---------------|----------------|---|-------|--------|
| ThO2(s) + 2<H2O> = Th+4 + 4<OH-1> |    |       |               |                |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 22 | 0     | Inf. Dilution | lgK:-49.7(3SF) | 0 | CRV   | 2556   |
| 2                                 | 25 | 0     | Inf. Dilution | lgK:-54.2(3SF) | 1 | EOR   |        |
| 3                                 | 25 | 1     | Unknown       | lgK:-47.6(3SF) | 3 | ENS   | 773    |

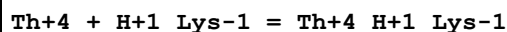
| Reaction No. 60643                      |    |       |               |                  |   |       |        |
|-----------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| H+1(2)_PO4-3 + Th+4 = Th+4_H+1(2)_PO4-3 |    |       |               |                  |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                       | 25 | 0     | Inf. Dilution | lgK:4.54(3SF)    | 0 | EBU   | 6372   |
| 2                                       | 25 | 0     | Inf. Dilution | lgK:5.59(0.32SD) | 5 | CRV   | 25812  |
| 3                                       | 25 | 2     | Unknown       | lgK:3.96(3SF)    | 5 | CRV   | 2556   |

| Reaction No. 60644                            |    |       |               |               |   |       |        |
|-----------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4 + 2<H+1(2)_PO4-3> = Th+4_H+1(4)_PO4-3(2) |    |       |               |               |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:7.50(3SF) | 4 | EBU   | 6372   |
| 2                                             | 25 | 2     | Unknown       | lgK:7.5(2SF)  | 5 | CRV   | 2556   |

| Reaction No. 61496                        |   |       |        |             |   |       |        |
|-------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Th+2 + H+1_Adenine-1 = Th+2_H+1_Adenine-1 |   |       |        |             |   |       |        |
| No                                        | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

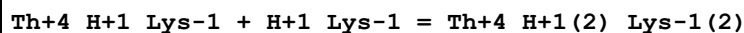
|   |    |     |      |                 |   |     |      |
|---|----|-----|------|-----------------|---|-----|------|
| 1 | 35 | 0.1 | KNO3 | lgK:10.3(0.1SD) | 0 | MPT | 1760 |
|---|----|-----|------|-----------------|---|-----|------|

## Reaction No. 61562



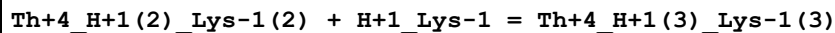
| No | t     | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|-------|-------|---------------|---------------|---|-------|--------|
| 1  | 20    | 0.1   | NaClO4        | lgK:8.50(3SF) | 3 | MGL   | 6000   |
| 2  | 25    | 0     | Inf. Dilution | lgK:8.7(2SF)  | 1 | ESC   |        |
| 3  | 30    | 0.1   | NaClO4        | lgK:8.52(3SF) | 3 | MGL   | 6000   |
| 4  | 40    | 0.1   | NaClO4        | lgK:8.60(3SF) | 3 | MGL   | 6000   |
| 5  | 20:40 |       |               | dH:1.4(2SF)   | 3 | MTD   | 6000   |

## Reaction No. 61563



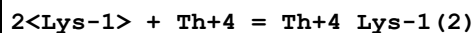
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO4        | lgK:8.20(3SF) | 3 | MGL   | 6000   |
| 2  | 25 | 0     | Inf. Dilution | lgK:8.4(2SF)  | 1 | ESC   |        |
| 3  | 30 | 0.1   | NaClO4        | lgK:8.25(3SF) | 3 | MGL   | 6000   |
| 4  | 40 | 0.1   | NaClO4        | lgK:8.29(3SF) | 3 | MGL   | 6000   |

## Reaction No. 61564



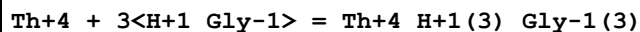
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 0.1   | NaClO4        | lgK:5.00(3SF) | 3 | MGL   | 6000   |
| 2  | 25 | 0     | Inf. Dilution | lgK:5.5(2SF)  | 1 | ESC   |        |
| 3  | 30 | 0.1   | NaClO4        | lgK:5.69(3SF) | 3 | MGL   | 6000   |
| 4  | 40 | 0.1   | NaClO4        | lgK:5.99(3SF) | 3 | MGL   | 6000   |

## Reaction No. 61588



| No                      | t     | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|-------------------------|-------|-------|---------------|---------------|---|-------|--------|
| 1                       | 25    | 0     | Inf. Dilution | lgK:20.0(3SF) | 1 | EES   |        |
| Note (1): Est from Fe+3 |       |       |               |               |   |       |        |
| 2                       | 20:40 |       |               | dH:1.4(2SF)   | 0 | MGL   | 6000   |

## Reaction No. 61597



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|   |    |   |         |                |   |     |     |
|---|----|---|---------|----------------|---|-----|-----|
| 1 | 25 | 1 | Unknown | lgK:5.54 (3SF) | 6 | CRV | 552 |
| 2 | 25 | 1 | Unknown | dH:2.6 (2SF)   | 6 | CRV | 552 |

| Reaction No. 62206                  |    |       |         |                |   |       |        |
|-------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Th+4 + H+1_EDDA-2 = Th+4_H+1_EDDA-2 |    |       |         |                |   |       |        |
| No                                  | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 1     | Unknown | lgK:6.54 (3SF) | 6 | CRV   | 552    |
| 2                                   | 25 | 1     | Unknown | dH:-1.6 (2SF)  | 6 | CRV   | 552    |

| Reaction No. 62207                          |    |       |         |                |   |       |        |
|---------------------------------------------|----|-------|---------|----------------|---|-------|--------|
| Th+4 + H+1 (2)_EDDA-2 = Th+4_H+1 (2)_EDDA-2 |    |       |         |                |   |       |        |
| No                                          | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 1     | Unknown | lgK:2.37 (3SF) | 6 | CRV   | 552    |
| 2                                           | 25 | 1     | Unknown | dH:2.7 (2SF)   | 6 | CRV   | 552    |

| Reaction No. 62376                          |    |       |               |                |   |       |        |
|---------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Bis2OHEtEDDA-2 + Th+4 = Th+4_Bis2OHEtEDDA-2 |    |       |               |                |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:13.0 (3SF) | 1 | ESC   |        |
| 2                                           | 30 | 0.1   | Unknown       | lgK:12.8 (3SF) | 6 | CRV   | 552    |

| Reaction No. 62377                                                   |    |       |               |               |   |       |        |
|----------------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_OH-1 (2)_Bis2OHEtEDDA-2 + 2<H+1> = Th+4_Bis2OHEtEDDA-2 + 2<H2O> |    |       |               |               |   |       |        |
| No                                                                   | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0     | Inf. Dilution | lgK:8.0 (2SF) | 1 | ESC   |        |
| 2                                                                    | 30 | 0.1   | Unknown       | lgK:7.8 (2SF) | 6 | CRV   | 552    |

| Reaction No. 62506                |    |       |         |                |   |       |        |
|-----------------------------------|----|-------|---------|----------------|---|-------|--------|
| Th+4_IDA-2 + H+1 = Th+4_H+1_IDA-2 |    |       |         |                |   |       |        |
| No                                | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 25 | 1     | Unknown | lgK:2.49 (3SF) | 6 | CRV   | 552    |
| 2                                 | 25 | 1     | Unknown | dH:-8.3 (2SF)  | 6 | CRV   | 552    |

| Reaction No. 65191                     |    |       |               |                  |   |       |        |
|----------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th+4 + 5<H2O> = Th+4_OH-1 (5) + 5<H+1> |    |       |               |                  |   |       |        |
| No                                     | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                      | 25 | 0     | Inf. Dilution | lgK:-17.81 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65192                                                   |    |       |               |                 |   |       |        |
|----------------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| $\text{Th+4} + 6\text{<H2O>} = \text{Th+4\_OH-1(6)} + 6\text{<H+1>}$ |    |       |               |                 |   |       |        |
| No                                                                   | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0     | Inf. Dilution | lgK:-22.42(4SF) | 0 | EBU   | 6372   |
| 2                                                                    | 25 | 0     | Inf. Dilution | lgK:-49.62(4SF) | 2 | CRV   | 32224  |

| Reaction No. 65193                                  |    |       |               |                |   |       |        |
|-----------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $5\text{<F-1>} + \text{Th+4} = \text{Th+4\_F-1(5)}$ |    |       |               |                |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:27.97(4SF) | 0 | EBU   | 6372   |
| 2                                                   | 25 | 0     | Inf. Dilution | lgK:24.76(4SF) | 3 | EOD   | 8120   |

| Reaction No. 65194                                  |    |       |               |                |   |       |        |
|-----------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $6\text{<F-1>} + \text{Th+4} = \text{Th+4\_F-1(6)}$ |    |       |               |                |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:30.30(4SF) | 0 | EBU   | 6372   |
| 2                                                   | 25 | 0     | Inf. Dilution | lgK:25.56(4SF) | 4 | EOD   | 8120   |

| Reaction No. 65195                                    |    |       |               |               |   |       |        |
|-------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $6\text{<Cl-1>} + \text{Th+4} = \text{Th+4\_Cl-1(6)}$ |    |       |               |               |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                     | 25 | 0     | Inf. Dilution | lgK:4.33(3SF) | 1 | EBU   | 6372   |

| Reaction No. 65196                                    |    |       |               |               |   |       |        |
|-------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $3\text{<Br-1>} + \text{Th+4} = \text{Th+4\_Br-1(3)}$ |    |       |               |               |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                     | 25 | 0     | Inf. Dilution | lgK:1.44(3SF) | 1 | EBU   | 6372   |

| Reaction No. 65197                                    |    |       |               |               |   |       |        |
|-------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $4\text{<Br-1>} + \text{Th+4} = \text{Th+4\_Br-1(4)}$ |    |       |               |               |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                     | 25 | 0     | Inf. Dilution | lgK:1.58(3SF) | 1 | EBU   | 6372   |

| Reaction No. 65198                                    |    |       |               |               |   |       |        |
|-------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $5\text{<Br-1>} + \text{Th+4} = \text{Th+4\_Br-1(5)}$ |    |       |               |               |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                     | 25 | 0     | Inf. Dilution | lgK:1.62(3SF) | 1 | EBU   | 6372   |



| Reaction No. 65199                                    |    |       |               |               |   |       |        |
|-------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $6\text{<Br-1>} + \text{Th+4} = \text{Th+4\_Br-1(6)}$ |    |       |               |               |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                     | 25 | 0     | Inf. Dilution | lgK:1.58(3SF) | 1 | EBU   | 6372   |

| Reaction No. 65200                            |    |       |               |               |   |       |        |
|-----------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $\text{I-1} + \text{Th+4} = \text{Th+4\_I-1}$ |    |       |               |               |   |       |        |
| No                                            | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                             | 25 | 0     | Inf. Dilution | lgK:0.46(2SF) | 1 | EBU   | 6372   |

| Reaction No. 65201                                  |    |       |               |               |   |       |        |
|-----------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $2\text{<I-1>} + \text{Th+4} = \text{Th+4\_I-1(2)}$ |    |       |               |               |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:0.61(2SF) | 1 | EBU   | 6372   |

| Reaction No. 65202                                  |    |       |               |               |   |       |        |
|-----------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $3\text{<I-1>} + \text{Th+4} = \text{Th+4\_I-1(3)}$ |    |       |               |               |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:0.59(2SF) | 1 | EBU   | 6372   |

| Reaction No. 65203                                  |    |       |               |               |   |       |        |
|-----------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $4\text{<I-1>} + \text{Th+4} = \text{Th+4\_I-1(4)}$ |    |       |               |               |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:0.44(2SF) | 1 | EBU   | 6372   |

| Reaction No. 65204                                  |    |       |               |               |   |       |        |
|-----------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| $5\text{<I-1>} + \text{Th+4} = \text{Th+4\_I-1(5)}$ |    |       |               |               |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:0.20(2SF) | 1 | EBU   | 6372   |

| Reaction No. 65205                                  |    |       |               |                |   |       |        |
|-----------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $6\text{<I-1>} + \text{Th+4} = \text{Th+4\_I-1(6)}$ |    |       |               |                |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:-0.13(2SF) | 1 | EBU   | 6372   |

| Reaction No. 65206                                      |   |       |        |             |   |       |        |
|---------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| $4\text{<IO3-1>} + \text{Th+4} = \text{Th+4\_IO3-1(4)}$ |   |       |        |             |   |       |        |
| No                                                      | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |   |               |                |   |     |      |
|---|----|---|---------------|----------------|---|-----|------|
| 1 | 25 | 0 | Inf. Dilution | lgK:9.22 (3SF) | 1 | EBU | 6372 |
|---|----|---|---------------|----------------|---|-----|------|

|                                  |    |       |               |                 |   |       |        |
|----------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Reaction No. 65207               |    |       |               |                 |   |       |        |
| 5<IO3-1> + Th+4 = Th+4_IO3-1 (5) |    |       |               |                 |   |       |        |
| No                               | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                | 25 | 0     | Inf. Dilution | lgK:10.69 (4SF) | 1 | EBU   | 6372   |

|                                  |    |       |               |                 |   |       |        |
|----------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Reaction No. 65208               |    |       |               |                 |   |       |        |
| 5<SO4-2> + Th+4 = Th+4_SO4-2 (5) |    |       |               |                 |   |       |        |
| No                               | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                | 25 | 0     | Inf. Dilution | lgK:12.05 (4SF) | 1 | EBU   | 6372   |

|                                  |    |       |               |                 |   |       |        |
|----------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Reaction No. 65209               |    |       |               |                 |   |       |        |
| 6<SO4-2> + Th+4 = Th+4_SO4-2 (6) |    |       |               |                 |   |       |        |
| No                               | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                | 25 | 0     | Inf. Dilution | lgK:11.43 (4SF) | 1 | EBU   | 6372   |

|                       |    |       |               |                |   |       |        |
|-----------------------|----|-------|---------------|----------------|---|-------|--------|
| Reaction No. 65210    |    |       |               |                |   |       |        |
| NH3 + Th+4 = Th+4_NH3 |    |       |               |                |   |       |        |
| No                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                     | 25 | 0     | Inf. Dilution | lgK:7.07 (3SF) | 1 | EBU   | 6372   |
| 2                     | 25 | 5     | Unknown       | lgK:0.4 (1SF)  | 0 | EES   | 1548   |

|                              |    |       |               |                 |   |       |        |
|------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Reaction No. 65211           |    |       |               |                 |   |       |        |
| 2<NH3> + Th+4 = Th+4_NH3 (2) |    |       |               |                 |   |       |        |
| No                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                            | 25 | 0     | Inf. Dilution | lgK:13.69 (4SF) | 1 | EBU   | 6372   |

|                              |    |       |               |                 |   |       |        |
|------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Reaction No. 65212           |    |       |               |                 |   |       |        |
| 3<NH3> + Th+4 = Th+4_NH3 (3) |    |       |               |                 |   |       |        |
| No                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                            | 25 | 0     | Inf. Dilution | lgK:19.98 (4SF) | 1 | EBU   | 6372   |

|                              |    |       |               |                 |   |       |        |
|------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Reaction No. 65213           |    |       |               |                 |   |       |        |
| 4<NH3> + Th+4 = Th+4_NH3 (4) |    |       |               |                 |   |       |        |
| No                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                            | 25 | 0     | Inf. Dilution | lgK:26.00 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65214          |    |       |               |                |   |       |        |
|-----------------------------|----|-------|---------------|----------------|---|-------|--------|
| 5<NH3> + Th+4 = Th+4_NH3(5) |    |       |               |                |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:31.78(4SF) | 1 | EBU   | 6372   |

| Reaction No. 65215          |    |       |               |                |   |       |        |
|-----------------------------|----|-------|---------------|----------------|---|-------|--------|
| 6<NH3> + Th+4 = Th+4_NH3(6) |    |       |               |                |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:37.32(4SF) | 1 | EBU   | 6372   |

| Reaction No. 65216              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 5<NO3-1> + Th+4 = Th+4_NO3-1(5) |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:3.39(3SF) | 1 | EBU   | 6372   |

| Reaction No. 65217              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 6<NO3-1> + Th+4 = Th+4_NO3-1(6) |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:2.97(3SF) | 1 | EBU   | 6372   |

| Reaction No. 65218                         |    |       |               |                |   |       |        |
|--------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4 + 4<H+1_PO4-3> = Th+4_H+1(4)_PO4-3(4) |    |       |               |                |   |       |        |
| No                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 25 | 0     | Inf. Dilution | lgK:40.72(4SF) | 1 | EBU   | 6372   |

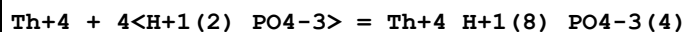
| Reaction No. 65219                         |    |       |               |                |   |       |        |
|--------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4 + 5<H+1_PO4-3> = Th+4_H+1(5)_PO4-3(5) |    |       |               |                |   |       |        |
| No                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 25 | 0     | Inf. Dilution | lgK:47.70(4SF) | 1 | EBU   | 6372   |

| Reaction No. 65220                         |    |       |               |                |   |       |        |
|--------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4 + 6<H+1_PO4-3> = Th+4_H+1(6)_PO4-3(6) |    |       |               |                |   |       |        |
| No                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                          | 25 | 0     | Inf. Dilution | lgK:53.51(4SF) | 1 | EBU   | 6372   |

| Reaction No. 65221                            |   |       |        |             |   |       |        |
|-----------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Th+4 + 3<H+1(2)_PO4-3> = Th+4_H+1(6)_PO4-3(3) |   |       |        |             |   |       |        |
| No                                            | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

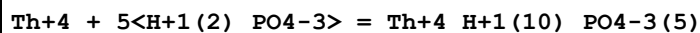
|   |    |   |               |                |   |     |      |
|---|----|---|---------------|----------------|---|-----|------|
| 1 | 25 | 0 | Inf. Dilution | lgK:9.10 (3SF) | 1 | EBU | 6372 |
|---|----|---|---------------|----------------|---|-----|------|

## Reaction No. 65222



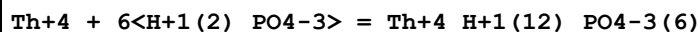
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:9.47 (3SF) | 1 | EBU   | 6372   |

## Reaction No. 65223



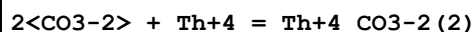
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:8.65 (3SF) | 1 | EBU   | 6372   |

## Reaction No. 65224



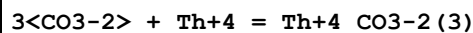
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.69 (3SF) | 1 | EBU   | 6372   |

## Reaction No. 65225



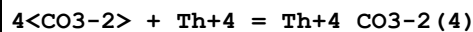
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:15. (2SF)   | 1 | EES   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:21.71 (4SF) | 0 | EBU   | 6372   |

## Reaction No. 65226



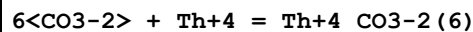
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:18. (2SF)   | 1 | EES   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:31.43 (4SF) | 0 | EBU   | 6372   |

## Reaction No. 65227



| No | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:40.59 (4SF)     | 0 | EBU   | 6372   |
| 2  | 25 | 0     | Inf. Dilution | lgK:31.60 (0.200SD) | 4 | CRV   | 32222  |

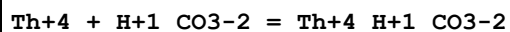
## Reaction No. 65228



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

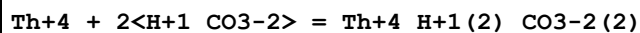
|   |    |   |               |                 |   |     |      |
|---|----|---|---------------|-----------------|---|-----|------|
| 1 | 25 | 0 | Inf. Dilution | lgK:57.41 (4SF) | 1 | EBU | 6372 |
|---|----|---|---------------|-----------------|---|-----|------|

## Reaction No. 65229



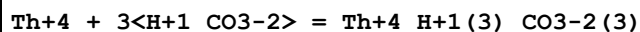
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.56 (3SF) | 1 | EBU   | 6372   |

## Reaction No. 65230



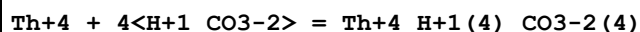
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:12.02 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65231



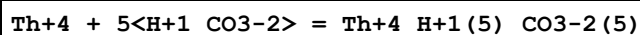
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:16.63 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65232



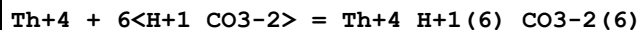
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:20.50 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65233



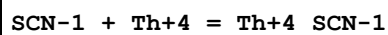
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:23.69 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65234



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:26.22 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65235



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.28 (3SF)     | 1 | EBU   | 6372   |
| 2  | 25 | 0     | Inf. Dilution | lgK:2.000 (0.25SD) | 5 | CRV   | 27322  |
| 3  | 25 | 1     | Unknown       | lgK:1.08 (3SF)     | 4 | CRV   | 552    |

|                   |    |   |               |               |   |     |       |
|-------------------|----|---|---------------|---------------|---|-----|-------|
| 4                 | 25 | 0 | Inf. Dilution | dG:-2.3(2SF)  | 3 | EOD | 32225 |
| Note (4): rxn 307 |    |   |               |               |   |     |       |
| 5                 | 25 | 0 | Inf. Dilution | dG:-2.26(3SF) | 3 | EOD | 32225 |
| Note (5): rxn 308 |    |   |               |               |   |     |       |

| Reaction No. 65236              |    |       |               |                  |   |       |        |
|---------------------------------|----|-------|---------------|------------------|---|-------|--------|
| 2<SCN-1> + Th+4 = Th+4_SCN-1(2) |    |       |               |                  |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:4.14(3SF)    | 0 | EBU   | 6372   |
| 2                               | 25 | 0     | Inf. Dilution | lgK:3.400(0.4SD) | 5 | CRV   | 27322  |

| Reaction No. 65237              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 3<SCN-1> + Th+4 = Th+4_SCN-1(3) |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:5.72(3SF) | 1 | EBU   | 6372   |
| 2                               | 25 | 1     | Unknown       | lgK:1.78(3SF) | 0 | CRV   | 552    |

| Reaction No. 65238              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 4<SCN-1> + Th+4 = Th+4_SCN-1(4) |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:7.06(3SF) | 1 | EBU   | 6372   |

| Reaction No. 65239              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 5<SCN-1> + Th+4 = Th+4_SCN-1(5) |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:8.20(3SF) | 1 | EBU   | 6372   |

| Reaction No. 65240              |    |       |               |               |   |       |        |
|---------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 6<SCN-1> + Th+4 = Th+4_SCN-1(6) |    |       |               |               |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:9.15(3SF) | 1 | EBU   | 6372   |

| Reaction No. 65241                    |    |       |               |                |   |       |        |
|---------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 6<Acetic-1> + Th+4 = Th+4_Acetic-1(6) |    |       |               |                |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:21.36(4SF) | 1 | EBU   | 6372   |

| Reaction No. 65242 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Benzoic-1 + Th+4 = Th+4_Benzoic-1 |    |       |               |                |   |       |        |
|-----------------------------------|----|-------|---------------|----------------|---|-------|--------|
| No                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:6.20 (3SF) | 1 | EBU   | 6372   |

| Reaction No. 65243                       |    |       |               |                 |   |       |        |
|------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<Benzoic-1> + Th+4 = Th+4_Benzoic-1 (2) |    |       |               |                 |   |       |        |
| No                                       | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                        | 25 | 0     | Inf. Dilution | lgK:11.51 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65244                       |    |       |               |                 |   |       |        |
|------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 3<Benzoic-1> + Th+4 = Th+4_Benzoic-1 (3) |    |       |               |                 |   |       |        |
| No                                       | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                        | 25 | 0     | Inf. Dilution | lgK:16.20 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65245                       |    |       |               |                 |   |       |        |
|------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 4<Benzoic-1> + Th+4 = Th+4_Benzoic-1 (4) |    |       |               |                 |   |       |        |
| No                                       | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                        | 25 | 0     | Inf. Dilution | lgK:20.36 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65246                       |    |       |               |                 |   |       |        |
|------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 5<Benzoic-1> + Th+4 = Th+4_Benzoic-1 (5) |    |       |               |                 |   |       |        |
| No                                       | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                        | 25 | 0     | Inf. Dilution | lgK:24.05 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65247                       |    |       |               |                 |   |       |        |
|------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 6<Benzoic-1> + Th+4 = Th+4_Benzoic-1 (6) |    |       |               |                 |   |       |        |
| No                                       | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                        | 25 | 0     | Inf. Dilution | lgK:27.30 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65248               |    |       |               |                 |   |       |        |
|----------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<Cat-2> + Th+4 = Th+4_Cat-2 (2) |    |       |               |                 |   |       |        |
| No                               | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                | 25 | 0     | Inf. Dilution | lgK:47.25 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65249               |    |       |               |                 |   |       |        |
|----------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 3<Cat-2> + Th+4 = Th+4_Cat-2 (3) |    |       |               |                 |   |       |        |
| No                               | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                | 25 | 0     | Inf. Dilution | lgK:69.72 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65250          |    |       |               |                 |   |       |        |
|-----------------------------|----|-------|---------------|-----------------|---|-------|--------|
| EtDiAm + Th+4 = Th+4_EtDiAm |    |       |               |                 |   |       |        |
| No                          | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                           | 25 | 0     | Inf. Dilution | lgK:16.58 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65251                |    |       |               |                 |   |       |        |
|-----------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<EtDiAm> + Th+4 = Th+4_EtDiAm(2) |    |       |               |                 |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:32.02 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65252                |    |       |               |                 |   |       |        |
|-----------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 3<EtDiAm> + Th+4 = Th+4_EtDiAm(3) |    |       |               |                 |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:47.02 (4SF) | 1 | EBU   | 6372   |

| Reaction No. 65253                                           |    |       |               |                 |   |       |        |
|--------------------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 2<Gly-1> + Th+4 = Th+4_Gly-1(2)                              |    |       |               |                 |   |       |        |
| No                                                           | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                            | 25 | 0     | Inf. Dilution | lgK:14.2 (3SF)  | 1 | ELC   |        |
| 2                                                            | 25 | 0     | Inf. Dilution | lgK:18.77 (4SF) | 0 | EBU   | 6372   |
| 3                                                            | 25 | 1     | NaClO4        | lgK:4.21 (3SF)  | 0 | MGL   | 3841   |
| 4                                                            | 35 | 0.01  | NaClO4        | lgK:11.64 (4SF) | 0 | MEP   | 11516  |
| Note (4): Yadava H et al., Bull. Soc. Chim. Fr., 1984, , 314 |    |       |               |                 |   |       |        |

| Reaction No. 65254              |    |       |               |                 |   |       |        |
|---------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| 3<Gly-1> + Th+4 = Th+4_Gly-1(3) |    |       |               |                 |   |       |        |
| No                              | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 25 | 0     | Inf. Dilution | lgK:27.21 (4SF) | 1 | EBU   | 6372   |

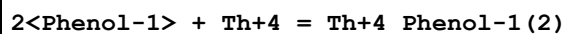
| Reaction No. 65255                        |    |       |               |                   |   |       |        |
|-------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| 5<Glycolic-1> + Th+4 = Th+4_Glycolic-1(5) |    |       |               |                   |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | Inf. Dilution | lgK:15.87 (4SF)   | 1 | EBU   | 6372   |
| 2                                         | 25 | 1     | NaClO4        | dH:-8.00 (3SF) kJ | 5 | MCL   | 7703   |

| Reaction No. 65256                        |  |  |  |  |  |  |  |
|-------------------------------------------|--|--|--|--|--|--|--|
| 6<Glycolic-1> + Th+4 = Th+4_Glycolic-1(6) |  |  |  |  |  |  |  |



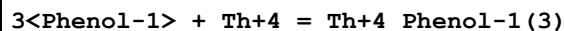
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:16.86 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65257



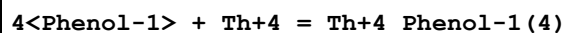
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:16.95 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65258



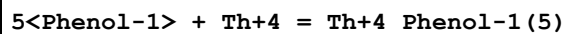
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:24.57 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65259



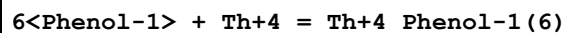
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:31.70 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65260



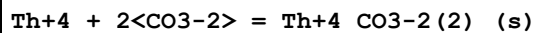
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:38.37 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65261



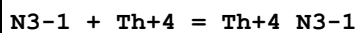
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:44.61 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65509



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:13.59 (4SF) | 1 | EBU   | 6372   |

## Reaction No. 65781



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:4.440 (0.32SD) | 5 | CRV   | 27322  |
| 2  | 25 | 3     | Unknown       | lgK:3.1 (2SF)      | 6 | CRV   | 552    |

| Reaction No. 65782                                    |    |       |               |                   |   |       |        |
|-------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $2\text{<N3-1>} + \text{Th+4} = \text{Th+4\_N3-1(2)}$ |    |       |               |                   |   |       |        |
| No                                                    | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                     | 25 | 0     | Inf. Dilution | lgK:8.590(0.32SD) | 5 | CRV   | 27322  |
| 2                                                     | 25 | 3     | Unknown       | lgK:6.37(3SF)     | 6 | CRV   | 552    |

| Reaction No. 66010                                           |    |       |         |                 |   |       |        |
|--------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| $\text{Th+4\_IO3-1(4)\_(s)} = \text{Th+4} + 4\text{<IO3-1>}$ |    |       |         |                 |   |       |        |
| No                                                           | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                            | 25 | 0.5   | Unknown | lgK:-14.62(4SF) | 6 | CRV   | 552    |

| Reaction No. 66179                                                                                                                                         |    |       |         |                 |   |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------|---------|-----------------|---|-------|--------|
| $2\text{<Th+4\_CDTA-4>} + 2\text{<H2O>} = \text{Th+4(2)\_OH-1(2)\_CDTA-4(2)} + 2\text{<H+1>}$                                                              |    |       |         |                 |   |       |        |
| Note: All secondary values have had their signs corrected; the original paper gives log K = +10.84 for $[\text{ThA}]^2/[\text{Th2A2OH2}]\cdot[\text{H}]^2$ |    |       |         |                 |   |       |        |
| No                                                                                                                                                         | t  | I-Str | Medium  | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                                                                                                                          | 25 | 0     | KNO3    | lgK:-10.84(4SF) | 4 | MGL   | 8247   |
| 2                                                                                                                                                          | 25 | 0.1   | KNO3    | lgK:-10.84(4SF) | 1 | MGL   | 999    |
| 3                                                                                                                                                          | 25 | 0.1   | KNO3    | lgK:-10.84(4SF) | 1 | EOD   | 22560  |
| Note (3): Bogucki & Martell, JACS, 1958, 80, 4170                                                                                                          |    |       |         |                 |   |       |        |
| 4                                                                                                                                                          | 25 | 0.1   | Unknown | lgK:-10.84(4SF) | 1 | CRV   | 552    |
| Note (4): Reproduced in [98SMM_7271]                                                                                                                       |    |       |         |                 |   |       |        |

| Reaction No. 67238                                                      |     |       |               |                |   |       |        |
|-------------------------------------------------------------------------|-----|-------|---------------|----------------|---|-------|--------|
| $\text{N2(g)} + 0.5\text{<O2(g)>} + 2\text{<Th(s)>} = \text{Th2N2O(s)}$ |     |       |               |                |   |       |        |
| No                                                                      | t   | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                       | 25  | 0     | Inf. Dilution | lgK:221.5(4SF) | 4 | ENS   | 6422   |
| 2                                                                       | 27  | 0     | Inf. Dilution | lgK:220.1(4SF) | 4 | ENS   | 6422   |
| 3                                                                       | 127 | 0     | Inf. Dilution | lgK:161.5(4SF) | 4 | ENS   | 6422   |
| 4                                                                       | 227 | 0     | Inf. Dilution | lgK:126.3(4SF) | 4 | ENS   | 6422   |
| 5                                                                       | 327 | 0     | Inf. Dilution | lgK:102.9(4SF) | 4 | ENS   | 6422   |
| 6                                                                       | 25  | 0     | Inf. Dilution | dG:-302.2(4SF) | 4 | EFE   | 6422   |
| 7                                                                       | 25  | 0     | Inf. Dilution | dH:-321.9(4SF) | 4 | MTD   | 6422   |
| 8                                                                       | 127 | 0     | Inf. Dilution | dH:-321.7(4SF) | 4 | MTD   | 6422   |
| 9                                                                       | 227 | 0     | Inf. Dilution | dH:-321.4(4SF) | 4 | MTD   | 6422   |
| 10                                                                      | 327 | 0     | Inf. Dilution | dH:-321.0(4SF) | 4 | MTD   | 6422   |

| Reaction No. 67239    |     |       |               |                 |   |       |        |
|-----------------------|-----|-------|---------------|-----------------|---|-------|--------|
| P(s) + Th(s) = ThP(s) |     |       |               |                 |   |       |        |
| No                    | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                     | 25  | 0     | Inf. Dilution | lgK:59.77 (4SF) | 4 | ENS   | 6422   |
| 2                     | 27  | 0     | Inf. Dilution | lgK:59.39 (4SF) | 4 | ENS   | 6422   |
| 3                     | 127 | 0     | Inf. Dilution | lgK:44.20 (4SF) | 4 | ENS   | 6422   |
| 4                     | 227 | 0     | Inf. Dilution | lgK:35.08 (4SF) | 4 | ENS   | 6422   |
| 5                     | 327 | 0     | Inf. Dilution | lgK:28.99 (4SF) | 4 | ENS   | 6422   |
| 6                     | 25  | 0     | Inf. Dilution | dG:-81.53 (4SF) | 4 | EFE   | 6422   |
| 7                     | 25  | 0     | Inf. Dilution | dH:-83.20 (4SF) | 4 | MTD   | 6422   |
| 8                     | 127 | 0     | Inf. Dilution | dH:-83.48 (4SF) | 4 | MTD   | 6422   |
| 9                     | 227 | 0     | Inf. Dilution | dH:-83.56 (4SF) | 4 | MTD   | 6422   |
| 10                    | 327 | 0     | Inf. Dilution | dH:-83.62 (4SF) | 4 | MTD   | 6422   |

| Reaction No. 67250            |     |       |               |                 |   |       |        |
|-------------------------------|-----|-------|---------------|-----------------|---|-------|--------|
| 3<Th(s)> + 4<P(s)> = Th3P4(s) |     |       |               |                 |   |       |        |
| No                            | t   | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                             | 25  | 0     | Inf. Dilution | lgK:194.7 (4SF) | 4 | ENS   | 6422   |
| 2                             | 27  | 0     | Inf. Dilution | lgK:193.5 (4SF) | 4 | ENS   | 6422   |
| 3                             | 127 | 0     | Inf. Dilution | lgK:143.7 (4SF) | 4 | ENS   | 6422   |
| 4                             | 227 | 0     | Inf. Dilution | lgK:113.7 (4SF) | 4 | ENS   | 6422   |
| 5                             | 327 | 0     | Inf. Dilution | lgK:93.73 (4SF) | 4 | ENS   | 6422   |
| 6                             | 25  | 0     | Inf. Dilution | dG:-265.6 (4SF) | 4 | EFE   | 6422   |
| 7                             | 25  | 0     | Inf. Dilution | dH:-272.9 (4SF) | 4 | MTD   | 6422   |
| 8                             | 127 | 0     | Inf. Dilution | dH:-273.9 (4SF) | 4 | MTD   | 6422   |
| 9                             | 227 | 0     | Inf. Dilution | dH:-274.2 (4SF) | 4 | MTD   | 6422   |
| 10                            | 327 | 0     | Inf. Dilution | dH:-274.4 (4SF) | 4 | MTD   | 6422   |

| Reaction No. 67765                        |    |       |        |                |   |       |        |
|-------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4 + OH-1_HOEDTA-3 = Th+4_OH-1_HOEDTA-3 |    |       |        |                |   |       |        |
| No                                        | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                         | 25 | 0.1   | KNO3   | lgK:15.4 (3SF) | 0 | MGL   | 6508   |

| Reaction No. 67766                                           |    |       |        |               |   |       |        |
|--------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_OH-1(3)_HOEDTA-3 + 2<H+1> = Th+4_OH-1_HOEDTA-3 + 2<H2O> |    |       |        |               |   |       |        |
| No                                                           | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                            | 25 | 0.1   | KNO3   | lgK:8.2 (2SF) | 3 | MGL   | 6508   |

| Reaction No. 67767                      |    |       |        |               |   |       |        |
|-----------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4 + OH-1_HIMDA-2 = Th+4_OH-1_HIMDA-2 |    |       |        |               |   |       |        |
| No                                      | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                       | 25 | 0.1   | KNO3   | lgK:10.7(3SF) | 0 | MGL   | 6508   |

| Reaction No. 67769                              |    |       |        |               |   |       |        |
|-------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4 + OH-1(2)_HPEDDA-4 = Th+4_OH-1(2)_HPEDDA-4 |    |       |        |               |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                               | 25 | 0.1   | KNO3   | lgK:12.8(3SF) | 0 | MGL   | 6508   |

| Reaction No. 67770                                                       |    |       |        |              |   |       |        |
|--------------------------------------------------------------------------|----|-------|--------|--------------|---|-------|--------|
| Th+4_OH-1(4)_HPEDDA-4 + 2<H+1> = Th+4_OH-1(2)_HPEDDA-4 + 2<H2O>          |    |       |        |              |   |       |        |
| Note: Weighted out coz its thermodynamically isolated; not nec. bad data |    |       |        |              |   |       |        |
| No                                                                       | t  | I-Str | Medium | Value (Dev)  | W | Tech. | Ref. # |
| 1                                                                        | 25 | 0.1   | KNO3   | lgK:7.8(2SF) | 0 | MGL   | 6508   |

| Reaction No. 69392                              |    |       |        |                 |   |       |        |
|-------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| 2ThenTriFAcone-1 + Th+4 = Th+4_2ThenTriFAcone-1 |    |       |        |                 |   |       |        |
| No                                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                               | 25 | 5     | NaCl   | lgK:7.1(0.05SD) | 5 | MPS   | 7144   |

| Reaction No. 69393              |    |       |        |                 |   |       |        |
|---------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Phenanth + Th+4 = Th+4_Phenanth |    |       |        |                 |   |       |        |
| No                              | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                               | 25 | 5     | NaCl   | lgK:3.8(0.05SD) | 5 | MPS   | 7144   |

| Reaction No. 69553                                                   |    |       |        |                |   |       |        |
|----------------------------------------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4_F-1(4)_H2O(4)_(s) + 2<H+1> = Th+4_F-1(2) + 2<H+1_F-1> + 4<H2O > |    |       |        |                |   |       |        |
| No                                                                   | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                    | 25 | 0.5   | NaClO4 | lgK:-7.23(3SF) | 5 | MRX   | 463    |

| Reaction No. 69554                      |    |       |               |                  |   |       |        |
|-----------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th+4 + H+1(3)_PO4-3 = Th+4_H+1(3)_PO4-3 |    |       |               |                  |   |       |        |
| No                                      | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                       | 25 | 0     | Inf. Dilution | lgK:1.89(0.16SD) | 5 | CRV   | 27322  |
| 2                                       | 25 | 0     | Inf. Dilution | lgK:1.89(0.31SD) | 3 | EOD   | 32233  |
| 3                                       | 25 | 2     | NaClO4        | lgK:1.89(3SF)    | 5 | MDS   | 366    |

| Reaction No. 69555                                                                             |    |       |               |                   |   |       |        |
|------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Th+4} + \text{H+1(3)}_{\text{PO4-3}} = \text{Th+4\_H+1(2)}_{\text{PO4-3}} + \text{H+1}$ |    |       |               |                   |   |       |        |
| No                                                                                             | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                              | 25 | 0     | Inf. Dilution | lgK:3.450(0.16SD) | 5 | CRV   | 27322  |
| 2                                                                                              | 25 | 0     | Inf. Dilution | lgK:3.45(0.32SD)  | 3 | EOD   | 32233  |
| 3                                                                                              | 25 | 2     | NaClO4        | lgK:2.18(3SF)     | 5 | MDS   | 366    |

| Reaction No. 69556                                                                                          |    |       |               |                   |   |       |        |
|-------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Th+4} + 2\text{<H+1(3)}_{\text{PO4-3}}\text{>} = \text{Th+4\_H+1(5)}_{\text{PO4-3(2)}} + \text{H+1}$ |    |       |               |                   |   |       |        |
| No                                                                                                          | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                           | 25 | 0     | Inf. Dilution | lgK:5.420(0.16SD) | 5 | CRV   | 27322  |
| 2                                                                                                           | 25 | 0     | Inf. Dilution | lgK:5.42(3SF)     | 3 | EOD   | 32233  |
| 3                                                                                                           | 25 | 2     | NaClO4        | lgK:4.15(3SF)     | 5 | MDS   | 366    |

| Reaction No. 69557                                                                                             |    |       |               |                   |   |       |        |
|----------------------------------------------------------------------------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| $\text{Th+4} + 2\text{<H+1(3)}_{\text{PO4-3}}\text{>} = \text{Th+4\_H+1(4)}_{\text{PO4-3(2)}} + 2\text{<H+1>}$ |    |       |               |                   |   |       |        |
| No                                                                                                             | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                                                                                              | 25 | 0     | Inf. Dilution | lgK:3.1(2SF)      | 1 | ELC   |        |
| 2                                                                                                              | 25 | 0     | Inf. Dilution | lgK:6.200(0.16SD) | 0 | CRV   | 27322  |
| 3                                                                                                              | 25 | 0     | Inf. Dilution | lgK:6.20(3SF)     | 0 | EOD   | 32233  |
| 4                                                                                                              | 25 | 2     | NaClO4        | lgK:3.90(3SF)     | 3 | MDS   | 366    |

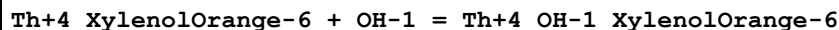
| Reaction No. 69859                                                                                         |    |       |               |                |   |       |        |
|------------------------------------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Th+4} + \text{H+1(6)}_{\text{XylenolOrange-6}} = \text{Th+4\_H+1\_XylenolOrange-6} + 5\text{<H+1>}$ |    |       |               |                |   |       |        |
| No                                                                                                         | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                                                          | 25 | 0     | Inf. Dilution | lgK:-6.8(2SF)  | 1 | ELC   |        |
| 2                                                                                                          | 35 | 0.1   | HNO3          | lgR:-8.66(3SF) | 3 | MGL   | 2332   |

| Reaction No. 69860                                                              |    |       |               |                |   |       |        |
|---------------------------------------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| $\text{Th+4} + \text{H+1\_XylenolOrange-6} = \text{Th+4\_H+1\_XylenolOrange-6}$ |    |       |               |                |   |       |        |
| No                                                                              | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                                                               | 25 | 0     | Inf. Dilution | lgK:19.8(3SF)  | 1 | ESC   |        |
| 2                                                                               | 35 | 0.1   | HNO3          | lgK:19.54(4SF) | 5 | MGL   | 2332   |

| Reaction No. 69861                                                              |  |  |  |  |  |  |  |
|---------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| $\text{Th+4\_H+1\_XylenolOrange-6} = \text{Th+4\_XylenolOrange-6} + \text{H+1}$ |  |  |  |  |  |  |  |

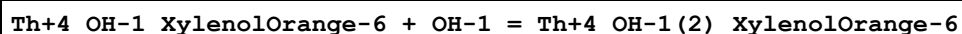
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-6.0 (2SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | HNO3          | lgR:-6.10 (3SF) | 5 | MGL   | 2332   |

## Reaction No. 69862



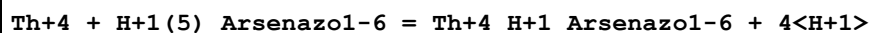
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:4.5 (2SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | HNO3          | lgK:4.30 (3SF) | 5 | MGL   | 2332   |

## Reaction No. 69863



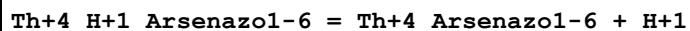
| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:3.5 (2SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | HNO3          | lgK:3.20 (3SF) | 5 | MGL   | 2332   |

## Reaction No. 69864



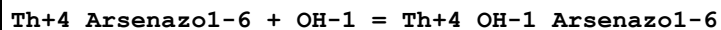
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-6.1 (2SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | HNO3          | lgR:-6.35 (3SF) | 3 | MGL   | 2332   |

## Reaction No. 69865



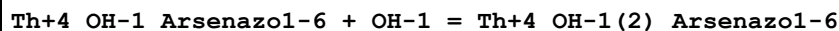
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-5.7 (2SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | HNO3          | lgR:-6.00 (3SF) | 5 | MGL   | 2332   |

## Reaction No. 69866



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:4.6 (2SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | HNO3          | lgK:4.45 (3SF) | 5 | MGL   | 2332   |

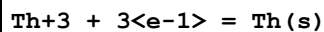
## Reaction No. 69867



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|   |    |     |               |               |   |     |      |
|---|----|-----|---------------|---------------|---|-----|------|
| 1 | 25 | 0   | Inf. Dilution | lgK:3.6(2SF)  | 1 | ESC |      |
| 2 | 35 | 0.1 | HNO3          | lgK:3.45(3SF) | 5 | MGL | 2332 |

## Reaction No. 70268



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-64.3(3SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | E:-1.17(3SF)   | 0 | ENS   | 7276   |

## Reaction No. 70943



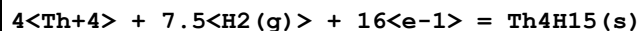
| No | t  | I-Str | Medium        | Value (Dev)  | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | E:-2.48(3SF) | 5 | CRV   | 6330   |
| 2  | 25 | 0     | Inf. Dilution | E:-2.48(3SF) | 4 | CRV   | 22519  |

## Reaction No. 70944



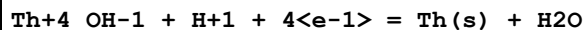
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-121.7(4SF) | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | E:-1.789(4SF)   | 0 | CRV   | 6330   |
| 3  | 25 | 0     | Inf. Dilution | E:-1.789(4SF)   | 0 | CRV   | 22519  |

## Reaction No. 72152



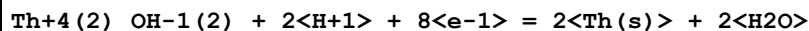
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | E:-1.468(4SF)     | 4 | CRV   | 7761   |
| 2  | 25 | 0     | Inf. Dilution | dH:2231.7(5SF) kJ | 4 | CRV   | 7761   |

## Reaction No. 72153



| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | E:-1.779(4SF)    | 0 | CRV   | 7761   |
| 2  | 25 | 0     | Inf. Dilution | dH:722.3(4SF) kJ | 2 | CRV   | 7761   |

## Reaction No. 72154



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

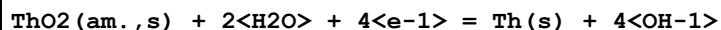
|   |    |   |               |                 |   |     |      |
|---|----|---|---------------|-----------------|---|-----|------|
| 1 | 25 | 0 | Inf. Dilution | lgK:-241.1(4SF) | 1 | ELC |      |
| 2 | 25 | 0 | Inf. Dilution | E:-1.78(3SF)    | 0 | CRV | 7761 |

## Reaction No. 72155



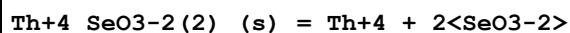
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | E:-2.627(4SF)    | 5 | CRV   | 7761   |
| 2  | 25 | 0     | Inf. Dilution | dH:878.0(4SF) kJ | 5 | CRV   | 7761   |

## Reaction No. 72156



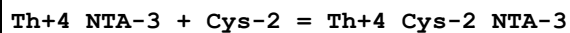
| No | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
|----|----|-------|---------------|------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-171.9(4SF)  | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | E:-2.55(3SF)     | 0 | CRV   | 7761   |
| 3  | 25 | 0     | Inf. Dilution | dH:844.9(4SF) kJ | 2 | CRV   | 7761   |

## Reaction No. 72962



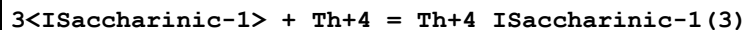
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 20 |       |               | lgK:-19.87(4SF) | 3 | MSL   | 12931  |
| 2  | 25 | 0     | Inf. Dilution | lgK:-19.87(4SF) | 3 | EOD   | 32222  |

## Reaction No. 73215



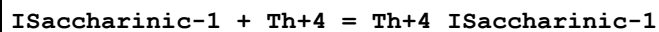
| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.7(2SF)  | 1 | ESC   |        |
| 2  | 35 | 0.1   | Unknown       | lgK:6.49(3SF) | 0 | MEP   | 6223   |

## Reaction No. 74670



| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 15 | 1     | NaClO4 | lgK:19.64(0.34SD) | 0 | MSE   | 22899  |
| 2  | 25 | 1     | NaClO4 | lgK:21.30(0.26SD) | 2 | MSE   | 22899  |
| 3  | 35 | 1     | NaClO4 | lgK:20.59(0.32SD) | 0 | MSE   | 22899  |

## Reaction No. 74671





| No | t  | I-Str | Medium | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|--------|-------------------|---|-------|--------|
| 1  | 15 | 1     | NaClO4 | lgK:12.88(3.31SD) | 5 | MSE   | 22899  |
| 2  | 25 | 1     | NaClO4 | lgK:12.56(5.01SD) | 5 | MSE   | 22899  |
| 3  | 35 | 1     | NaClO4 | lgK:12.93(4.16SD) | 5 | MSE   | 22899  |

| Reaction No. 74756                                                     |    |       |        |               |   |       |        |
|------------------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Leu-1 + Leu-1 = Th+4_Leu-1(2)                                     |    |       |        |               |   |       |        |
| No                                                                     | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                                      | 25 | 0.1   | KNO3   | lgK:5.89(3SF) | 3 | MGL   | 11516  |
| Note (1): Nourmand M et al., J. Chem. Soc., Dalton Trans., 1983, ,1529 |    |       |        |               |   |       |        |

| Reaction No. 75036          |      |       |               |                  |   |       |        |
|-----------------------------|------|-------|---------------|------------------|---|-------|--------|
| CrO4-2 + Th+4 = Th+4_CrO4-2 |      |       |               |                  |   |       |        |
| No                          | t    | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                           | 0.01 | 0     | Inf. Dilution | lgK:8.4604(5SF)  | 3 | EOD   | 22958  |
| 2                           | 25   | 0     | Inf. Dilution | lgK:8.7081(5SF)  | 3 | EOD   | 22958  |
| 3                           | 60   | 0     | Inf. Dilution | lgK:9.3345(5SF)  | 3 | EOD   | 22958  |
| 4                           | 100  | 0     | Inf. Dilution | lgK:10.2240(6SF) | 3 | EOD   | 22958  |
| 5                           | 150  | 0     | Inf. Dilution | lgK:11.4840(6SF) | 3 | EOD   | 22958  |
| 6                           | 200  | 0     | Inf. Dilution | lgK:12.883(5SF)  | 3 | EOD   | 22958  |

| Reaction No. 75647                 |    |       |        |               |   |       |        |
|------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Ile-1 + Ile-1 = Th+4_Ile-1(2) |    |       |        |               |   |       |        |
| No                                 | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                  | 25 | 0.1   | KNO3   | lgK:5.96(3SF) | 5 | MGL   | 1115   |

| Reaction No. 75670                    |    |       |               |               |   |       |        |
|---------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_Alal-1 + Alal-1 = Th+4_Alal-1(2) |    |       |               |               |   |       |        |
| No                                    | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                     | 25 | 0     | Inf. Dilution | lgK:6.0(2SF)  | 1 | ELC   |        |
| 2                                     | 25 | 0.1   | KNO3          | lgK:7.33(3SF) | 0 | MGL   | 1115   |

| Reaction No. 75680                 |    |       |        |               |   |       |        |
|------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Met-1 + Met-1 = Th+4_Met-1(2) |    |       |        |               |   |       |        |
| No                                 | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                  | 25 | 0.1   | KNO3   | lgK:6.66(3SF) | 5 | MGL   | 1115   |

| Reaction No. 75691                  |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4_Phe-1 + Phe-1 = Th+4_Phe-1 (2) |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KNO3   | lgK:6.67 (3SF) | 6 | MGL   | 1115   |

| Reaction No. 75723                  |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4_Ser-1 + Ser-1 = Th+4_Ser-1 (2) |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KNO3   | lgK:8.50 (3SF) | 4 | MGL   | 1115   |

| Reaction No. 75729                  |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4_Thr-1 + Thr-1 = Th+4_Thr-1 (2) |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KNO3   | lgK:6.80 (3SF) | 5 | MGL   | 1115   |

| Reaction No. 75736                  |    |       |        |                |   |       |        |
|-------------------------------------|----|-------|--------|----------------|---|-------|--------|
| Th+4_Cys-2 + Cys-2 = Th+4_Cys-2 (2) |    |       |        |                |   |       |        |
| No                                  | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                                   | 25 | 0.1   | KNO3   | lgK:7.29 (3SF) | 4 | MGL   | 1115   |

| Reaction No. 75787               |    |       |        |                 |   |       |        |
|----------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Th+4 + 2<Asn-1> = Th+4_Asn-1 (2) |    |       |        |                 |   |       |        |
| No                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                | 25 | 0.1   | NaClO4 | lgK:16.05 (4SF) | 3 | MGL   | 22630  |

| Reaction No. 75810                               |    |       |        |                 |   |       |        |
|--------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Th+4 + 2<Glu-2> = Th+4_Glu-2 (2)                 |    |       |        |                 |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                | 25 | 0.1   | NaClO4 | lgK:17.63 (4SF) | 4 | MGL   | 11516  |
| Note (1): Singh M et al., Talanta, 1972, 19, 699 |    |       |        |                 |   |       |        |

| Reaction No. 75840                               |    |       |        |                 |   |       |        |
|--------------------------------------------------|----|-------|--------|-----------------|---|-------|--------|
| Th+4 + 2<Asp-2> = Th+4_Asp-2 (2)                 |    |       |        |                 |   |       |        |
| No                                               | t  | I-Str | Medium | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                | 25 | 0.1   | NaClO4 | lgK:17.80 (4SF) | 3 | MGL   | 11516  |
| Note (1): Singh M et al., Talanta, 1972, 19, 699 |    |       |        |                 |   |       |        |

| Reaction No. 75853 |  |  |  |  |  |  |  |
|--------------------|--|--|--|--|--|--|--|
|--------------------|--|--|--|--|--|--|--|

| Th+4 + 2<Gln-1> = Th+4_Gln-1(2) |    |       |        |                |   |       |        |
|---------------------------------|----|-------|--------|----------------|---|-------|--------|
| No                              | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
| 1                               | 25 | 0.1   | NaClO4 | lgK:15.91(4SF) | 6 | MGL   | 22630  |

| Reaction No. 75864                                           |    |       |        |               |   |       |        |
|--------------------------------------------------------------|----|-------|--------|---------------|---|-------|--------|
| Th+4_Gly-1 + Gly-1 = Th+4_Gly-1(2)                           |    |       |        |               |   |       |        |
| No                                                           | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                            | 25 | 1     | NaClO4 | lgK:1.66(3SF) | 5 | MGL   | 3841   |
| 2                                                            | 35 | 0.01  | NaClO4 | lgK:3.82(3SF) | 5 | MEP   | 11516  |
| Note (2): Yadava H et al., Bull. Soc. Chim. Fr., 1984, , 314 |    |       |        |               |   |       |        |

| Reaction No. 75905                                      |    |       |         |               |   |       |        |
|---------------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Th+4_Citric-3 + Citric-3 = Th+4_Citric-3(2)             |    |       |         |               |   |       |        |
| No                                                      | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                       | 25 | 0.1   | NaCl    | lgK:9.53(3SF) | 2 | MGL   | 3900   |
| 2                                                       | 25 | 0.5   | Unknown | lgK:7.97(3SF) | 2 | MGL   | 11516  |
| Note (2): Nebel D et al., Z. Phys. Chem., 1966, 233, 73 |    |       |         |               |   |       |        |

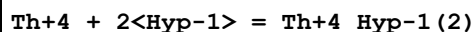
| Reaction No. 75926                          |    |       |               |               |   |       |        |
|---------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_Lactic-1 + Lactic-1 = Th+4_Lactic-1(2) |    |       |               |               |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                           | 20 | 1     | NaClO4        | lgK:3.57(3SF) | 5 | MEF   | 3132   |
| 2                                           | 25 | 0     | Inf. Dilution | lgK:3.5(2SF)  | 1 | ESC   |        |

| Reaction No. 75950                                             |    |       |               |               |   |       |        |
|----------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_Oxalic-2 + Oxalic-2 = Th+4_Oxalic-2(2)                    |    |       |               |               |   |       |        |
| No                                                             | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                              | 23 | 0.5   | NaNO3         | lgK:6.98(3SF) | 0 | MSP   | 11516  |
| Note (1): Grebennikova V et al., Radiokhim., 1970, 12, 279     |    |       |               |               |   |       |        |
| 2                                                              | 25 | 0     | Inf. Dilution | lgK:9.2(2SF)  | 1 | ELC   |        |
| 3                                                              | 25 | 1     | NH4ClO4       | lgK:8.57(3SF) | 0 | MSL   | 11516  |
| Note (3): Moskvina A et al., Zhur. Neorg. Khim., 1967, 12, 359 |    |       |               |               |   |       |        |
| 4                                                              | 25 | 1     | NaClO4        | lgK:6.26(3SF) | 3 | MDS   | 4706   |

| Reaction No. 76016                                   |   |       |        |             |   |       |        |
|------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Th+4_Salicylic-2 + Salicylic-2 = Th+4_Salicylic-2(2) |   |       |        |             |   |       |        |
| No                                                   | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

|   |    |     |        |               |   |     |     |
|---|----|-----|--------|---------------|---|-----|-----|
| 1 | 25 | 0.1 | NaClO4 | lgK:3.35(3SF) | 5 | MDS | 106 |
|---|----|-----|--------|---------------|---|-----|-----|

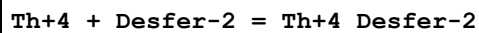
## Reaction No. 76111



| No | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------|---------------|---|-------|--------|
| 1  | 25 | 0.5   | Unknown | lgK:8.23(3SF) | 5 | MEF   | 11516  |

Note (1): Sergeev G et al., Radiokhim., 1973, 15, 618

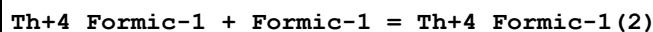
## Reaction No. 76217



| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 25 | 0.1   | KCl    | lgK:18.9(3SF) | 0 | MGL   | 11516  |

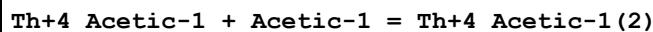
Note (1): wrong NDP Whisenhunt D et al., Inorg. Chem., 1996, 35, 4128

## Reaction No. 76249



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:2.06(3SF) | 5 | MEF   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:2.0(2SF)  | 1 | ESC   |        |

## Reaction No. 76269

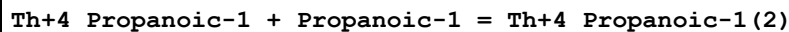


| No | t  | I-Str | Medium | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|--------|---------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4 | lgK:3.03(3SF) | 3 | MEF   | 2109   |
| 2  | 25 | 0.5   | KNO3   | lgK:0.05(1SF) | 0 | MEF   | 11516  |

Note (2): Sergeev G et al., Radiokhim., 1970, 12, 392

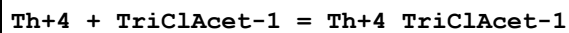
|   |    |   |        |               |   |     |      |
|---|----|---|--------|---------------|---|-----|------|
| 3 | 25 | 1 | NaClO4 | lgK:3.11(3SF) | 3 | MCL | 3099 |
|---|----|---|--------|---------------|---|-----|------|

## Reaction No. 76299



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 20 | 1     | NaClO4        | lgK:3.31(3SF) | 5 | MEF   | 2109   |
| 2  | 25 | 0     | Inf. Dilution | lgK:3.1(2SF)  | 1 | ESC   |        |

## Reaction No. 76473



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

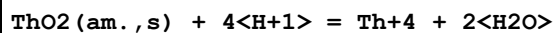
|   |    |     |        |                |   |     |      |
|---|----|-----|--------|----------------|---|-----|------|
| 1 | 25 | 0.5 | NaClO4 | lgK:1.62 (3SF) | 5 | MDS | 7943 |
|---|----|-----|--------|----------------|---|-----|------|

## Reaction No. 76474



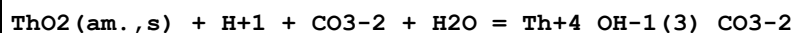
| No | t  | I-Str | Medium | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|--------|----------------|---|-------|--------|
| 1  | 25 | 0.5   | NaClO4 | lgK:1.20 (3SF) | 5 | MDS   | 7943   |

## Reaction No. 76816



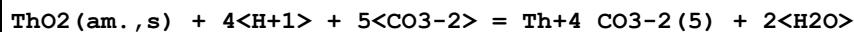
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.31 (0.3MD)  | 3 | EDH   | 4285   |
| 2  | 25 | 0.5   | NaClO4        | lgK:9.37 (0.13MD) | 5 | MSL   | 4285   |

## Reaction No. 76817



| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:6.78 (0.3MD)  | 3 | EDH   | 4285   |
| 2  | 25 | 0.5   | NaClO4        | lgK:6.11 (0.19MD) | 5 | MSL   | 4285   |

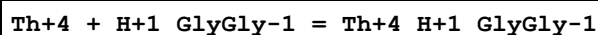
## Reaction No. 76818



| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:38.6 (3SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:39.64 (0.4MD)  | 0 | EDH   | 4285   |
| 3  | 25 | 0.5   | NaClO4        | lgK:42.12 (0.32MD) | 0 | MSL   | 4285   |

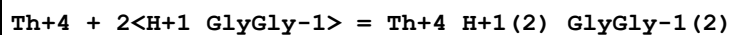
Note (3): Rxn balanced with H2O

## Reaction No. 76879



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 1     | Unknown | lgK:3.10 (3SF) | 3 | CRV   | 7271   |
| 2  | 25 | 1     | Unknown | dH:1.6 (2SF)   | 3 | CRV   | 7271   |

## Reaction No. 76880



| No | t  | I-Str | Medium  | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------|----------------|---|-------|--------|
| 1  | 25 | 1     | Unknown | lgK:5.40 (3SF) | 3 | CRV   | 7271   |

|   |    |   |         |             |   |     |      |
|---|----|---|---------|-------------|---|-----|------|
| 2 | 25 | 1 | Unknown | dH:3.2(2SF) | 3 | CRV | 7271 |
|---|----|---|---------|-------------|---|-----|------|

| Reaction No. 76881                               |    |       |         |               |   |       |        |
|--------------------------------------------------|----|-------|---------|---------------|---|-------|--------|
| Th+4 + 3<H+1_GlyGly-1> = Th+4_H+1(3)_GlyGly-1(3) |    |       |         |               |   |       |        |
| No                                               | t  | I-Str | Medium  | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                | 25 | 1     | Unknown | lgK:6.89(3SF) | 3 | CRV   | 7271   |
| 2                                                | 25 | 1     | Unknown | dH:4.6(2SF)   | 3 | CRV   | 7271   |

| Reaction No. 77293                                        |    |       |               |               |   |       |        |
|-----------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_CDTA-4 + Glycolic-1 = Th+4_Glycolic-1_CDTA-4         |    |       |               |               |   |       |        |
| No                                                        | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                         | 25 | 0     | Inf. Dilution | lgK:6.0(2SF)  | 1 | EAE   |        |
| 2                                                         | 35 | 0.1   | KNO3          | lgK:5.88(3SF) | 4 | EOD   | 22560  |
| Note (2): Pachauri & Tandon, Monat. Chem., 1976, 107, 991 |    |       |               |               |   |       |        |

| Reaction No. 77294                                        |    |       |               |               |   |       |        |
|-----------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| Th+4_CDTA-4 + Malic-2 = Th+4_Malic-2_CDTA-4               |    |       |               |               |   |       |        |
| No                                                        | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                         | 25 | 0     | Inf. Dilution | lgK:4.9(2SF)  | 1 | EES   |        |
| 2                                                         | 35 | 0.1   | KNO3          | lgK:4.79(3SF) | 4 | EOD   | 22560  |
| Note (2): Pachauri & Tandon, Monat. Chem., 1976, 107, 991 |    |       |               |               |   |       |        |

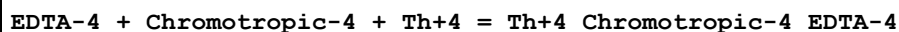
| Reaction No. 77295                                             |    |       |               |               |   |       |        |
|----------------------------------------------------------------|----|-------|---------------|---------------|---|-------|--------|
| 2<Th+4_CDTA-4> + 2<OH-1> = Th+4(2)_OH-1(2)_CDTA-4(2)           |    |       |               |               |   |       |        |
| No                                                             | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
| 1                                                              | 20 | 0.1   | NaClO4        | lgK:5.70(3SF) | 0 | EOD   | 22560  |
| Note (1): Bottari & Anderegg, Helv. Chim. Acta, 1967, 50, 2349 |    |       |               |               |   |       |        |
| 2                                                              | 25 | 0     | Inf. Dilution | lgK:17.1(3SF) | 1 | ELC   |        |

| Reaction No. 77905                              |    |       |               |                |   |       |        |
|-------------------------------------------------|----|-------|---------------|----------------|---|-------|--------|
| 3<Gly-1> + Th+4 + 3<H+1> = Th+4_H+1(3)_Gly-1(3) |    |       |               |                |   |       |        |
| No                                              | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                               | 25 | 0     | Inf. Dilution | lgK:34.93(4SF) | 1 | ELC   |        |

| Reaction No. 77947                      |   |       |        |             |   |       |        |
|-----------------------------------------|---|-------|--------|-------------|---|-------|--------|
| EDTA-4 + Th+4 + OH-1 = Th+4_OH-1_EDTA-4 |   |       |        |             |   |       |        |
| No                                      | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |

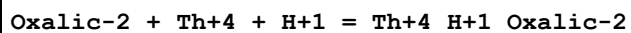
|   |    |   |               |                 |   |     |  |
|---|----|---|---------------|-----------------|---|-----|--|
| 1 | 25 | 0 | Inf. Dilution | lgK:33.37 (4SF) | 1 | ELC |  |
|---|----|---|---------------|-----------------|---|-----|--|

## Reaction No. 77961



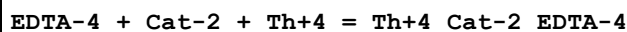
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:40.29 (4SF) | 1 | ELC   |        |

## Reaction No. 78006



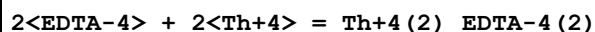
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:12.29 (4SF) | 1 | ELC   |        |

## Reaction No. 78387



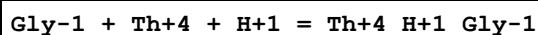
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:39.53 (4SF) | 1 | ELC   |        |

## Reaction No. 78524



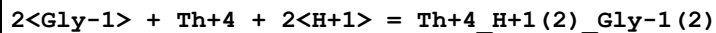
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:57.56 (4SF) | 1 | ELC   |        |

## Reaction No. 78559



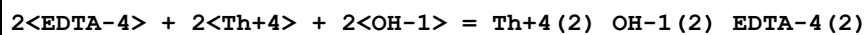
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:12.35 (4SF) | 1 | ELC   |        |

## Reaction No. 78574



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:23.8 (4SF) | 1 | ELC   |        |

## Reaction No. 78633



| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:70.77 (4SF) | 1 | ELC   |        |

## Reaction No. 80411

| Th+4 + 2<SO4-2> + 9<H2O> = Th+4_SO4-2(2)_H2O(9)_(s) |    |       |               |                     |   |       |        |
|-----------------------------------------------------|----|-------|---------------|---------------------|---|-------|--------|
| No                                                  | t  | I-Str | Medium        | Value (Dev)         | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:11.250(0.048SD) | 5 | CRV   | 27322  |

| Reaction No. 80412                                |    |       |               |                    |   |       |        |
|---------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Th+4_SeO3-2(2)_H2O_(s) = Th+4_SeO3-2(2)_(s) + H2O |    |       |               |                    |   |       |        |
| No                                                | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                 | 25 | 0     | Inf. Dilution | lgK:-6.499(0.28SD) | 5 | CRV   | 27322  |
| 2                                                 | 25 | 0     | Inf. Dilution | dH:94.100(1.5SD)kJ | 5 | CRV   | 27322  |

| Reaction No. 80413                          |    |       |               |                    |   |       |        |
|---------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Th+4 + 4<CO3-2> + OH-1 = Th+4_OH-1_CO3-2(4) |    |       |               |                    |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:35.600(0.25SD) | 5 | CRV   | 27322  |

| Reaction No. 80414                          |    |       |               |                   |   |       |        |
|---------------------------------------------|----|-------|---------------|-------------------|---|-------|--------|
| Th+4 + CO3-2 + 4<OH-1> = Th+4_OH-1(4)_CO3-2 |    |       |               |                   |   |       |        |
| No                                          | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
| 1                                           | 25 | 0     | Inf. Dilution | lgK:40.400(0.3SD) | 5 | CRV   | 27322  |

| Reaction No. 80415                                                      |    |       |               |                    |   |       |        |
|-------------------------------------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Th+4_Na+1(6)_CO3-2(5)_H2O(12)_(s) = 6<Na+1> + Th+4 + 5<CO3-2> + 12<H2O> |    |       |               |                    |   |       |        |
| No                                                                      | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                                                       | 25 | 0     | Inf. Dilution | lgK:-42.200(0.4SD) | 5 | CRV   | 27322  |

| Reaction No. 80416                        |    |       |               |                    |   |       |        |
|-------------------------------------------|----|-------|---------------|--------------------|---|-------|--------|
| Th+4 + 4<OH-1> = ThO2(am.aged,s) + 2<H2O> |    |       |               |                    |   |       |        |
| No                                        | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
| 1                                         | 25 | 0     | Inf. Dilution | lgK:47.500(0.45SD) | 5 | CRV   | 27322  |

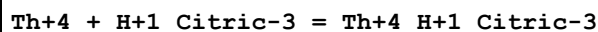
| Reaction No. 80587                              |    |       |               |                  |   |       |        |
|-------------------------------------------------|----|-------|---------------|------------------|---|-------|--------|
| Th+4 + 8<OH-1> + 4<Ca+2> = Th+4_Ca+2(4)_OH-1(8) |    |       |               |                  |   |       |        |
| No                                              | t  | I-Str | Medium        | Value (Dev)      | W | Tech. | Ref. # |
| 1                                               | 25 | 0     | Inf. Dilution | lgK:49.3(0.45SD) | 5 | CRV   | 31301  |

| Reaction No. 80588                                                            |   |       |        |             |   |       |        |
|-------------------------------------------------------------------------------|---|-------|--------|-------------|---|-------|--------|
| Th+4 + 3<H+1(2)_SiH2O4-2> + 3<H2O> = Th+4_H+1(3)_OH-1(3)_SiH2O4-2(3) + 6<H+1> |   |       |        |             |   |       |        |
| No                                                                            | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |



|   |    |   |               |                   |   |     |       |
|---|----|---|---------------|-------------------|---|-----|-------|
| 1 | 25 | 0 | Inf. Dilution | lgK:-27.8(0.35SD) | 5 | CRV | 31301 |
|---|----|---|---------------|-------------------|---|-----|-------|

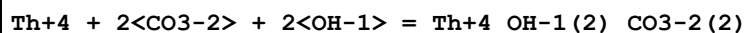
## Reaction No. 80638



| No | t  | I-Str | Medium        | Value (Dev)   | W | Tech. | Ref. # |
|----|----|-------|---------------|---------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:7.32(3SF) | 1 | CRV   | 31301  |

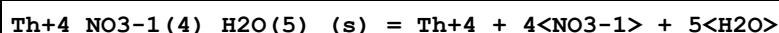
Note (1): Suspect B110

## Reaction No. 80740



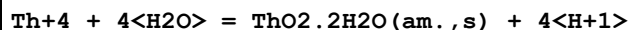
| No | t  | I-Str | Medium        | Value (Dev)        | W | Tech. | Ref. # |
|----|----|-------|---------------|--------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:36.800(0.25SD) | 5 | CRV   | 27322  |
| 2  | 25 | 0     | Inf. Dilution | lgK:36.8(3SF)      | 2 | EOD   | 29848  |

## Reaction No. 80875



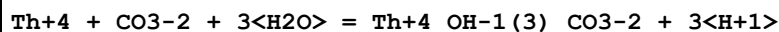
| No | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
|----|----|-------|---------------|-----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:2.0470(5SF) | 7 | MSL   | 31759  |

## Reaction No. 81098



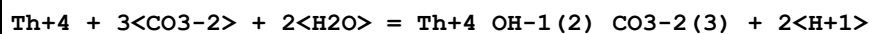
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-9.00(1.00SD) | 4 | CRV   | 32222  |

## Reaction No. 81099



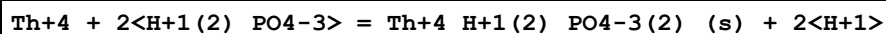
| No | t  | I-Str | Medium        | Value (Dev)       | W | Tech. | Ref. # |
|----|----|-------|---------------|-------------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:-0.9(1SF)     | 1 | ELC   |        |
| 2  | 25 | 0     | Inf. Dilution | lgK:-1.90(1.00SD) | 0 | CRV   | 32222  |

## Reaction No. 81100



| No | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
|----|----|-------|---------------|----------------|---|-------|--------|
| 1  | 25 | 0     | Inf. Dilution | lgK:10.20(4SF) | 4 | CRV   | 32222  |

## Reaction No. 81101



| No | t | I-Str | Medium | Value (Dev) | W | Tech. | Ref. # |
|----|---|-------|--------|-------------|---|-------|--------|
|----|---|-------|--------|-------------|---|-------|--------|

|   |    |   |               |                   |   |     |       |
|---|----|---|---------------|-------------------|---|-----|-------|
| 1 | 25 | 0 | Inf. Dilution | lgK:15.98(0.50SD) | 4 | CRV | 32222 |
|---|----|---|---------------|-------------------|---|-----|-------|

| Reaction No. 81185                |    |       |               |                |   |       |        |
|-----------------------------------|----|-------|---------------|----------------|---|-------|--------|
| Th+4_Cl-1(4)_(s) = Th+4 + 4<Cl-1> |    |       |               |                |   |       |        |
| No                                | t  | I-Str | Medium        | Value (Dev)    | W | Tech. | Ref. # |
| 1                                 | 25 | 0     | Inf. Dilution | lgK:23.68(4SF) | 4 | CRV   | 32224  |
| 2                                 | 25 | 0     | Inf. Dilution | dH:-60.0(3SF)  | 3 | EOD   | 32225  |
| Note (2): rxn 141                 |    |       |               |                |   |       |        |

| Reaction No. 81620                                  |    |       |               |                 |   |       |        |
|-----------------------------------------------------|----|-------|---------------|-----------------|---|-------|--------|
| Th+4_F-1(4)_H2O(2.5)_(s) = Th+4 + 4<F-1> + 2.5<H2O> |    |       |               |                 |   |       |        |
| No                                                  | t  | I-Str | Medium        | Value (Dev)     | W | Tech. | Ref. # |
| 1                                                   | 25 | 0     | Inf. Dilution | lgK:-33.67(4SF) | 2 | CRV   | 32224  |
| Note (1): source indicates uncertainty              |    |       |               |                 |   |       |        |

## Data Assessment

| Weight | Assessment                                    |
|--------|-----------------------------------------------|
| 9      | highest accuracy                              |
| 8      | multi-determined; excellent dependability     |
| 7      | trustworthy; outstanding single determination |
| 6      | better than average reliability               |
| 5      | average single determination                  |
| 4      | some doubt                                    |
| 3      | poorly known                                  |
| 2      | guess to achieve consistency (or just as bad) |
| 1      | no information but consistent                 |
| 0      | problematic/inconsistent                      |

## Techniques

| Abbrev | Method                              |
|--------|-------------------------------------|
| CMN    | Several experimental values (crit.) |

|     |                                         |
|-----|-----------------------------------------|
| CRV | Critical review                         |
| EAE | Automated extrapolation                 |
| EBU | Brown's Unified Theory                  |
| EDH | Debye-Hueckel corrections               |
| EDT | Debye-Hueckel & Temperature corrections |
| EES | Personal estimate                       |
| EFE | Free Energy relations                   |
| EIU | Critical IUPAC review (estim.)          |
| ELC | Linear Combination of Reactions         |
| EMS | Mean Spherical Approximation            |
| ENS | Not specified                           |
| EOD | Calculated from data of others          |
| EOR | Other Reaction Constants                |
| ESC | Standard conditions anchor              |
| EVK | Variation of lgK with T                 |
| MAG | Silver electrode e.m.f.                 |
| MAX | Anion exchange                          |
| MCE | Cation exchange                         |
| MCL | Calorimetry                             |
| MCN | Conductivity                            |
| MDG | Combination of Thermodynamic Data       |
| MDS | Distribution between two phases         |
| MEF | Electromotoric force, not specified     |
| MEP | Electrophoresis                         |
| MGL | Glass electrode                         |
| MHG | Emf with amalgam electrode              |
| MIS | Ion selective electrode                 |
| MIX | Ion exchange                            |
| MJM | Job's continuous variation              |
| MKN | Rate of reaction                        |
| MPH | pH method, not specified                |
| MPL | Polarography                            |
| MPS | Potentiometry and Spectrophotometry     |
| MPT | Potentiometry                           |
| MQH | Quinhydrone electrode                   |
| MRM | Raman spectra                           |

|     |                          |
|-----|--------------------------|
| MRX | Emf with redox electrode |
| MSE | Solvent Extraction       |
| MSL | Solubility               |
| MSP | Spectroscopy             |
| MTD | Temperature difference   |

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